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Bachelor's Thesis

Approximation and Optimization of Linear Continuous Time Recurrent Neural Networks

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Notation

The following table gives an overview of the notation used in this thesis. Some symbols have different meanings in different chapters. In Chapter 2, σ denotes the activation function, while $\sigma(W)$ denotes the spectrum of a matrix. The symbol λ denotes eigenvalues and also the Lebesgue measure in Chapter 4. In Chapter 6, ω denotes the memory parameter, while $\omega(g;\delta)$ denotes the modulus of continuity in Chapter 5.

Symbol	Meaning
General notation	
$\sigma(W)$	Spectrum of a matrix W
e_i	i-th standard basis vector
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra on $\mathbb{R}$
λ	Lebesgue measure (Chapter 4)
Norms	
$\|v\|_2$	Euclidean norm of a vector $v \in \mathbb{R}^q$ or $v \in \mathbb{C}^q$
$\|A\|_2$	Induced Euclidean operator norm of a matrix A, see (3.2)
$\|v\|_\infty$	Maximum norm of a vector $v \in \mathbb{R}^d$
$\|v\|_1$	ℓ^1-norm of a vector $v \in \mathbb{R}^d$
$\|x\|_\mathcal{X}$	Supremum norm on $\mathcal{X}$, $\|x\|_\mathcal{X} = \sup_{t\in\mathbb{R}} \|x_t\|_\infty$
$\|H\|$	Operator norm of a continuous linear functional H (Definition 3.1)
$\|f\|_{L^1([0,\infty))}$	L^1-norm of a function $f\colon [0,\infty) \to \mathbb{R}^d$, $\|f\|_{L^1([0,\infty))} = \int_0^\infty \|f(s)\|_1 \,\mathrm{d}s$ (Definition 4.7). For a scalar function on an interval I, such as $\mathbb{R}$ or $[0,1]$, $\|f\|_{L^1(I)} = \int_I \|f(s)\| \,\mathrm{d}s$
$\|f\|_{L^2(I)}$	L^2-norm on an interval I, such as $[0,\infty)$ or $\mathbb{R}$, $\|f\|_{L^2(I)} = \left(\int_I \|f(s)\|_2^2 \,\mathrm{d}s\right)^{1/2}$, where $\|f(s)\|_2 = \|f(s)\|$ for scalar f
$\|g\|_{\infty,[a,b]}$	Supremum norm of a bounded function g on $[a,b]$
$\|\nu\|$	Total variation measure of a finite signed measure ν (Definition 4.4)
$\|\nu\|_1(A)$	Sum of the total variations of the components of a vector valued signed measure ν on a Borel set A
Spaces and dimensions	
d	Dimension of the input space
m	Hidden dimension or width of the recurrent neural network

Symbol	Meaning
$\mathcal{X}$	Input space, $\mathcal{X} = C_0(\mathbb{R}, \mathbb{R}^d)$, of continuous functions that vanish at infinity
$\mathcal{Y}$	Output space, $\mathcal{Y} = C_b(\mathbb{R}, \mathbb{R})$, of bounded continuous functions
$L^p(I, \mathbb{R}^d)$, $L^p(I)$	For $p \in \{1, 2\}$, space of measurable functions $f\colon I \to \mathbb{R}^d$ with finite L^p-norm on an interval I, identified if they agree Lebesgue almost everywhere. We write $L^p(I)$ if $d = 1$
$\mathcal{P}_n$	Space of real polynomials of degree at most n
Discrete time RNNs	
T	Length of the input sequence
x_k, y_k	Input and target at time step k
h_k, $\widehat{y}_k$	Hidden state and prediction at time step k (Definition 2.1)
σ	Activation function, applied componentwise
$b \in \mathbb{R}^m$	Bias vector
$\theta = (W, U, b, c) \in \Theta$	Shared parameters of the discrete time RNN
$\mathcal{L}(\theta)$	Time averaged squared loss, see (2.1)
D_k	Diagonal matrix of activation derivatives at time step k
η_r	Learning rate in step r of gradient descent
Inputs and temporal functionals	
$x \in \mathcal{X}$	Input path
$x_t \in \mathbb{R}^d$	Input value at time t
$x^{(\tau)}$	Input shifted by τ, $x^{(\tau)}_s = x_{s-\tau}$ (Definition 3.4)
H_t	Target functional $H_t\colon \mathcal{X} \to \mathbb{R}$
$\{H_t\}_{t\in\mathbb{R}}$	Time indexed family of target functionals
$\mathcal{H}$	Class of continuous linear, causal, regular, and time homogeneous target functional families (Definition 3.5)
$\widehat{H}_t$	Functional generated by a stable linear continuous time RNN
$\widehat{\mathcal{H}}_m$	Class of functional families generated by stable linear continuous time RNNs with hidden dimension m
$\widehat{\mathcal{H}}$	Class of all stable linear continuous time RNN functional families, $\widehat{\mathcal{H}} = \bigcup_{m=1}^{\infty} \widehat{\mathcal{H}}_m$
$\widetilde{H}_t$	Kernel induced extension of H_t to step inputs (Proposition 4.12)
Linear continuous time RNN	
$h_t \in \mathbb{R}^m$	Hidden state at time t
$W \in \mathbb{R}^{m\times m}$	Recurrent matrix

Symbol	Meaning
$U \in \mathbb{R}^{m\times d}$	Input matrix
$c \in \mathbb{R}^m$	Linear readout vector
$\mathcal{W}_m$	Set of stable recurrent matrices, $$\mathcal{W}_m = \left\{ W \in \mathbb{R}^{m\times m} \,\middle|\, \max_{\lambda\in\sigma(W)} \operatorname{Re}\lambda < 0 \right\}$$
W_m, U_m, c_m	Parameters of the width m RNNs in the inverse approximation results (Theorem 5.11)
w_m	Largest real part of the eigenvalues of W_m, $w_m = \max_{\lambda\in\sigma(W_m)} \operatorname{Re}\lambda$
Memory kernels and step responses	
ρ	Target memory kernel
ρ_i	i-th component of the target memory kernel
$\widehat{\rho}$	Memory kernel generated by a stable linear continuous time RNN, $\widehat{\rho}(s) = U^\top e^{W^\top s} c$ (Definition 4.9)
$\mathbf{e}_i$	Step input associated with the i-th standard basis vector, $\mathbf{e}_{i,t} = e_i \mathbf{1}_{\{t\geq 0\}}$
y_i	Step response, $y_i(t) = \widetilde{H}_t(\mathbf{e}_i) = \int_0^t \rho_i(s)\,\mathrm{d}s$ for $t \geq 0$
$y_i^{(k)}$	Derivative of order k of the i-th step response
Approximation	
ε	Approximation tolerance
α	Smoothness order
β	Exponential memory decay rate
γ	Bound on exponentially weighted derivatives
$E_n(f; a, b)$	Error of the best uniform polynomial approximation of degree at most n (Definition 5.6)
$\omega(g; \delta)$	Modulus of continuity of g (Definition 5.7)
Optimization	
τ	Optimization time
$a = (a_1, \ldots, a_m)^\top$	Coefficients of the exponential modes
$w = (w_1, \ldots, w_m)^\top$	Positive exponential decay rates
$\vartheta = (a, w)$	Parameter vector of the reduced optimization problem
$\widehat{\rho}_{a,w}$	Reduced exponential sum kernel, $\widehat{\rho}_{a,w}(t) = \sum_{i=1}^m a_i e^{-w_i t}$
$J(a, w)$	Reduced kernel loss, see (6.3)
ω	Memory parameter, the long term component of the target is shifted by $1/\omega$
$\bar{\rho}$	Short term component of the target, see (6.9)

Symbol	Meaning
m^*	Number of exponential modes of $\bar{\rho}$
ρ_0, $\rho_{0,\omega}$	Long term component and its shift, $\rho_{0,\omega}(t) = \rho_0(t - 1/\omega)$
ρ_ω	Target kernel with long memory, $\rho_\omega = \bar{\rho} + \rho_{0,\omega}$, see (6.8)
J_ω	Kernel loss for the target ρ_ω, see (6.10)
$\overline{J}$	Short term kernel loss, see (6.18)
$\Delta_{n,\omega}(w)$	Overlap terms, see (6.14)
$\vartheta(\tau)$, $\vartheta_\omega(\tau)$	Parameter trajectory under gradient flow
ϑ_0	Initial parameter vector that represents $\bar{\rho}$ exactly
τ_0, τ_0'	Parameter and loss escape times for the tolerance δ (Theorem 6.5)

1 Introduction

Recurrent neural networks have a hidden state that carries information from one time step to the next. In theory, this state can retain information for arbitrarily long times [GBC16, Ch. 10]. In practice, RNNs can already struggle with simple tasks that require remembering information over many time steps, as shown by Bengio et al. [BSF94, pp. 158–159].

A common explanation is given by vanishing and exploding gradients [BSF94; PMB13]. The gradient has to be propagated backwards through many recurrent steps and can become very small or very large. This describes a difficulty of gradient based training, but it does not separate properties of the model from properties of the training procedure. There are two different questions behind this problem. Can an RNN approximate a target with long memory, and can gradient based training find suitable parameters?

Classical approximation results show that recurrent neural networks can approximate dynamical systems. For example, Schäfer and Zimmermann prove a universal approximation result for discrete time RNNs, but their result does not give an approximation rate [SZ06]. Li et al. study a different setting, where the target is described by a family of linear functionals [LHEL22, pp. 7–9]. They study approximation and optimization for linear continuous time RNNs and introduce the term *curse of memory* to describe the difficulties caused by long memory [LHEL22]. The main reference for this thesis is the paper *Approximation and Optimization Theory for Linear Continuous-Time Recurrent Neural Networks* by Li et al. [LHEL22].

To study these questions separately, Li et al. replace the discrete time RNN by a linear continuous time model [LHEL22, pp. 6–9]. We follow this reduction. Its purpose is not to describe general RNNs as realistically as possible. It gives a setting in which the memory structure can be studied directly using ordinary differential equations and functional analysis [cf. LHEL22, p. 6] .

Chapters 2 and 3 introduce the models used in the analysis. The main results of this thesis concern the approximation and optimization of linear continuous time RNNs. First, we study the representation of linear temporal functionals by integrable memory kernels in Chapter 4 [LHEL22, pp. 15–16, Lem. 15]. This representation allows us to study both approximation and optimization through the same mathematical object. In Chapter 5, we present the universal approximation result of Li et al. and study approximation rates [LHEL22, Thms. 7 and 10]. For sufficiently smooth targets with exponentially decaying memory, the approximation error decreases at a rate of order $m^{-\alpha}$, where m is the RNN width [LHEL22, p. 12, Thm. 10]. We also study inverse approximation and show that the inverse approximation theorem of Li et al. [LHEL22, pp. 12–13, Thm. 11] does not hold for general nonnormal recurrent matrices. Under the additional assumption of normal recurrent matrices, we prove a modified version of the theorem. Finally, Chapter 6 studies the optimization problem. We present the result of Li et al. showing that gradient flow can have exponentially long plateaus as the memory length increases [LHEL22, Thm. 20]. These results show that long memory does not prevent approximation, but it can make approximation and optimization more difficult. The appendix contains the detailed BPTT derivation and the numerical validation of the plateau experiment.

Our own contribution is to give detailed proofs of several results from Li et al. and fill in steps that are only briefly explained in their paper [LHEL22]. In the approximation

theory chapter, we work out the proofs, including the induction arguments and the treatment of boundary cases. We also introduce the measure theoretic foundations needed for the memory kernel representation and give a detailed proof of this result (Lemma 4.8). Furthermore, we define an extension of the functionals to step inputs, which do not belong to the original input space $\mathcal{X}$ (Proposition 4.12). An important part of our own work is the construction of a counterexample to the inverse approximation theorem for nonnormal recurrent matrices (Proposition 5.12). We then prove a modified version of the theorem for normal recurrent matrices (Theorem 5.14). In addition, we implement numerical experiments to illustrate vanishing gradients, approximation rates, and optimization plateaus.

2 Recurrent Neural Networks

Recurrent neural networks (RNNs) are neural networks with a hidden state that can retain information from the past and the same parameters are shared across different time steps [GBC16, Ch. 10, pp. 370–371]. This chapter introduces discrete time RNNs as motivation for the continuous time models studied later. We first describe their training by backpropagation through time and then discuss vanishing gradients.

The main references for this chapter are Goodfellow et al. [GBC16, Chs. 4, 5, 8 and 10], Pascanu et al. [PMB13], and Bengio et al. [BSF94].

This chapter mainly presents known results on discrete time RNNs. We make some changes to the RNN model of Goodfellow et al. by using a scalar linear output, a general activation function, and a time averaged squared loss [cf. GBC16, pp. 373–374]. We also give a detailed proof of the BPTT gradients for this model in Appendix 8.1. In Example 2.3, we implement a simple experiment to illustrate the memory problem of RNNs. Finally, we calculate how gradients vanish in linear RNNs with symmetric recurrent matrices.

2.1 Discrete Time Recurrent Neural Networks

We use a scalar output variant of the recurrent neural network described by Goodfellow et al. [GBC16, Ch. 10, pp. 373–374, Fig. 10.3 and Eqs. (10.8)–(10.11)].

Definition 2.1 (Discrete time RNN)**.** Let $d, m, T \in \mathbb{N}_+$, let $x_1, \dots, x_T \in \mathbb{R}^d$ be an input sequence, and let $h_0 \in \mathbb{R}^m$ be the initial hidden state. Let $\sigma : \mathbb{R} \to \mathbb{R}$ be an activation function. We use the same symbol for its componentwise extension $\sigma : \mathbb{R}^m \to \mathbb{R}^m$, defined by

$$(z_1, \dots, z_m)^\top \mapsto \big(\sigma(z_1), \dots, \sigma(z_m)\big)^\top.$$

Let $W \in \mathbb{R}^{m \times m}$, $U \in \mathbb{R}^{m \times d}$, and $b, c \in \mathbb{R}^m$. Define the state transition map

$$F : \mathbb{R}^m \times \mathbb{R}^d \to \mathbb{R}^m, \qquad F(h, x) \coloneqq \sigma(Wh + Ux + b),$$

and the output map

$$G : \mathbb{R}^m \to \mathbb{R}, \qquad G(h) \coloneqq c^\top h.$$

The discrete time recurrent neural network is defined recursively by

$$h_k = F(h_{k-1}, x_k), \qquad \widehat{y}_k = G(h_k), \qquad k = 1, \dots, T.$$

Goodfellow et al. [GBC16, p. 374, Eqs. (10.8)–(10.11)] use the hyperbolic tangent as the activation function and a vector valued output with an output bias followed by a softmax transformation. In contrast, we allow a general activation function and use the scalar linear output $\widehat{y}_k = c^\top h_k$ without an output bias. This output form is also used for the continuous time RNNs studied in the following chapters. The formulation using the maps F and G is our own way of presenting this model.

By the recursive update, h_k depends on earlier hidden states and can therefore carry information from earlier inputs [PMB13, Sec. 1].

2.2 Backpropagation Through Time (BPTT)

We first formulate the training problem. Backpropagation through time (BPTT) then computes the gradient of the resulting loss with respect to the RNN parameters by applying the chain rule backwards through the sequence [GBC16, pp. 378–380].

Consider one training sequence $x_1, \dots, x_T \in \mathbb{R}^d$ with corresponding targets $y_1, \dots, y_T \in \mathbb{R}$. The same shared parameters are used at every time step, so the network is described by a single tuple

$$\theta = (W, U, b, c) \in \Theta, \qquad \Theta \coloneqq \mathbb{R}^{m\times m} \times \mathbb{R}^{m\times d} \times \mathbb{R}^m \times \mathbb{R}^m.$$

For each $\theta \in \Theta$, with $h_0 \in \mathbb{R}^m$ fixed, Definition 2.1 determines hidden states $h_k(\theta) \in \mathbb{R}^m$ and predictions $\widehat{y}_k(\theta) \in \mathbb{R}$, $k = 1, \dots, T$.

The aim of training is to choose the parameters such that the predictions are close to the corresponding targets. For example, x_k may contain weather measurements recorded on day k, such as temperature, humidity, and air pressure, while y_k is the temperature on the following day.

A standard loss for regression is the squared error [GBC16, p. 106]. We therefore define $\ell : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ by

$$\ell(\widehat{y}, y) \coloneqq (\widehat{y} - y)^2.$$

Goodfellow et al. [cf. GBC16, p. 374, Eqs. (10.12)–(10.14)] form the total loss by summing the loss contributions over the time steps. We use the same structure but scale the sum by $1/T$. For one training sequence, we therefore define the time averaged loss $\mathcal{L} : \Theta \to [0, \infty)$ by

$$\mathcal{L}(\theta) \coloneqq \frac{1}{T} \sum_{k=1}^{T} \ell\big(\widehat{y}_k(\theta), y_k\big) = \frac{1}{T} \sum_{k=1}^{T} \big(\widehat{y}_k(\theta) - y_k\big)^2. \tag{2.1}$$

For fixed T, the factor $1/T$ only rescales the loss and its gradient and does not change its minimizers. The choice of the squared error loss and the factor $1/T$ is our own.

If several training sequences are available, the corresponding sequence losses can be averaged over the training set. In the following, we consider one training sequence to keep the notation simple.

For the gradient derivation below, we assume $\sigma \in C^1(\mathbb{R})$. Together with the squared error loss, this makes $\mathcal{L}$ differentiable with respect to the parameters.

Training aims to minimize $\mathcal{L}$ over Θ. Since $\mathcal{L}$ is in general nonconvex, gradient based methods need not find a global minimizer [GBC16, Sec. 8.2, pp. 279–281].

Training consists of a forward pass, in which the hidden states, predictions, and loss are computed, and a backward pass, in which the gradient of $\mathcal{L}$ with respect to the shared parameters is determined [GBC16, pp. 374–380].

2.2.1 Forward Pass

Before the first training iteration, the model parameters are initialized. A common choice is to initialize the entries of W, U, and c randomly from a Gaussian or uniform distribution, while b is initialized by a fixed constant [GBC16, p. 298].

During the forward pass, the RNN is evaluated from $k = 1$ to $k = T$. Starting from the fixed initial state h_0, we compute

$$h_k(\theta) = \sigma(W h_{k-1}(\theta) + U x_k + b), \qquad \widehat{y}_k(\theta) = c^\top h_k(\theta), \qquad k = 1, \dots, T.$$

After the predictions have been computed, the loss $\mathcal{L}(\theta)$ is evaluated according to (2.1).

2.2.2 Backward Pass

The following proposition gives the BPTT gradients. In this subsection, we suppress the dependence on θ and write h_k, a_k, and $\widehat{y}_k$. For $k = 1, \dots, T$, define

$$a_k \coloneqq W h_{k-1} + U x_k + b \in \mathbb{R}^m,$$

and

$$D_k \coloneqq \operatorname{diag}\big(\sigma'((a_k)_1), \dots, \sigma'((a_k)_m)\big) \in \mathbb{R}^{m \times m}.$$

We write $\nabla_{h_k} \mathcal{L}(\theta) \in \mathbb{R}^m$ for the gradient of the loss with respect to the hidden state h_k. It describes the total effect of h_k on the loss, both directly at time k and indirectly through later hidden states [Wer90, p. 1552, Eq. (7)]. A precise definition is given in Appendix 8.1.

For a scalar valued function of a matrix parameter, we use $\nabla_W \mathcal{L}$ for the matrix of its scalar partial derivatives, and analogously for U.

Goodfellow et al. derive the BPTT gradients for an RNN with a hyperbolic tangent activation, a softmax output, and the negative log-likelihood loss [GBC16, pp. 379–380, Eqs. (10.17)–(10.28)]. We adapt this derivation to the RNN defined above, with a general activation function $\sigma \in C^1(\mathbb{R})$, a scalar linear output, and the squared error loss.

Proposition 2.2 (BPTT gradients [cf. GBC16, Sec. 10.2.2, pp. 378–380])**.** *Assume that $\sigma \in C^1(\mathbb{R})$. The terminal hidden-state gradient is*

$$\nabla_{h_T} \mathcal{L}(\theta) = \frac{2}{T}\big(c^\top h_T - y_T\big) c \in \mathbb{R}^m.$$

For $k = T-1, \dots, 1$,

$$\nabla_{h_k} \mathcal{L}(\theta) = \frac{2}{T}\big(c^\top h_k - y_k\big) c + W^\top D_{k+1} \nabla_{h_{k+1}} \mathcal{L}(\theta) \in \mathbb{R}^m.$$

The parameter gradients are

$$\nabla_W \mathcal{L}(\theta) = \sum_{k=1}^{T} D_k \nabla_{h_k} \mathcal{L}(\theta) h_{k-1}^\top \in \mathbb{R}^{m \times m},$$

$$\nabla_U \mathcal{L}(\theta) = \sum_{k=1}^{T} D_k \nabla_{h_k} \mathcal{L}(\theta) x_k^\top \in \mathbb{R}^{m \times d},$$

$$\nabla_b \mathcal{L}(\theta) = \sum_{k=1}^{T} D_k \nabla_{h_k} \mathcal{L}(\theta) \in \mathbb{R}^m,$$

and

$$\nabla_c \mathcal{L}(\theta) = \frac{2}{T} \sum_{k=1}^{T} \big(c^\top h_k - y_k\big) h_k \in \mathbb{R}^m.$$

Proof. We give a detailed proof in Appendix 8.1. We use separate parameter copies for each time step, as in Goodfellow et al. [GBC16, p. 380]. Our own work is the detailed calculation of the gradients for the RNN defined in this chapter. In the appendix, we also define the gradient with respect to the hidden state using the loss from time step k onwards. We show the chain rule calculations step by step. □

The backward pass starts with $\nabla_{h_T}\mathcal{L}(\theta)$ and evaluates the hidden state recursion backwards from $k = T$ to $k = 1$. The resulting hidden state gradients are then used to compute the parameter gradients.

2.2.3 Parameter Update

The gradient points in the direction of the steepest local increase of the loss. Gradient descent therefore updates the parameters in the opposite direction [GBC16, Sec. 4.3, p. 83]. We collect the parameter gradients from Proposition 2.2 in

$$\nabla_\theta \mathcal{L}(\theta) \coloneqq \big(\nabla_W \mathcal{L}(\theta), \nabla_U \mathcal{L}(\theta), \nabla_b \mathcal{L}(\theta), \nabla_c \mathcal{L}(\theta)\big) \in \Theta.$$

Starting from an initial parameter value $\theta_0 \in \Theta$, gradient descent generates the iterates

$$\theta_{r+1} = \theta_r - \eta_r \nabla_\theta \mathcal{L}(\theta_r), \qquad r = 0, 1, \ldots,$$

where $\eta_r > 0$ is the learning rate and the operations are understood componentwise in Θ [GBC16, Sec. 4.3, pp. 83–84].

The forward pass, backward pass, and parameter update are repeated until a stopping criterion is reached [GBC16, Sec. 8.3.1, p. 291, Alg. 8.1]. As noted above, the loss is in general non convex, so gradient descent does not guarantee convergence to a global minimizer [GBC16, pp. 281–283].

2.2.4 Vanishing and Exploding Gradients

For long term dependencies, the learning signal has to propagate backwards across many time steps. During this propagation, the gradient can become very small or very large. These two cases are called vanishing and exploding gradients [PMB13, Sec. 2]. Bengio et al. show that gradient based learning becomes more difficult as the temporal distance between relevant inputs and outputs increases [BSF94, pp. 157–161].

Example 2.3 (Copy-first memory task). We consider a simple memory task based on the long term dependency setting of Bengio et al. [BSF94, pp. 158–159]. We set $d = 1$. The input sequence is given by

$$x_1, \ldots, x_T, \qquad x_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1),$$

and the target at the final time step is

$$y_T = x_1.$$

The RNN processes the complete sequence and predicts y_T from its final hidden state h_T. It therefore has to retain the information from x_1 over $T - 1$ time steps. In this experiment, only the squared error of the final prediction,

$$\big(\widehat{y}_T - y_T\big)^2,$$

is used as the loss.

We train the same one layer RNN for

$$T \in \{4, 10, 20, 50\}.$$

The RNN has hidden dimension $m = 16$ and uses a tanh activation. It is trained for 20 epochs with the Adam optimizer [KB15], learning rate 0.01, and batch size 32. We use 1000 training sequences and 200 test sequences. Figure 1 shows the test mean squared error. The experiment is implemented using PyTorch [Pas+19]. We choose the network parameters and training settings, implement the experiment, and create Figure 1 ourselves. The implementation is available in `experiments/01_copy_first_memory.py`.

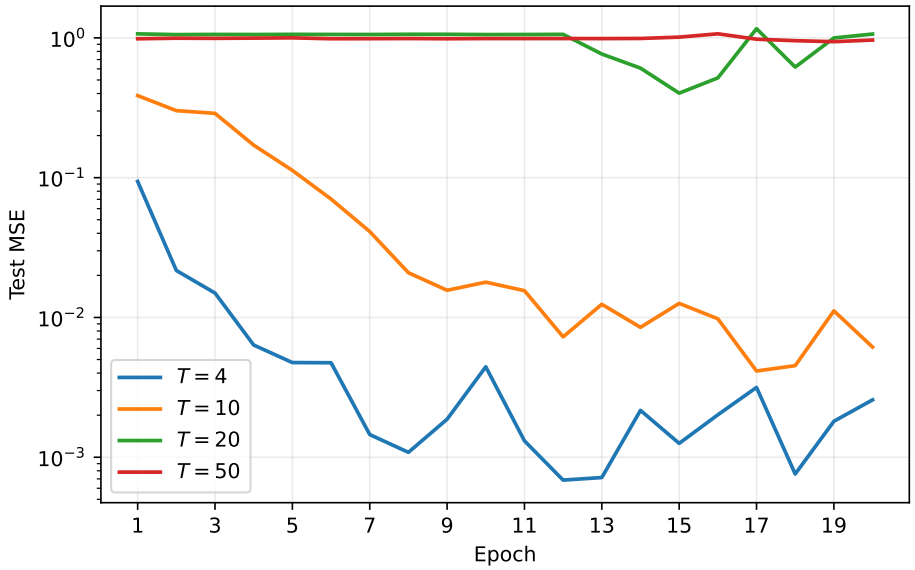


Figure 1: Test mean squared error on the copy-first memory task for different sequence lengths T. The curves show one training run.

A predictor that always returns zero has expected mean squared error

$$\mathbb{E}[x_1^2] = 1.$$

For $T = 4$ and $T = 10$, the RNN learns the task well. For $T = 20$, the test error decreases during parts of training but remains unstable and returns to a value close to one. For $T = 50$, the error remains close to one throughout most of the training run. The experiment illustrates that even this simple memory task becomes more difficult as the temporal distance increases.

We now return to the general setting of Definition 2.1. The restriction $d = 1$ was only used for the copy-first memory task.

To see where vanishing and exploding gradients come from, consider the loss contribution at time step n, defined by $\mathcal{L}_n \colon \Theta \to [0, \infty)$,

$$\mathcal{L}_n(\theta) = \frac{1}{T}\left(c^\top h_n - y_n\right)^2.$$

For $k < n$, we use $\nabla_{h_k} \mathcal{L}_n(\theta)$ for the gradient of this single loss contribution with respect to h_k. The loss $\mathcal{L}_n$ depends on h_k through the intermediate hidden states $h_{k+1}, \ldots, h_n$. The multivariable chain rule gives

$$\nabla_{h_k} \mathcal{L}_n(\theta) = W^\top D_{k+1} \nabla_{h_{k+1}} \mathcal{L}_n(\theta).$$

Applying this relation repeatedly gives

$$\nabla_{h_k}\mathcal{L}_n(\theta) = W^\top D_{k+1} W^\top D_{k+2} \cdots W^\top D_n \nabla_{h_n}\mathcal{L}_n(\theta).$$

This is the temporal gradient structure described by Pascanu et al. [PMB13, Secs. 1.1–2.1].

In the following, $\|\cdot\|_2$ denotes the Euclidean norm for vectors and the induced operator norm for matrices. For $A \in \mathbb{R}^{p\times q}$,

$$\|A\|_2 \coloneqq \sup_{\substack{v\in\mathbb{R}^q \\ \|v\|_2=1}} \|Av\|_2.$$

Repeated application of

$$\|Av\|_2 \leq \|A\|_2 \|v\|_2$$

gives

$$\|\nabla_{h_k}\mathcal{L}_n(\theta)\|_2 \leq \prod_{s=k+1}^{n} \left\|W^\top D_s\right\|_2 \|\nabla_{h_n}\mathcal{L}_n(\theta)\|_2 .$$

We now give our own calculation to illustrate the vanishing gradient problem for linear RNNs with symmetric recurrent matrices. This is a special case of the general analysis of Pascanu et al. [PMB13, Sec. 2.1].

For a linear RNN, $\sigma(z) = z$ and $D_s = I_m$. We then get

$$\nabla_{h_k}\mathcal{L}_n(\theta) = (W^\top)^{n-k} \nabla_{h_n}\mathcal{L}_n(\theta).$$

If W is symmetric with eigenvalues $\lambda_1, \ldots, \lambda_m$, then the spectral theorem [FS25, Sec. 6.7.2, p. 364] and the orthogonal invariance of the Euclidean norm give

$$\|W^\top\|_2 = \max_{1\leq j\leq m} |\lambda_j|.$$

The norm estimate above gives

$$\|\nabla_{h_k}\mathcal{L}_n(\theta)\|_2 \leq \left(\max_{1\leq j\leq m} |\lambda_j|\right)^{n-k} \|\nabla_{h_n}\mathcal{L}_n(\theta)\|_2 .$$

Let

$$q \coloneqq \max_{\lambda\in\sigma(W)} |\lambda|.$$

If $q < 1$, then, because W is symmetric,

$$\left\|(W^\top)^{n-k}\right\|_2 = q^{n-k} \longrightarrow 0$$

as $n - k \to \infty$. Since $q < 1$, the gradient transport factor decays exponentially with the temporal distance $n - k$. This illustrates the vanishing gradient problem even for linear RNNs.

Pascanu et al. show more generally for linear RNNs that

$$\max_{\lambda\in\sigma(W)} |\lambda| < 1$$

is sufficient for long term gradient contributions to vanish, while

$$\max_{\lambda\in\sigma(W)} |\lambda| > 1$$

is necessary for them to grow exponentially [PMB13, Sec. 2.1].

3 Linear Continuous Time RNNs and Functionals

Chapter 2 introduced recurrent neural networks in discrete time. Following Li et al., we now study a linear continuous time model, where the activation function is the identity and there is no bias term [LHEL22, Secs. 3–4, pp. 5–10]. The main reference for this chapter is Li et al. [LHEL22, Secs. 3–4, pp. 5–10]. For linear differential equations and the matrix exponential, we use Teschl [Tes12, Ch. 3].

We first define the input and output spaces and the properties of the functionals acting on them. We then introduce linear continuous time RNNs and derive their infinite past representation. Finally, we study the structural properties of these functionals.

Our own work in this chapter consists of several detailed derivations and proofs. We derive the solution of the state equation step by step and explain the properties of the matrix exponential used in the proof. We also give a detailed proof of the exponential decay estimate with explicit constants. Furthermore, we prove the infinite past representation and the structural properties of RNN functionals. Li et al. give the infinite past representation formally and state the structural properties without proof [LHEL22, pp. 9–10, Eq. (18) and Prop. 6]. We also explain the connection between discrete and continuous time RNNs.

3.1 Input and Output Spaces

We consider continuous input signals

$$x\colon \mathbb{R} \to \mathbb{R}^d, \qquad t \mapsto x_t.$$

Following Li et al., we define the input space by

$$\mathcal{X} \coloneqq C_0(\mathbb{R}, \mathbb{R}^d),$$

where $C_0(\mathbb{R}, \mathbb{R}^d)$ denotes the space of continuous functions that vanish at infinity [LHEL22, p. 6, Eq. (3)]. Every $x \in \mathcal{X}$ satisfies

$$\lim_{|t|\to\infty} \|x_t\|_\infty = 0.$$

Here $\|\cdot\|_\infty$ is the maximum norm on $\mathbb{R}^d$.

We equip $\mathcal{X}$ with the supremum norm

$$\|x\|_{\mathcal{X}} \coloneqq \sup_{t\in\mathbb{R}} \|x_t\|_\infty$$

[LHEL22, p. 6, Eq. (4)].

For the output space, we use

$$\mathcal{Y} \coloneqq C_b(\mathbb{R}, \mathbb{R}),$$

the space of bounded continuous scalar valued functions [LHEL22, p. 6, Eq. (5)].

3.2 Continuous Time Setting

We describe the target relation by a family of functionals

$$H_t \colon \mathcal{X} \to \mathbb{R}, \qquad y_t = H_t(x).$$

For every $x \in \mathcal{X}$, we write

$$H(x) \coloneqq \big(H_t(x)\big)_{t\in\mathbb{R}}$$

[LHEL22, pp. 6–7, Eqs. (6)–(7)].

Following Li et al., we assume for the moment that the output is continuous in time,

$$\lim_{\delta\to 0} H_{t+\delta}(x) = H_t(x)$$

for every $t \in \mathbb{R}$ and $x \in \mathcal{X}$ [LHEL22, p. 7, Eq. (8)]. Together with boundedness, this implies that

$$H(x) \in \mathcal{Y}.$$

The continuity condition will later follow from the properties imposed on the target functional family.

Definition 3.1 (Continuous linear functional)**.** A functional $H \colon \mathcal{X} \to \mathbb{R}$ is linear if

$$H(\lambda x + \mu z) = \lambda H(x) + \mu H(z)$$

for all $x, z \in \mathcal{X}$ and $\lambda, \mu \in \mathbb{R}$. A linear functional is continuous if its operator norm

$$\|H\| \coloneqq \sup_{\|x\|_{\mathcal{X}}\le 1} |H(x)|$$

is finite [LHEL22, p. 8, Eqs. (12)–(14)].

Definition 3.2 (Causal functional)**.** A functional H_t is causal if

$$x_r = z_r \quad \text{for all } r \le t \qquad \Longrightarrow \qquad H_t(x) = H_t(z)$$

for all $x, z \in \mathcal{X}$ [LHEL22, p. 7, Eq. (11)].

A causal functional at time t therefore depends only on the input values up to time t. We next define the regularity property used below.

Definition 3.3 (Regular functionals [LHEL22, p. 8, Def. 4, Eq. (15)])**.** We call a functional

$$H \colon \mathcal{X} \to \mathbb{R}$$

regular if, for every sequence

$$\big\{x^{(n)} \in \mathcal{X} : n \in \mathbb{N}_+\big\}$$

such that

$$\sup_n \big\|x^{(n)}\big\|_{\mathcal{X}} \le 1$$

and

$$x_t^{(n)} \xrightarrow[n\to\infty]{} 0 \qquad \text{in } \mathbb{R}^d$$

for almost every $t \in \mathbb{R}$ with respect to the Lebesgue measure, we have

$$\lim_{n\to\infty} H\big(x^{(n)}\big) = 0.$$

We say that the family

$$\{H_t\}_{t\in\mathbb{R}}$$

is regular if each H_t is regular.

Regularity excludes dependence on input values concentrated in arbitrarily small time intervals [LHEL22, p. 8]. This property is later used to get an integral kernel representation of the target functional.

Definition 3.4 (Time homogeneous family)**.** For $\tau \in \mathbb{R}$, define the shifted input $x^{(\tau)}$ by

$$x_s^{(\tau)} \coloneqq x_{s-\tau}.$$

Here, $x^{(\tau)}$ is a time shifted input, while $x^{(n)}$ is the n-th element of a sequence. The shift preserves the input space and its norm:

$$x^{(\tau)} \in \mathcal{X}, \qquad \|x^{(\tau)}\|_{\mathcal{X}} = \|x\|_{\mathcal{X}}.$$

A family $\{H_t\}_{t\in\mathbb{R}}$ is time homogeneous if

$$H_t(x) = H_{t+\tau}\big(x^{(\tau)}\big)$$

for every $t, \tau \in \mathbb{R}$ and $x \in \mathcal{X}$ [LHEL22, p. 8, Eq. (16)].

Time homogeneity means that shifting the input and the evaluation time by the same amount does not change the input output relation. The relation therefore depends on relative time differences and not on the choice of the time origin [LHEL22, p. 8].

We collect the target functionals with the properties introduced above in the following class, following the assumptions of Li et al. [cf. LHEL22, p. 10, Thm. 7].

Definition 3.5 (Target functional class)**.** We define

$$\mathcal{H} \coloneqq \left\{ \{H_t\}_{t\in\mathbb{R}} \;\middle|\; \begin{array}{l} H_t \text{ is continuous linear, causal, and regular for every } t \in \mathbb{R}, \\ \{H_t\}_{t\in\mathbb{R}} \text{ is time homogeneous} \end{array} \right\}.$$

3.3 Linear Continuous Time RNNs

We now introduce the linear continuous time RNN used in the following analysis.

Definition 3.6 (Linear continuous time RNN [LHEL22, p. 9, Eq. (17)])**.** Let $m \in \mathbb{N}_+$ be the hidden dimension and let

$$W \in \mathbb{R}^{m\times m}, \qquad U \in \mathbb{R}^{m\times d}, \qquad c \in \mathbb{R}^m.$$

For an input $x \in \mathcal{X}$, with

$$x\colon \mathbb{R} \to \mathbb{R}^d,$$

a linear continuous time RNN has a hidden state

$$h\colon \mathbb{R} \to \mathbb{R}^m, \qquad t \mapsto h_t,$$

that satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}h_t = Wh_t + Ux_t, \qquad \widehat{y}_t = c^\top h_t. \tag{3.1}$$

The output is a scalar valued function

$$\widehat{y}\colon \mathbb{R} \to \mathbb{R}, \qquad t \mapsto \widehat{y}_t.$$

The matrix W determines the recurrent dynamics, U maps the input to the hidden state, and c maps the hidden state to the scalar output.

To get an explicit representation of the hidden state, we first solve the linear state equation. For this, we introduce the matrix exponential and the properties needed below.

Here and below, $\|\cdot\|_2$ denotes the Euclidean norm on $\mathbb{C}^q$ and, for $A \in \mathbb{C}^{p\times q}$, the induced operator norm

$$\|A\|_2 \coloneqq \sup_{\substack{v\in\mathbb{C}^q \\ \|v\|_2=1}} \|Av\|_2 \tag{3.2}$$

[cf. Tes12, p. 60, Eq. (3.8)]. For a real matrix $A \in \mathbb{R}^{p\times q}$, this definition gives the same induced operator norm as over $\mathbb{R}$. Every real unit vector is also a complex unit vector, so

$$\sup_{\substack{x\in\mathbb{R}^q \\ \|x\|_2=1}} \|Ax\|_2 \le \sup_{\substack{z\in\mathbb{C}^q \\ \|z\|_2=1}} \|Az\|_2.$$

For the reverse inequality, write $z = x + \mathrm{i}y$ with $x, y \in \mathbb{R}^q$. Let

$$M \coloneqq \sup_{\substack{x\in\mathbb{R}^q \\ \|x\|_2=1}} \|Ax\|_2.$$

Because A, x, and y are real,

$$Az = A(x + \mathrm{i}y) = Ax + \mathrm{i}Ay.$$

Using the standard complex inner product on $\mathbb{C}^p$,

$$\begin{aligned}
\|Az\|_2^2 &= \langle Ax + \mathrm{i}Ay, Ax + \mathrm{i}Ay\rangle \\
&= \|Ax\|_2^2 + \|Ay\|_2^2 + \mathrm{i}\langle Ax, Ay\rangle - \mathrm{i}\langle Ay, Ax\rangle \\
&= \|Ax\|_2^2 + \|Ay\|_2^2 \\
&\le M^2\|x\|_2^2 + M^2\|y\|_2^2 \\
&= M^2\|z\|_2^2.
\end{aligned}$$

For $\|z\|_2 = 1$, we get $\|Az\|_2 \le M$. The real and complex induced 2-norms therefore have the same value. This comparison is our own detailed calculation. We need it because the proof of Lemma 3.7 uses the complex Jordan form.

For a matrix $A \in \mathbb{C}^{m\times m}$, the matrix exponential is defined by

$$e^A \coloneqq \sum_{n=0}^{\infty} \frac{A^n}{n!}.$$

The series of norms converges, because

$$\sum_{n=0}^{\infty} \left\| \frac{A^n}{n!} \right\|_2 \le \sum_{n=0}^{\infty} \frac{\|A\|_2^n}{n!} = e^{\|A\|_2} < \infty.$$

Since $\mathbb{C}^{m\times m}$ is complete with respect to this norm, the matrix series converges [Tes12, Sec. 3.1, pp. 60–61, Eqs. (3.7)–(3.10)].

For $W \in \mathbb{R}^{m\times m}$ and $t \in \mathbb{R}$, this gives

$$e^{-Wt} = \sum_{n=0}^{\infty} \frac{(-t)^n W^n}{n!}.$$

We define the derivative of

$$\mathbb{R} \to \mathbb{R}^{m\times m}, \qquad t \mapsto e^{-Wt},$$

by

$$\frac{\mathrm{d}}{\mathrm{d}t} e^{-Wt} := \left(\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-Wt}\right)_{ij}\right)_{i,j=1}^{m}.$$

For $1 \le i, j \le m$,

$$\left(e^{-Wt}\right)_{ij} = \sum_{n=0}^{\infty} \frac{(-t)^n (W^n)_{ij}}{n!}.$$

Using the Cauchy–Schwarz inequality, the definition of the induced operator norm, and $\|e_i\|_2 = \|e_j\|_2 = 1$, we get

$$\begin{aligned} |(W^n)_{ij}| &= |e_i^\top W^n e_j| \\ &\le \|e_i\|_2 \|W^n e_j\|_2 \\ &\le \|W^n\|_2 \\ &\le \|W\|_2^n. \end{aligned}$$

Therefore,

$$\sum_{n=0}^{\infty} \left|\frac{(-t)^n (W^n)_{ij}}{n!}\right| \le \sum_{n=0}^{\infty} \frac{|t|^n \|W\|_2^n}{n!} = e^{|t|\|W\|_2} < \infty.$$

The power series has infinite radius of convergence and can therefore be differentiated term by term [FL23, pp. 371–372, Thm. 21.6]. We get

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} e^{-Wt} &= \sum_{n=1}^{\infty} \frac{n(-t)^{n-1}(-1) W^n}{n!} \\ &= -W \sum_{n=1}^{\infty} \frac{(-t)^{n-1} W^{n-1}}{(n-1)!} \\ &= -W e^{-Wt}. \end{aligned} \tag{3.3}$$

Because W commutes with every power W^n,

$$\begin{aligned} W e^{-Wt} &= W \sum_{n=0}^{\infty} \frac{(-t)^n W^n}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(-t)^n W^{n+1}}{n!} \\ &= \sum_{n=0}^{\infty} \frac{(-t)^n W^n W}{n!} \\ &= e^{-Wt} W. \end{aligned}$$

Therefore W commutes with e^{-Wt}. Our own contribution here is the detailed derivation of the derivative and the commutation property. We use the power series definition of the matrix exponential from Teschl [Tes12, Sec. 3.1, pp. 60–61] and show the calculations step by step.

Let $t_0 \in \mathbb{R}$ be an initial time and let $h_{t_0} \in \mathbb{R}^m$ be the corresponding initial state.

We start from

$$\frac{\mathrm{d}}{\mathrm{d}t} h_t = W h_t + U x_t.$$

Multiplying by e^{-Wt} from the left gives

$$e^{-Wt} \frac{\mathrm{d}}{\mathrm{d}t} h_t - e^{-Wt} W h_t = e^{-Wt} U x_t.$$

Since $e^{-Wt} W = W e^{-Wt}$ and

$$\frac{\mathrm{d}}{\mathrm{d}t} e^{-Wt} = -W e^{-Wt},$$

the left hand side is

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(e^{-Wt} h_t \right).$$

We get

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(e^{-Wt} h_t \right) = e^{-Wt} U x_t.$$

All $\mathbb{R}^m$ valued integrals in this chapter are taken componentwise. Integrating from t_0 to t gives

$$e^{-Wt} h_t - e^{-Wt_0} h_{t_0} = \int_{t_0}^{t} e^{-Ws} U x_s \, \mathrm{d}s.$$

Multiplying by e^{Wt} from the left gives

$$h_t = e^{W(t-t_0)} h_{t_0} + \int_{t_0}^{t} e^{W(t-s)} U x_s \, \mathrm{d}s, \qquad t \geq t_0. \tag{3.4}$$

This expression is the solution of the inhomogeneous linear system (3.1) with initial value h_{t_0} [Tes12, p. 84, Thm. 3.12].

Applying the output map $c^\top$ gives

$$\widehat{y}_t = c^\top e^{W(t-t_0)} h_{t_0} + \int_{t_0}^{t} c^\top e^{W(t-s)} U x_s \, \mathrm{d}s.$$

This gives the solution of (3.1) for the initial state h_{t_0}.

The continuous time model is an idealization of the discrete time RNN introduced in Chapter 2. Li et al. consider this setting because the discrete iterative structure is difficult to analyze directly [LHEL22, pp. 5–6].

In discrete time, the hidden state is updated at separate time steps. In continuous time, its evolution is described by a differential equation. For linear discrete RNNs, powers of the recurrent matrix describe how earlier states and inputs affect later states. In continuous time, the matrix exponential $e^{W(t-s)}$ plays a similar role, and the integral in (3.4) replaces the sum over past inputs.

The continuous time setting allows us to use tools from differential equations, functional analysis, and approximation theory [cf. LHEL22, p. 6]. This makes it easier to study how past inputs affect the output of the RNN.

3.4 Stability and Infinite Past Solutions

The solution in (3.4) depends on the initial time t_0 and the initial state h_{t_0}. To include the complete input history, we let $t_0 \to -\infty$. For this limit, we require the hidden state dynamics to be stable.

Our goal is to represent the output of the RNN only in terms of the input history. This removes the dependence on an arbitrary finite initial state and gives a time independent description of the input output relation. The stability condition allows us to take the limit $t_0 \to -\infty$ and leads to an integral representation of the RNN output. This representation will later be used to study approximation properties of linear continuous time RNNs.

3.4.1 Stability of the Recurrent Dynamics

We define the set of stable recurrent matrices by

$$\mathcal{W}_m := \left\{ W \in \mathbb{R}^{m\times m} \;\middle|\; \max_{\lambda\in\sigma(W)} \operatorname{Re}(\lambda) < 0 \right\},$$

where $\sigma(W)$ denotes the spectrum of W [LHEL22, p. 9, Eq. (19)].

We first prove a lemma that we use later to control the matrix exponential e^{Ws} for large s.

Lemma 3.7 (Exponential decay of stable matrix exponentials [Tes12, p. 71, Cor. 3.6, Eq. (3.44)])**.** *Let $W \in \mathcal{W}_m$. Then there exist constants $C > 0$ and $\alpha > 0$ such that*

$$\|e^{Ws}\|_2 \le Ce^{-\alpha s}, \qquad s \ge 0.$$

In particular,

$$\lim_{s\to\infty} \|e^{Ws}\|_2 = 0.$$

Proof. We follow the proof using the Jordan form in Teschl [Tes12, pp. 70–71, Eq. (3.42) and Cor. 3.6]. Our own contribution is the detailed calculation of the bounds and the explicit construction of the constants $M_{i,k}$, C_i, and C.

Let $W \in \mathcal{W}_m$. Viewing W as a matrix in $\mathbb{C}^{m\times m}$, the Jordan canonical form [Tes12, p. 61, Thm. 3.2] gives an invertible matrix $P \in \mathbb{C}^{m\times m}$ and Jordan blocks $J_1, \dots, J_q$ such that

$$W = PJP^{-1}, \qquad J = \operatorname{diag}(J_1, \dots, J_q).$$

For $i = 1, \dots, q$, let r_i be the size of the Jordan block J_i. Then

$$J_i = \lambda_i I_{r_i} + N_i,$$

where $\lambda_i \in \mathbb{C}$ is an eigenvalue of W and $N_i \in \mathbb{C}^{r_i\times r_i}$ is nilpotent with

$$N_i^{r_i} = 0.$$

Because $\lambda_i I_{r_i}$ and N_i commute, the product formula for matrix exponentials gives

$$\begin{aligned} e^{J_i s} &= e^{\lambda_i s I_{r_i}} e^{N_i s} \\ &= e^{\lambda_i s} I_{r_i} e^{N_i s} \\ &= e^{\lambda_i s} e^{N_i s} \end{aligned}$$

[Tes12, p. 61, Lem. 3.1, Eq. (3.12)]. The identity

$$e^{\lambda_i s I_{r_i}} = e^{\lambda_i s} I_{r_i}$$

follows directly from the power series definition. Using the power series definition of the matrix exponential,

$$e^{N_i s} = \sum_{k=0}^{\infty} \frac{s^k N_i^k}{k!}.$$

Since $N_i^{r_i} = 0$, all terms with $k \geq r_i$ vanish. We get

$$e^{J_i s} = e^{\lambda_i s} \sum_{k=0}^{r_i - 1} \frac{s^k N_i^k}{k!}.$$

Since $W \in \mathcal{W}_m$, every eigenvalue satisfies

$$\operatorname{Re}(\lambda_i) < 0.$$

Choose

$$0 < \alpha < - \max_{\lambda \in \sigma(W)} \operatorname{Re}(\lambda).$$

Then

$$\operatorname{Re}(\lambda_i) + \alpha < 0$$

for every eigenvalue λ_i. Define

$$a_i \coloneqq -\big(\operatorname{Re}(\lambda_i) + \alpha\big) > 0.$$

For $k = 0$, $e^{-a_i s} \leq 1$ for $s \geq 0$, so we set $M_{i,0} = 1$. For $k \geq 1$, define $f_{i,k} \colon [0, \infty) \to [0, \infty)$ by

$$f_{i,k}(s) \coloneqq s^k e^{-a_i s}, \qquad s \geq 0.$$

Its derivative is

$$f'_{i,k}(s) = s^{k-1} e^{-a_i s} \left(k - a_i s\right).$$

The derivative is positive for $0 < s < k/a_i$ and negative for $s > k/a_i$. The maximum is attained at $s = k/a_i$, and

$$s^k e^{-a_i s} \leq M_{i,k} \coloneqq \left(\frac{k}{a_i e}\right)^k, \qquad s \geq 0.$$

Since

$$|e^{\lambda_i s}| = e^{\operatorname{Re}(\lambda_i) s},$$

using absolute homogeneity and the triangle inequality for the operator norm, we get

$$\begin{aligned}
\|e^{J_i s}\|_2 &\leq e^{\operatorname{Re}(\lambda_i) s} \sum_{k=0}^{r_i - 1} \frac{s^k \|N_i^k\|_2}{k!} \\
&= e^{-\alpha s} \sum_{k=0}^{r_i - 1} \frac{s^k e^{(\operatorname{Re}(\lambda_i) + \alpha) s} \|N_i^k\|_2}{k!} \\
&= e^{-\alpha s} \sum_{k=0}^{r_i - 1} \frac{s^k e^{-a_i s} \|N_i^k\|_2}{k!} \\
&\leq e^{-\alpha s} \sum_{k=0}^{r_i - 1} \frac{M_{i,k} \|N_i^k\|_2}{k!}.
\end{aligned}$$

Set

$$C_i \coloneqq \sum_{k=0}^{r_i-1} \frac{M_{i,k}\|N_i^k\|_2}{k!}.$$

Then

$$\|e^{J_i s}\|_2 \leq C_i e^{-\alpha s}.$$

Since powers of a block diagonal matrix are block diagonal, the power series definition of the matrix exponential gives

$$e^{Js} = \operatorname{diag}\left(e^{J_1 s}, \ldots, e^{J_q s}\right).$$

For a block vector $v = (v_1, \ldots, v_q) \in \mathbb{C}^m$, where $v_i \in \mathbb{C}^{r_i}$ for $i = 1, \ldots, q$, we have

$$\begin{aligned}\|e^{Js}v\|_2^2 &= \sum_{i=1}^{q} \|e^{J_i s} v_i\|_2^2 \\ &\leq \left(\max_{1\leq i\leq q} \|e^{J_i s}\|_2^2\right) \sum_{i=1}^{q} \|v_i\|_2^2.\end{aligned}$$

Taking the supremum over all $\|v\|_2 = 1$ gives

$$\|e^{Js}\|_2 \leq \max_{1\leq i\leq q} \|e^{J_i s}\|_2.$$

Since

$$\|e^{J_i s}\|_2 \leq C_i e^{-\alpha s}$$

for every $i \in \{1, \ldots, q\}$, we get

$$\|e^{Js}\|_2 \leq \left(\max_{1\leq i\leq q} C_i\right) e^{-\alpha s}.$$

Set

$$C_J \coloneqq \max_{1\leq i\leq q} C_i.$$

Then

$$\|e^{Js}\|_2 \leq C_J e^{-\alpha s}.$$

Since $W = PJP^{-1}$,

$$W^n = PJ^nP^{-1}$$

for every $n \in \mathbb{N}_0$. The power series definition of the matrix exponential gives

$$e^{Ws} = Pe^{Js}P^{-1}.$$

Using the submultiplicativity of the operator norm,

$$\begin{aligned}\|e^{Ws}\|_2 &\leq \|P\|_2 \|e^{Js}\|_2 \|P^{-1}\|_2 \\ &\leq \|P\|_2 \|P^{-1}\|_2 C_J e^{-\alpha s}.\end{aligned}$$

Set

$$C \coloneqq \|P\|_2 \|P^{-1}\|_2 C_J.$$

This gives

$$\|e^{Ws}\|_2 \leq Ce^{-\alpha s}.$$

Since $Ce^{-\alpha s} \to 0$ as $s \to \infty$, we also have

$$\|e^{Ws}\|_2 \xrightarrow[s\to\infty]{} 0.$$

□

3.4.2 Infinite Past Representation

Proposition 3.8 (Infinite past representation of linear CT RNNs [cf. LHEL22, p. 9, Eq. (18)]). *Let $W \in \mathcal{W}_m$, $U \in \mathbb{R}^{m\times d}$, $c \in \mathbb{R}^m$, and $x \in \mathcal{X}$. Then the following statements hold.*

(i) Let $h^{(t_0)}\colon [t_0,\infty) \to \mathbb{R}^m$ denote the solution of (3.1) with initial condition

$$h_{t_0}^{(t_0)} = 0.$$

For every fixed $t \in \mathbb{R}$, $h_t^{(t_0)}$ converges as $t_0 \to -\infty$, and its limit is given by

$$h_t = \int_0^\infty e^{Ws} U x_{t-s}\,\mathrm{d}s.$$

(ii) The infinite past state h_t solves the state equation

$$\frac{\mathrm{d}}{\mathrm{d}t} h_t = W h_t + U x_t.$$

(iii) The corresponding output

$$\widehat{y}_t = c^\top h_t = \int_0^\infty c^\top e^{Ws} U x_{t-s}\,\mathrm{d}s$$

belongs to $\mathcal{Y}$.

(iv) The RNN defines a family of functionals

$$\widehat{H}_t\colon \mathcal{X} \to \mathbb{R}, \qquad \widehat{H}_t(x) = \int_0^\infty c^\top e^{Ws} U x_{t-s}\,\mathrm{d}s.$$

Proof. Li et al. write the infinite past representation formally by setting $h_{-\infty} = 0$ [LHEL22, p. 9, Eq. (18)]. The following proof is our own. We show that the limit $t_0 \to -\infty$ exists and prove the properties (ii)–(iv) step by step.

(i) For a finite initial time $t_0 < t$, choose

$$h_{t_0}^{(t_0)} = 0.$$

By (3.4),

$$h_t^{(t_0)} = \int_{t_0}^t e^{W(t-r)} U x_r\,\mathrm{d}r.$$

With $s = t - r$, this becomes

$$h_t^{(t_0)} = \int_0^{t-t_0} e^{Ws} U x_{t-s}\,\mathrm{d}s.$$

For every $s \geq 0$,

$$\begin{aligned}
\|x_{t-s}\|_2 &= \left(\sum_{j=1}^d |(x_{t-s})_j|^2\right)^{1/2} \\
&\leq \left(\sum_{j=1}^d \|x_{t-s}\|_\infty^2\right)^{1/2} \\
&= \sqrt{d}\,\|x_{t-s}\|_\infty \\
&\leq \sqrt{d}\,\|x\|_{\mathcal{X}}.
\end{aligned}$$

By Lemma 3.7,

$$\begin{aligned}\int_0^\infty \|e^{Ws}Ux_{t-s}\|_2 \,\mathrm{d}s &\le \int_0^\infty \|e^{Ws}\|_2\|U\|_2\|x_{t-s}\|_2 \,\mathrm{d}s \\ &\le C\sqrt{d}\,\|U\|_2\|x\|_{\mathcal{X}} \int_0^\infty e^{-\alpha s}\,\mathrm{d}s \\ &= \frac{C\sqrt{d}}{\alpha}\|U\|_2\|x\|_{\mathcal{X}} < \infty.\end{aligned}$$

The integrand is continuous and its norm is integrable over $[0,\infty)$. Since $\mathbb{R}^m$ is complete, the improper integral converges in $\mathbb{R}^m$, and

$$h_t \coloneqq \lim_{t_0\to-\infty} h_t^{(t_0)} = \int_0^\infty e^{Ws}Ux_{t-s}\,\mathrm{d}s.$$

(ii) Changing variables in the representation from part (i) gives

$$h_t = \int_{-\infty}^t e^{W(t-r)}Ux_r\,\mathrm{d}r.$$

The matrices Wt and $-Wr$ commute, because both are scalar multiples of W. The product formula for matrix exponentials [Tes12, p. 61, Lem. 3.1, Eq. (3.12)] gives

$$e^{W(t-r)} = e^{Wt}e^{-Wr}.$$

Therefore, define

$$G\colon \mathbb{R}\to\mathbb{R}^m, \qquad G(t) \coloneqq \int_{-\infty}^t e^{-Wr}Ux_r\,\mathrm{d}r.$$

Then

$$h_t = e^{Wt}G(t).$$

We first check that $G(t)$ is well defined. For $r \le 0$,

$$e^{-Wr} = e^{W(-r)}, \qquad -r \ge 0.$$

Lemma 3.7 gives

$$\|e^{-Wr}\|_2 \le Ce^{\alpha r}.$$

Using

$$\|x_r\|_2 \le \sqrt{d}\,\|x\|_{\mathcal{X}},$$

we get

$$\begin{aligned}\|e^{-Wr}Ux_r\|_2 &\le \|e^{-Wr}\|_2\|U\|_2\|x_r\|_2 \\ &\le C\sqrt{d}\,\|U\|_2\|x\|_{\mathcal{X}}e^{\alpha r}.\end{aligned}$$

The function on the right hand side is integrable over $(-\infty,0]$, because

$$\int_{-\infty}^0 e^{\alpha r}\,\mathrm{d}r = \frac{1}{\alpha} < \infty.$$

On every finite interval, the map $\mathbb{R}\to\mathbb{R}^m$,

$$r \longmapsto e^{-Wr}Ux_r$$

is continuous. Therefore, $G(t)$ exists for every $t \in \mathbb{R}$.

Fix $t \in \mathbb{R}$ and choose $a < t$. Then

$$G(t) = G(a) + \int_a^t e^{-Wr} U x_r \, \mathrm{d}r.$$

The first term is constant and the integrand in the second term is continuous. The fundamental theorem of calculus, applied componentwise, gives

$$G'(t) = e^{-Wt} U x_t.$$

Applying the product rule to $h_t = e^{Wt} G(t)$ gives

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} h_t &= W e^{Wt} G(t) + e^{Wt} G'(t) \\ &= W h_t + e^{Wt} e^{-Wt} U x_t \\ &= W h_t + U x_t. \end{aligned}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} e^{Wt} = W e^{Wt} \tag{3.5}$$

and Equation (3.5) follows from Equation (3.3) with W in place of $-W$.

(iii) Applying the output map gives

$$\widehat{y}_t = c^\top h_t = \int_0^\infty c^\top e^{Ws} U x_{t-s} \, \mathrm{d}s$$

[LHEL22, p. 9, Eq. (18)].

We first show that $\widehat{y}$ is bounded. By Lemma 3.7,

$$\begin{aligned} |\widehat{y}_t| &\le \int_0^\infty \|c\|_2 \|e^{Ws}\|_2 \|U\|_2 \|x_{t-s}\|_2 \, \mathrm{d}s \\ &\le C\sqrt{d} \, \|c\|_2 \|U\|_2 \|x\|_{\mathcal{X}} \int_0^\infty e^{-\alpha s} \, \mathrm{d}s \\ &= \frac{C\sqrt{d}}{\alpha} \|c\|_2 \|U\|_2 \|x\|_{\mathcal{X}}, \end{aligned}$$

where

$$\int_0^\infty e^{-\alpha s} \, \mathrm{d}s = \frac{1}{\alpha}.$$

The bound is independent of t, so

$$\sup_{t \in \mathbb{R}} |\widehat{y}_t| < \infty.$$

We next show that $\widehat{y}$ is continuous. Let $\varepsilon > 0$. Because $x \in C_0(\mathbb{R}, \mathbb{R}^d)$, there exists $R > 0$ such that

$$\|x_r\|_\infty < \frac{\varepsilon}{2} \qquad \text{for } |r| > R.$$

The interval $[-R-2, R+2]$ is compact by the Heine–Borel theorem. The restriction of x to this interval is uniformly continuous by the Heine–Cantor theorem. Therefore, there exists $\eta \in (0, 1)$ such that

$$|u - v| < \eta \quad \Longrightarrow \quad \|x_u - x_v\|_\infty < \varepsilon$$

for all $u, v \in [-R-2, R+2]$.

Let $|\delta| < \eta$. If $r \in [-R-1, R+1]$, then both r and $r+\delta$ belong to $[-R-2, R+2]$, so

$$\|x_{r+\delta} - x_r\|_\infty < \varepsilon.$$

If $r \notin [-R-1, R+1]$, then

$$|r| > R+1.$$

By the reverse triangle inequality,

$$|r+\delta| \geq |r| - |\delta| > (R+1) - 1 = R.$$

Therefore,

$$\begin{aligned} \|x_{r+\delta} - x_r\|_\infty &\leq \|x_{r+\delta}\|_\infty + \|x_r\|_\infty \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

For every $|\delta| < \eta$, we therefore have

$$\sup_{r\in\mathbb{R}} \|x_{r+\delta} - x_r\|_\infty \leq \varepsilon.$$

Thus,

$$\sup_{r\in\mathbb{R}} \|x_{r+\delta} - x_r\|_\infty \longrightarrow 0$$

as $\delta \to 0$. Again using Lemma 3.7, we get

$$\begin{aligned} |\widehat{y}_{t+\delta} - \widehat{y}_t| &\leq \int_0^\infty \|c\|_2 \|e^{Ws}\|_2 \|U\|_2 \|x_{t+\delta-s} - x_{t-s}\|_2 \,\mathrm{d}s \\ &\leq C\sqrt{d}\, \|c\|_2 \|U\|_2 \sup_{r\in\mathbb{R}} \|x_{r+\delta} - x_r\|_\infty \int_0^\infty e^{-\alpha s} \,\mathrm{d}s \\ &= \frac{C\sqrt{d}}{\alpha} \|c\|_2 \|U\|_2 \sup_{r\in\mathbb{R}} \|x_{r+\delta} - x_r\|_\infty \xrightarrow[\delta\to 0]{} 0. \end{aligned}$$

The output is bounded and continuous, so

$$\widehat{y} \in \mathcal{Y}.$$

(iv) For every $t \in \mathbb{R}$, define

$$\widehat{H}_t \colon \mathcal{X} \to \mathbb{R}, \qquad \widehat{H}_t(x) \coloneqq \int_0^\infty c^\top e^{Ws} U x_{t-s} \,\mathrm{d}s.$$

By part (iii), for every $x \in \mathcal{X}$,

$$\widehat{H}(x) \coloneqq \big(\widehat{H}_t(x)\big)_{t\in\mathbb{R}} \in \mathcal{Y}.$$

□

The infinite past representation makes the memory of the RNN explicit. The output

$$\widehat{y}_t = \int_0^\infty c^\top e^{Ws} U x_{t-s} \, \mathrm{d}s$$

is a weighted superposition of past inputs, where

$$c^\top e^{Ws} U$$

determines the influence of an input from s time units in the past. By Lemma 3.7, the norm of this weight decays exponentially for stable W. This representation motivates the memory kernel studied in Chapter 4.

3.4.3 RNN Functional Class and Structural Properties

We now collect all functionals that can be represented in this way. This gives the hypothesis class that we use later when comparing RNN functionals with target functionals.

For a fixed hidden dimension m, we define the corresponding hypothesis class by

$$\widehat{\mathcal{H}}_m \coloneqq \left\{ \{\widehat{H}_t\}_{t\in\mathbb{R}} \;\middle|\; \begin{array}{l} \text{there exist } W \in \mathcal{W}_m,\ U \in \mathbb{R}^{m\times d},\ c \in \mathbb{R}^m \text{ such that} \\ \widehat{H}_t(x) = \displaystyle\int_0^\infty c^\top e^{Ws} U x_{t-s} \, \mathrm{d}s \quad \text{for all } t \in \mathbb{R},\ x \in \mathcal{X} \end{array} \right\}.$$

We further define

$$\widehat{\mathcal{H}} \coloneqq \bigcup_{m=1}^{\infty} \widehat{\mathcal{H}}_m$$

[LHEL22, p. 9, Eq. (20)].

The class $\widehat{\mathcal{H}}_m$ contains the RNN functionals with hidden dimension m, while $\widehat{\mathcal{H}}$ allows arbitrary finite hidden dimensions.

We now check that the RNN functionals satisfy the properties introduced in Section 3.2. These properties will be used later in the approximation analysis.

Proposition 3.9 (Structural properties of linear CT RNN functionals [LHEL22, p. 10, Prop. 6])**.** *Let*

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}.$$

Then the following statements hold:

(i) For every $t \in \mathbb{R}$, $\widehat{H}_t$ is a continuous linear functional in the sense of Definition 3.1;

(ii) For every $t \in \mathbb{R}$, $\widehat{H}_t$ is causal in the sense of Definition 3.2;

(iii) For every $t \in \mathbb{R}$, $\widehat{H}_t$ is regular in the sense of Definition 3.3;

(iv) The family $\{\widehat{H}_t\}_{t\in\mathbb{R}}$ is time homogeneous in the sense of Definition 3.4.

Proof. Li et al. state this proposition without proof [LHEL22, p. 10, Prop. 6]. The following proof is our own. We verify linearity, continuity, causality, regularity, and time homogeneity directly from the integral representation.

Since

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}},$$

there exist a hidden dimension $m \geq 1$ and parameters

$$W \in \mathcal{W}_m, \qquad U \in \mathbb{R}^{m\times d}, \qquad c \in \mathbb{R}^m$$

such that

$$\widehat{H}_t(x) = \int_0^\infty c^\top e^{Ws} U x_{t-s}\,\mathrm{d}s$$

for every $t \in \mathbb{R}$ and $x \in \mathcal{X}$. By Lemma 3.7, there exist constants $C > 0$ and $\alpha > 0$ such that

$$\|e^{Ws}\|_2 \leq Ce^{-\alpha s}, \qquad s \geq 0.$$

(i) Let $t \in \mathbb{R}$ be fixed. We first prove linearity. Let $x, z \in \mathcal{X}$ and $\lambda, \mu \in \mathbb{R}$. Then

$$\begin{aligned}\widehat{H}_t(\lambda x + \mu z) &= \int_0^\infty c^\top e^{Ws} U\big(\lambda x_{t-s} + \mu z_{t-s}\big)\,\mathrm{d}s \\ &= \lambda \int_0^\infty c^\top e^{Ws} U x_{t-s}\,\mathrm{d}s + \mu \int_0^\infty c^\top e^{Ws} U z_{t-s}\,\mathrm{d}s \\ &= \lambda \widehat{H}_t(x) + \mu \widehat{H}_t(z).\end{aligned}$$

The functional $\widehat{H}_t$ is therefore linear.

We next prove continuity. For every $x \in \mathcal{X}$, the triangle inequality, the Cauchy Schwarz inequality, and the submultiplicativity of the operator norm give

$$\begin{aligned}|\widehat{H}_t(x)| &= \left|\int_0^\infty c^\top e^{Ws} U x_{t-s}\,\mathrm{d}s\right| \\ &\leq \int_0^\infty \left|c^\top e^{Ws} U x_{t-s}\right|\,\mathrm{d}s \\ &\leq \int_0^\infty \|c\|_2 \|e^{Ws}\|_2 \|U\|_2 \|x_{t-s}\|_2\,\mathrm{d}s.\end{aligned}$$

Since $x_{t-s} \in \mathbb{R}^d$,

$$\|x_{t-s}\|_2 \leq \sqrt{d}\,\|x_{t-s}\|_\infty \leq \sqrt{d}\,\|x\|_{\mathcal{X}}.$$

Using Lemma 3.7, we get

$$\begin{aligned}|\widehat{H}_t(x)| &\leq C\sqrt{d}\,\|c\|_2 \|U\|_2 \|x\|_{\mathcal{X}} \int_0^\infty e^{-\alpha s}\,\mathrm{d}s \\ &= \frac{C\sqrt{d}}{\alpha} \|c\|_2 \|U\|_2 \|x\|_{\mathcal{X}}.\end{aligned}$$

It follows that

$$\|\widehat{H}_t\| \leq \frac{C\sqrt{d}}{\alpha} \|c\|_2 \|U\|_2 < \infty.$$

Here, $\|\widehat{H}_t\|$ is the operator norm from Definition 3.1. The linear functional $\widehat{H}_t$ is therefore continuous in the sense of this definition.

(ii) Let $x, z \in \mathcal{X}$ satisfy

$$x_r = z_r \qquad \text{for all } r \leq t.$$

For every $s \geq 0$,

$$t - s \leq t,$$

so

$$x_{t-s} = z_{t-s}.$$

We then have

$$\begin{aligned} \widehat{H}_t(x) &= \int_0^\infty c^\top e^{Ws} U x_{t-s} \,\mathrm{d}s \\ &= \int_0^\infty c^\top e^{Ws} U z_{t-s} \,\mathrm{d}s \\ &= \widehat{H}_t(z). \end{aligned}$$

By Definition 3.2, $\widehat{H}_t$ is causal.

(iii) Let

$$\left\{x^{(n)} \in \mathcal{X} : n \in \mathbb{N}_+\right\}$$

satisfy

$$\sup_{n \in \mathbb{N}_+} \left\|x^{(n)}\right\|_{\mathcal{X}} \leq 1$$

and

$$x_r^{(n)} \xrightarrow[n\to\infty]{} 0 \qquad \text{in } \mathbb{R}^d$$

for Lebesgue almost every $r \in \mathbb{R}$.

The affine map $s \mapsto t - s$ preserves sets of Lebesgue measure zero. Therefore,

$$x_{t-s}^{(n)} \xrightarrow[n\to\infty]{} 0 \qquad \text{in } \mathbb{R}^d$$

for almost every $s \geq 0$. For every such s,

$$c^\top e^{Ws} U x_{t-s}^{(n)} \xrightarrow[n\to\infty]{} 0 \qquad \text{in } \mathbb{R}.$$

Moreover,

$$\left\|x_{t-s}^{(n)}\right\|_2 \leq \sqrt{d}\left\|x^{(n)}\right\|_{\mathcal{X}} \leq \sqrt{d}.$$

Using Lemma 3.7, we get

$$\begin{aligned} \left|c^\top e^{Ws} U x_{t-s}^{(n)}\right| &\leq \|c\|_2 \|e^{Ws}\|_2 \|U\|_2 \left\|x_{t-s}^{(n)}\right\|_2 \\ &\leq C\sqrt{d}\,\|c\|_2 \|U\|_2 e^{-\alpha s}. \end{aligned}$$

The function on the right hand side is independent of n and integrable over $[0, \infty)$, since

$$\int_0^\infty e^{-\alpha s} \,\mathrm{d}s = \frac{1}{\alpha} < \infty.$$

The dominated convergence theorem [Els18, p. 160, Thm. IV.5.2] gives

$$\begin{aligned} \lim_{n\to\infty} \widehat{H}_t\big(x^{(n)}\big) &= \lim_{n\to\infty} \int_0^\infty c^\top e^{Ws} U x_{t-s}^{(n)} \,\mathrm{d}s \\ &= \int_0^\infty \lim_{n\to\infty} c^\top e^{Ws} U x_{t-s}^{(n)} \,\mathrm{d}s \\ &= 0. \end{aligned}$$

By Definition 3.3, $\widehat{H}_t$ is regular. Since $t \in \mathbb{R}$ was arbitrary, this holds for every $t \in \mathbb{R}$.

(iv) Let $t, \tau \in \mathbb{R}$ and $x \in \mathcal{X}$. By the definition of the shifted input in Definition 3.4,

$$\begin{aligned} x^{(\tau)}_{t+\tau-s} &= x_{(t+\tau-s)-\tau} \\ &= x_{t-s}. \end{aligned}$$

We then get

$$\begin{aligned} \widehat{H}_{t+\tau}\big(x^{(\tau)}\big) &= \int_0^\infty c^\top e^{Ws} U x^{(\tau)}_{t+\tau-s} \,\mathrm{d}s \\ &= \int_0^\infty c^\top e^{Ws} U x_{t-s} \,\mathrm{d}s \\ &= \widehat{H}_t(x). \end{aligned}$$

This shows that

$$\widehat{H}_t(x) = \widehat{H}_{t+\tau}\big(x^{(\tau)}\big)$$

for every $t, \tau \in \mathbb{R}$ and $x \in \mathcal{X}$. By Definition 3.4, the family $\{\widehat{H}_t\}_{t\in\mathbb{R}}$ is time homogeneous. □

In particular,

$$\widehat{\mathcal{H}} \subseteq \mathcal{H}.$$

Every family generated by a stable linear continuous time RNN therefore satisfies the properties of the target class $\mathcal{H}$. In Chapter 5, we study whether families in $\mathcal{H}$ can be approximated by elements of $\widehat{\mathcal{H}}$.

4 The Memory Kernel Perspective

In Section 3.2, we introduced the target class $\mathcal{H}$. We now consider

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}.$$

In Section 3.4, we showed that a family

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m$$

generated by a stable linear continuous time RNN has the representation

$$\widehat{H}_t(x) = \int_0^\infty c^\top e^{Ws} U x_{t-s}\,\mathrm{d}s.$$

The goal of this chapter is to describe the target functionals

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}$$

in a form that can be compared directly with the RNN functionals.

The main reference for this chapter is Li et al. [LHEL22, Sec. 5, pp. 15–18]. For the measure theory used in the proofs, we use Rudin [Rud87, Ch. 6] and Bogachev [Bog07, Sec. 3.1].

Our own work in this chapter consists of several detailed proofs and additional results. We expand the proof of the memory kernel representation, including the causality argument, the construction of the reflected measure, the regularity argument, and the norm identity. We also prove that the RNN kernel is integrable. We state the error bound from Li et al. [LHEL22, p. 18, Eq. (61)] as a separate proposition and give a detailed proof. Finally, we prove the uniqueness of the memory kernel and define an extension to step inputs that are not contained in $\mathcal{X}$.

4.1 Signed Measures

Before stating the representation theorem, we introduce some measure theoretic concepts that are used later in the proof.

Definition 4.1 (Positive and negative sets [Bog07, proof of Thm. 3.1.1, p. 175])**.** Let ν be a finite signed measure on a measurable space $(X, \mathcal{A})$. A measurable set $P \in \mathcal{A}$ is called positive for ν if

$$\nu(A) \geq 0$$

for every measurable set $A \subseteq P$. Similarly, a measurable set $N \in \mathcal{A}$ is called negative for ν if

$$\nu(A) \leq 0$$

for every measurable set $A \subseteq N$.

Theorem 4.2 (Hahn decomposition [Bog07, Thm. 3.1.1, pp. 175–176])**.** *Let ν be a finite signed measure on $(X, \mathcal{A})$. Then there exist disjoint measurable sets $P, N \in \mathcal{A}$ such that*

$$X = P \cup N, \qquad P \cap N = \varnothing,$$

where P is positive and N is negative for ν.

Corollary 4.3 (Jordan decomposition [Bog07, Cor. 3.1.2 and Def. 3.1.4, p. 176])**.** *Let P, N be a Hahn decomposition of a finite signed measure ν. Define*

$$\nu^+(A) \coloneqq \nu(A \cap P), \qquad \nu^-(A) \coloneqq -\nu(A \cap N)$$

for every $A \in \mathcal{A}$. Then ν^+ and ν^- are finite positive measures and

$$\nu = \nu^+ - \nu^-.$$

The measures ν^+ and ν^- do not depend on the choice of the Hahn decomposition. They are called the positive and negative parts of ν, respectively.

Definition 4.4 (Total variation [Bog07, Def. 3.1.4, p. 176])**.** Let

$$\nu = \nu^+ - \nu^-$$

be the Jordan decomposition of a finite signed measure ν. The total variation measure of ν is defined by

$$|\nu| \coloneqq \nu^+ + \nu^-.$$

The number

$$|\nu|(X)$$

is called the total variation of ν.

Two simple examples illustrate this definition. For a Dirac measure δ_a,

$$|\delta_a| = \delta_a, \qquad |\delta_a|(\mathbb{R}) = 1.$$

Thus, a point mass has total variation one.

If a signed measure ν has a Lebesgue density $g \in L^1(\mathbb{R})$, that is,

$$\nu(A) = \int_A g(s)\,\mathrm{d}s,$$

then its total variation has density $|g|$ [Rud87, p. 125, Thm. 6.13], and therefore

$$|\nu|(\mathbb{R}) = \int_{\mathbb{R}} |g(s)|\,\mathrm{d}s = \|g\|_{L^1(\mathbb{R})}.$$

For every $f \in L^1(|\nu|)$, integration with respect to the signed measure ν is defined by

$$\int_X f\,\mathrm{d}\nu \coloneqq \int_X f\,\mathrm{d}\nu^+ - \int_X f\,\mathrm{d}\nu^-$$

[Bog07, Sec. 3.1, p. 177].

Corollary 4.5 (Total variation bound [cf. Bog07, Sec. 3.1, p. 177])**.** *Let ν be a finite signed measure and let f be $|\nu|$-integrable. Then*

$$\left| \int_X f\,\mathrm{d}\nu \right| \leq \int_X |f|\,\mathrm{d}|\nu|. \tag{4.1}$$

Proof. By the definition of integration with respect to a signed measure,

$$\int_X f \,\mathrm{d}\nu = \int_X f \,\mathrm{d}\nu^+ - \int_X f \,\mathrm{d}\nu^-.$$

The triangle inequality gives

$$\begin{aligned}\left|\int_X f \,\mathrm{d}\nu\right| &\le \int_X |f| \,\mathrm{d}\nu^+ + \int_X |f| \,\mathrm{d}\nu^- \\ &= \int_X |f| \,\mathrm{d}|\nu|,\end{aligned}$$

where the last equality follows from Definition 4.4. □

4.2 Measure Representation

Let

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}.$$

To describe how input values at different time points contribute to $H_t(x)$, we first represent H_t by a measure.

The scalar Riesz–Markov–Kakutani representation theorem is stated in Rudin [Rud87, p. 130, Thm. 6.19]. Applying this theorem to each component gives the vector valued form used below, as stated by Li et al. [LHEL22, p. 15, Thm. 14].

Theorem 4.6 (Riesz–Markov–Kakutani representation theorem [LHEL22, p. 15, Thm. 14])**.** *Let*

$$H\colon \mathcal{X} \to \mathbb{R}$$

be a continuous linear functional. Then there exists a unique vector valued measure

$$\mu\colon \mathcal{B}(\mathbb{R}) \to \mathbb{R}^d, \qquad \mu = (\mu_1, \dots, \mu_d)^\top,$$

whose components μ_i are finite regular signed Borel measures, such that

$$H(x) = \int_{\mathbb{R}} x_s^\top \,\mathrm{d}\mu(s) = \sum_{i=1}^{d} \int_{\mathbb{R}} x_{s,i} \,\mathrm{d}\mu_i(s)$$

for every $x \in \mathcal{X}$. Moreover,

$$\|H\| = \|\mu\|_1(\mathbb{R}), \qquad \|\mu\|_1(\mathbb{R}) \coloneqq \sum_{i=1}^{d} |\mu_i|(\mathbb{R}),$$

where $\|H\|$ is the operator norm from Definition 3.1 and $|\mu_i|$ is the total variation measure from Definition 4.4.

For a vector valued signed measure

$$\nu = (\nu_1, \dots, \nu_d)^\top$$

and a Borel set A, we use the notation

$$\|\nu\|_1(A) \coloneqq \sum_{i=1}^{d} |\nu_i|(A).$$

Thus, $\|\nu\|_1(A)$ is the sum of the total variations of the component measures on A, consistent with the notation in [LHEL22, p. 15, Eq. (41)].

Theorem 4.6 replaces the abstract functional H by a signed measure μ. Each component μ_i describes how values of the i-th input component at different time points contribute to the output.

For every $t \in \mathbb{R}$, the functional

$$H_t \colon \mathcal{X} \to \mathbb{R}$$

is linear and continuous by the definition of $\mathcal{H}$.

Therefore, Theorem 4.6 applies to each fixed t and gives a unique vector valued regular signed Borel measure

$$\mu_t \colon \mathcal{B}(\mathbb{R}) \to \mathbb{R}^d$$

such that

$$H_t(x) = \int_{\mathbb{R}} x_s^\top \, \mathrm{d}\mu_t(s)$$

for every $x \in \mathcal{X}$. We now use the structural properties from Section 3.2 to further restrict this measure representation.

4.3 Kernel Representation

Definition 4.7 (Vector valued L^1 norm [LHEL22, p. 15, Lem. 15])**.** For a vector

$$v = (v_1, \dots, v_d)^\top \in \mathbb{R}^d,$$

we use the ℓ^1-norm

$$\|v\|_1 \coloneqq \sum_{i=1}^{d} |v_i|.$$

For a measurable function

$$f \colon [0, \infty) \to \mathbb{R}^d, \qquad f(s) = \big(f_1(s), \dots, f_d(s)\big)^\top,$$

we define

$$\begin{aligned} \|f\|_{L^1([0,\infty))} &\coloneqq \int_0^\infty \|f(s)\|_1 \, \mathrm{d}s \\ &= \sum_{i=1}^{d} \int_0^\infty |f_i(s)| \, \mathrm{d}s. \end{aligned}$$

The space

$$L^1([0, \infty), \mathbb{R}^d)$$

consists of equivalence classes of measurable functions $f : [0, \infty) \to \mathbb{R}^d$ for which this norm is finite. Two functions are identified if they are equal Lebesgue almost everywhere.

The measure representation can still contain singular contributions such as point masses. We now show that the structural properties lead to an integrable memory kernel.

Lemma 4.8 (Representation by an integrable memory kernel [LHEL22, pp. 15–16, Lem. 15])**.** *Let*

$$\{H_t\}_{t\in\mathbb{R}}$$

be a family of continuous linear, causal, regular, and time homogeneous functionals on $\mathcal{X}$. Then there exists a measurable function

$$\rho\colon [0,\infty) \to \mathbb{R}^d$$

satisfying

$$\rho \in L^1([0,\infty),\mathbb{R}^d)$$

such that

$$H_t(x) = \int_0^\infty x_{t-s}^\top \rho(s)\,\mathrm{d}s$$

for every $t \in \mathbb{R}$ and $x \in \mathcal{X}$.

In particular,

$$\sup_{t\in\mathbb{R}} \sup_{\|x\|_{\mathcal{X}}\le 1} |H_t(x)| = \|\rho\|_{L^1([0,\infty))}$$

and, for every $x \in \mathcal{X}$, the map $\mathbb{R} \to \mathbb{R}$,

$$t \longmapsto H_t(x)$$

is continuous for every $x \in \mathcal{X}$.

Proof. We follow the proof of Li et al. [LHEL22, pp. 15–16, proof of Lem. 15] and give the arguments in detail. We prove the result in four steps.

(i) Causality restricts the representing measures to present and past inputs.

(ii) Time homogeneity reduces the family of representing measures to a single measure on $[0,\infty)$.

(iii) Regularity implies that this measure is absolutely continuous with respect to the Lebesgue measure. The Radon–Nikodym theorem then gives the memory kernel ρ.

(iv) We prove the norm identity and the continuity of $t \mapsto H_t(x)$.

Fix $t \in \mathbb{R}$. Since H_t is a continuous linear functional, Theorem 4.6 gives a unique vector valued regular signed Borel measure

$$\mu_t = (\mu_{t,1},\dots,\mu_{t,d})^\top$$

on $\mathbb{R}$ such that

$$\begin{aligned} H_t(x) &= \int_{\mathbb{R}} x_s^\top \,\mathrm{d}\mu_t(s) \\ &= \sum_{i=1}^d \int_{\mathbb{R}} x_{s,i}\,\mathrm{d}\mu_{t,i}(s) \end{aligned} \tag{4.2}$$

for every $x \in \mathcal{X}$. Moreover,

$$\|H_t\| = \sum_{i=1}^d |\mu_{t,i}|(\mathbb{R}),$$

where $\|H_t\|$ is the operator norm from Definition 3.1 and $|\mu_{t,i}|$ denotes the total variation measure of $\mu_{t,i}$ [LHEL22, p. 15, Eqs. (40)–(41)].

(i) We first use causality to restrict the representing measure to the past. Let $x \in \mathcal{X}$ satisfy

$$x_s = 0 \qquad \text{for every } s \le t.$$

The input x may therefore be nonzero only after the evaluation time t. Thus, x agrees with the zero input for every $s \le t$. By Definition 3.2, a causal functional cannot distinguish between two inputs that agree up to the evaluation time. Hence,

$$H_t(x) = H_t(0).$$

Since H_t is linear,

$$H_t(0) = 0.$$

Consequently,

$$H_t(x) = 0.$$

Using the measure representation, we can split the integral into its past and future parts:

$$H_t(x) = \int_{(-\infty,t]} x_s^\top \, \mathrm{d}\mu_t(s) + \int_{(t,\infty)} x_s^\top \, \mathrm{d}\mu_t(s).$$

Since $x_s = 0$ for every $s \le t$, the first integral vanishes automatically. Therefore,

$$0 = H_t(x) = \int_{(t,\infty)} x_s^\top \, \mathrm{d}\mu_t(s)$$

for every $x \in \mathcal{X}$ that vanishes up to time t.

Li et al. state that the integral over future inputs vanishes [LHEL22, p. 15, Eq. (44)]. We now prove that the representing measure itself vanishes on the future interval (t, ∞), that is,

$$\mu_t\big|_{(t,\infty)} = 0.$$

For this, let

$$f \in C_0((t,\infty), \mathbb{R}^d)$$

and define its extension by zero to $\mathbb{R}$ by

$$\widetilde{f}_s := \begin{cases} 0, & s \le t, \\ f_s, & s > t. \end{cases}$$

By the definition of $C_0((t,\infty), \mathbb{R}^d)$,

$$f_s \to 0 \quad \text{as } s \downarrow t \quad \text{and as } s \to \infty.$$

Therefore, the extension by zero is continuous at t and vanishes at infinity, so

$$\widetilde{f} \in C_0(\mathbb{R}, \mathbb{R}^d) = \mathcal{X}.$$

Moreover, $\widetilde{f}_s = 0$ for every $s \le t$. The result above therefore gives

$$0 = \int_{(t,\infty)} \widetilde{f}_s^\top \, \mathrm{d}\mu_t(s) = \int_{(t,\infty)} f_s^\top \, \mathrm{d}\mu_t(s)$$

for every

$$f \in C_0((t,\infty),\mathbb{R}^d).$$

The restriction of μ_t to (t,∞) therefore represents the zero functional on $C_0((t,\infty),\mathbb{R}^d)$. The zero measure also represents this functional. Since (t,∞), equipped with the standard topology, is a locally compact Hausdorff space, the scalar Riesz–Markov–Kakutani theorem applies componentwise on this space [Rud87, p. 130, Thm. 6.19]. Uniqueness of the representing measures gives

$$\mu_t\big|_{(t,\infty)} = 0.$$

Therefore, the representing measure assigns no mass to future input values, and

$$H_t(x) = \int_{(-\infty,t]} x_s^\top \,\mathrm{d}\mu_t(s)$$

for every $x \in \mathcal{X}$ [LHEL22, p. 15, Eq. (44)].

(ii) We next use time homogeneity. By Definition 3.4, for every $t,\tau \in \mathbb{R}$,

$$H_t(x) = H_{t+\tau}\big(x^{(\tau)}\big),$$

where

$$x_s^{(\tau)} = x_{s-\tau}.$$

Using the causal measure representation on both sides gives

$$\begin{aligned}\int_{(-\infty,t]} x_s^\top \,\mathrm{d}\mu_t(s) &= H_t(x)\\ &= H_{t+\tau}\big(x^{(\tau)}\big)\\ &= \int_{(-\infty,t+\tau]} \big(x_s^{(\tau)}\big)^\top \,\mathrm{d}\mu_{t+\tau}(s)\\ &= \int_{(-\infty,t+\tau]} x_{s-\tau}^\top \,\mathrm{d}\mu_{t+\tau}(s).\end{aligned} \tag{4.3}$$

The last equality follows directly from the definition of the shifted input [LHEL22, p. 15, Eq. (45)].

Setting $\tau = -t$ gives $t+\tau = 0$, and therefore

$$\begin{aligned}H_t(x) &= H_0\big(x^{(-t)}\big)\\ &= \int_{(-\infty,0]} \big(x_r^{(-t)}\big)^\top \,\mathrm{d}\mu_0(r)\\ &= \int_{(-\infty,0]} x_{t+r}^\top \,\mathrm{d}\mu_0(r).\end{aligned}$$

Thus, every H_t can be represented using the same measure μ_0. The variable $r \le 0$ describes time relative to the reference time 0. We now replace it by the nonnegative time lag $s = -r$.

Li et al. write the measure representation on $[0,\infty)$ [LHEL22, p. 16, Eq. (46)]. We give the construction of this measure in detail.

By step (i), μ_0 vanishes on $(0,\infty)$. We can therefore restrict each component $\mu_{0,i}$ to $(-\infty,0]$ without losing any mass. Define

$$R\colon (-\infty,0] \to [0,\infty), \qquad R(r) \coloneqq -r.$$

For $i \in \{1,\dots,d\}$, define the reflected signed measure μ_i on $[0,\infty)$ by

$$\mu_i(A) \coloneqq \mu_{0,i}\big(R^{-1}(A)\big),$$

for every Borel set $A \subseteq [0,\infty)$, and set

$$\mu \coloneqq (\mu_1,\dots,\mu_d)^\top.$$

For a positive measure ν, the integration formula for image measures states that

$$\int_{[0,\infty)} g(s)\,\mathrm{d}(\nu \circ R^{-1})(s) = \int_{(-\infty,0]} g(R(r))\,\mathrm{d}\nu(r)$$

for every integrable function g [Bog07, Thm. 3.6.1, p. 190]. Applying this formula to the positive and negative parts of the Jordan decomposition of $\mu_{0,i}$ and subtracting the two identities gives

$$\int_{[0,\infty)} g(s)\,\mathrm{d}\mu_i(s) = \int_{(-\infty,0]} g(-r)\,\mathrm{d}\mu_{0,i}(r).$$

For fixed $t \in \mathbb{R}$ and $i \in \{1,\dots,d\}$, define $g\colon [0,\infty) \to \mathbb{R}$ by

$$g(s) \coloneqq x_{t-s,i}.$$

This function is continuous and bounded because $x \in \mathcal{X}$, and the measures are finite, so the required integrals exist. Moreover,

$$g(-r) = x_{t-(-r),i} = x_{t+r,i}.$$

Applying the image measure formula componentwise therefore gives

$$\begin{aligned} H_t(x) &= \sum_{i=1}^{d} \int_{(-\infty,0]} x_{t+r,i}\,\mathrm{d}\mu_{0,i}(r) \\ &= \sum_{i=1}^{d} \int_{[0,\infty)} x_{t-s,i}\,\mathrm{d}\mu_i(s) \\ &= \int_{[0,\infty)} x_{t-s}^\top\,\mathrm{d}\mu(s). \end{aligned} \tag{4.4}$$

This holds for every $t \in \mathbb{R}$ and $x \in \mathcal{X}$ [LHEL22, p. 16, Eq. (46)].

The family $\{H_t\}_{t\in\mathbb{R}}$ is therefore represented by a single measure μ on $[0,\infty)$. The variable $s \geq 0$ describes the time lag between the past input x_{t-s} and the evaluation time t.

(iii) It remains to show that the common measure μ has a density with respect to the Lebesgue measure. Since μ is obtained from the representing measure of H_0, we use the regularity of H_0.

We follow the regularity argument of Li et al. [LHEL22, p. 16, Eq. (48)] and give the details of the proof. We argue by contradiction. If one component μ_i assigns nonzero mass to a Lebesgue null set, we construct continuous inputs that converge to zero almost everywhere, while their outputs do not converge to zero. This contradicts the regularity of H_0.

Compared with Li et al., we give a more detailed construction of these inputs [LHEL22, p. 16, proof of Lem. 15].

Fix $i \in \{1, \dots, d\}$, and let

$$E \subseteq [0, \infty)$$

be a Borel set satisfying

$$\lambda(E) = 0.$$

We show that

$$|\mu_i|(E) = 0.$$

By Theorem 4.2, let

$$[0, \infty) = P \cup N$$

be a Hahn decomposition of μ_i, where P is positive and N is negative. Let

$$\mu_i = \mu_i^+ - \mu_i^-$$

be the corresponding Jordan decomposition from Corollary 4.3.

We first prove that

$$\mu_i^+(E) = 0.$$

Suppose, for contradiction, that

$$\mu_i^+(E) > 0,$$

and define

$$E_P := E \cap P.$$

By Corollary 4.3,

$$\mu_i^+(E) = \mu_i(E \cap P) = \mu_i^+(E_P).$$

Set

$$\delta := \frac{\mu_i^+(E)}{2}.$$

We regard μ_i^+ as a Borel measure on $\mathbb{R}$ by extending it by zero outside $[0, \infty)$. By the regular approximation theorem, there exist a compact set $K \subseteq E_P$ and an open set $U \supseteq E_P$ such that

$$\mu_i^+(U \setminus K) < \delta$$

[Bog07, Thm. 1.4.8, pp. 15–16]. Since

$$E_P \setminus K \subseteq U \setminus K,$$

we obtain

$$\mu_i^+(E_P \setminus K) < \delta.$$

Consequently,

$$\begin{aligned} \mu_i^+(K) &= \mu_i^+(E_P) - \mu_i^+(E_P \setminus K) \\ &> \mu_i^+(E) - \delta \\ &= \frac{\mu_i^+(E)}{2} > 0. \end{aligned}$$

Moreover, since

$$K \subseteq E$$

and $\lambda(E) = 0$, monotonicity of the Lebesgue measure gives

$$\lambda(K) = 0.$$

For every $n \in \mathbb{N}_+$, define $\varphi_n \colon \mathbb{R} \to [0,1]$ by

$$\varphi_n(s) \coloneqq \max\bigl\{0, 1 - n\, d(s,K)\bigr\}, \qquad s \in \mathbb{R},$$

where

$$d(s,K) \coloneqq \inf_{u \in K} |s - u|$$

denotes the distance from s to K. Since the distance function is continuous, each φ_n is continuous. Furthermore,

$$0 \le \varphi_n(s) \le 1$$

for every $s \in \mathbb{R}$, and

$$\varphi_n(s) = 1 \qquad \text{for } s \in K.$$

The function φ_n vanishes outside the closed neighbourhood

$$\left\{ s \in \mathbb{R} : d(s,K) \le \frac{1}{n} \right\}.$$

The compactness of K implies that this neighbourhood is bounded, and continuity of the distance function $s \mapsto d(s,K)$ implies that it is closed. The neighbourhood is therefore compact, so

$$\varphi_n \in C_0(\mathbb{R}).$$

We next determine the pointwise limit. If $s \in K$, then

$$d(s,K) = 0,$$

and therefore

$$\varphi_n(s) = 1$$

for every n. If $s \notin K$, then K is closed and hence

$$d(s,K) > 0.$$

For all sufficiently large n, we then have

$$n\, d(s,K) \ge 1,$$

so that

$$\varphi_n(s) = 0.$$

It follows that

$$\varphi_n(s) \xrightarrow[n\to\infty]{} \mathbf{1}_K(s)$$

for every $s \in \mathbb{R}$.

We now use these functions to construct inputs that test only the i-th component of the measure. Let $e_i \in \mathbb{R}^d$ denote the i-th standard basis vector, and define

$$x_r^{(n)} \coloneqq \varphi_n(-r) e_i, \qquad r \in \mathbb{R}.$$

Equivalently,

$$x_{r,j}^{(n)} = 0 \qquad \text{for } j \neq i,$$

and

$$x_{r,i}^{(n)} = \varphi_n(-r).$$

The reflected function $r \mapsto \varphi_n(-r)$ is continuous and vanishes outside a compact set because φ_n has these properties, so

$$x^{(n)} \in C_0(\mathbb{R}, \mathbb{R}^d) = \mathcal{X}.$$

The bound $0 \leq \varphi_n \leq 1$ gives

$$\sup_{n \in \mathbb{N}_+} \left\| x^{(n)} \right\|_{\mathcal{X}} \leq 1.$$

For every $r \in \mathbb{R}$,

$$x_r^{(n)} \xrightarrow[n\to\infty]{} \mathbf{1}_K(-r) e_i = \mathbf{1}_{-K}(r) e_i,$$

where

$$-K \coloneqq \{-s : s \in K\}.$$

Since reflection preserves Lebesgue measure and $\lambda(K) = 0$, we also have

$$\lambda(-K) = 0.$$

Therefore,

$$x_r^{(n)} \xrightarrow[n\to\infty]{} 0$$

for Lebesgue-almost every $r \in \mathbb{R}$.

The sequence thus satisfies both assumptions of Definition 3.3. By regularity of H_0,

$$H_0\big(x^{(n)}\big) \xrightarrow[n\to\infty]{} 0.$$

On the other hand, for every $s \geq 0$,

$$x_{-s}^{(n)} = \varphi_n(s) e_i.$$

Using the measure representation, we therefore obtain

$$\begin{aligned} H_0\big(x^{(n)}\big) &= \int_{[0,\infty)} \left(x_{-s}^{(n)}\right)^\top \mathrm{d}\mu(s) \\ &= \int_{[0,\infty)} \varphi_n(s) \, \mathrm{d}\mu_i(s). \end{aligned}$$

We have already shown that

$$\varphi_n(s) \xrightarrow[n\to\infty]{} \mathbf{1}_K(s)$$

for every $s \in [0, \infty)$. Therefore,

$$|\varphi_n(s) - \mathbf{1}_K(s)| \xrightarrow[n\to\infty]{} 0.$$

Moreover, since

$$0 \leq \varphi_n(s) \leq 1 \qquad \text{and} \qquad 0 \leq \mathbf{1}_K(s) \leq 1,$$

we have

$$|\varphi_n(s) - \mathbf{1}_K(s)| \leq 1.$$

The constant function 1 is integrable with respect to the positive measure $|\mu_i|$, because

$$|\mu_i|([0,\infty)) < \infty.$$

By the dominated convergence theorem [Els18, p. 160, Thm. IV.5.2],

$$\begin{aligned}\lim_{n\to\infty} \int_{[0,\infty)} |\varphi_n(s) - \mathbf{1}_K(s)| \,\mathrm{d}|\mu_i|(s) &= \int_{[0,\infty)} \lim_{n\to\infty} |\varphi_n(s) - \mathbf{1}_K(s)| \,\mathrm{d}|\mu_i|(s) \\ &= 0.\end{aligned}$$

Using (4.1),

$$\begin{aligned}\left| \int_{[0,\infty)} \varphi_n(s) \,\mathrm{d}\mu_i(s) - \int_{[0,\infty)} \mathbf{1}_K(s) \,\mathrm{d}\mu_i(s) \right| &\leq \int_{[0,\infty)} |\varphi_n(s) - \mathbf{1}_K(s)| \,\mathrm{d}|\mu_i|(s) \\ &\xrightarrow[n\to\infty]{} 0.\end{aligned}$$

Using

$$H_0\big(x^{(n)}\big) = \int_{[0,\infty)} \varphi_n(s) \,\mathrm{d}\mu_i(s)$$

and

$$\int_{[0,\infty)} \mathbf{1}_K(s) \,\mathrm{d}\mu_i(s) = \mu_i(K),$$

we get

$$H_0\big(x^{(n)}\big) \xrightarrow[n\to\infty]{} \mu_i(K).$$

Since $K \subseteq E_P \subseteq P$, Corollary 4.3 gives

$$\mu_i^-(K) = 0.$$

Therefore,

$$\mu_i(K) = \mu_i^+(K) > 0.$$

We have thus obtained both

$$H_0\big(x^{(n)}\big) \xrightarrow[n\to\infty]{} 0$$

from regularity and

$$H_0\big(x^{(n)}\big) \xrightarrow[n\to\infty]{} \mu_i^+(K) > 0$$

from the measure representation. This is a contradiction. Therefore,

$$\mu_i^+(E) = 0.$$

The same argument applied to the negative part μ_i^-, using

$$E_N \coloneqq E \cap N,$$

shows that

$$\mu_i^-(E) = 0.$$

By Definition 4.4,

$$\begin{aligned}|\mu_i|(E) &= \mu_i^+(E) + \mu_i^-(E)\\ &= 0.\end{aligned}$$

Since E was an arbitrary Lebesgue-null Borel set,

$$|\mu_i| \ll \lambda.$$

By Definition 4.4,

$$|\mu_i| = \mu_i^+ + \mu_i^-.$$

Since μ_i^+ and μ_i^- are positive measures, for every Borel set $A \subseteq [0,\infty)$,

$$0 \le \mu_i^\pm(A) \le |\mu_i|(A).$$

Therefore,

$$|\mu_i| \ll \lambda$$

implies

$$\mu_i^+ \ll \lambda \qquad \text{and} \qquad \mu_i^- \ll \lambda.$$

The Radon–Nikodym theorem gives nonnegative measurable functions $\rho_i^\pm\colon [0,\infty) \to [0,\infty)$, unique up to equality λ-almost everywhere, such that

$$\mu_i^\pm(A) = \int_A \rho_i^\pm(s)\,\mathrm{d}s$$

for every Borel set $A \subseteq [0,\infty)$ [Rud87, Thm. 6.10, p. 121].

Define

$$\rho_i \coloneqq \rho_i^+ - \rho_i^-.$$

Since

$$\mu_i = \mu_i^+ - \mu_i^-,$$

we have

$$\mu_i(A) = \int_A \rho_i(s)\,\mathrm{d}s$$

for every Borel set $A \subseteq [0,\infty)$.

Let P, N be the Hahn decomposition used above. Since

$$\mu_i^-(P) = 0 \qquad \text{and} \qquad \mu_i^+(N) = 0,$$

and $\rho_i^-, \rho_i^+ \ge 0$, we have

$$\rho_i^- = 0 \quad \lambda\text{-almost everywhere on } P,$$

and

$$\rho_i^+ = 0 \quad \lambda\text{-almost everywhere on } N.$$

Therefore,

$$|\rho_i| = \rho_i^+ + \rho_i^- \qquad \lambda\text{-almost everywhere.}$$

By Definition 4.4,

$$|\mu_i| = \mu_i^+ + \mu_i^-.$$

Since μ_i^+ and μ_i^- have densities ρ_i^+ and ρ_i^-, respectively, for every Borel set $A \subseteq [0,\infty)$,

$$\begin{aligned} |\mu_i|(A) &= \mu_i^+(A) + \mu_i^-(A) \\ &= \int_A \left(\rho_i^+(s) + \rho_i^-(s)\right) \mathrm{d}s. \end{aligned}$$

Thus, $|\mu_i|$ has density $\rho_i^+ + \rho_i^-$. Since

$$\rho_i^+ + \rho_i^- = |\rho_i| \qquad \lambda\text{-almost everywhere,}$$

the total variation measure $|\mu_i|$ has density $|\rho_i|$.

Since $|\mu_i|$ is finite,

$$\begin{aligned} \int_0^\infty |\rho_i(s)| \,\mathrm{d}s &= |\mu_i|([0,\infty)) \\ &< \infty. \end{aligned}$$

Hence,

$$\rho_i \in L^1([0,\infty)).$$

Define

$$\rho \coloneqq (\rho_1, \ldots, \rho_d)^\top.$$

Applying the integration formula for measures with densities [Rud87, p. 23, Thm. 1.29] to the positive and negative parts of f, separately for μ_i^+ and μ_i^-, and subtracting gives, for every bounded real valued Borel function f,

$$\int_{[0,\infty)} f(s) \,\mathrm{d}\mu_i(s) = \int_0^\infty f(s)\rho_i(s) \,\mathrm{d}s.$$

For fixed $t \in \mathbb{R}$, we apply this identity with

$$f(s) = x_{t-s,i}.$$

Since $x \in \mathcal{X}$, this function is continuous and bounded. Applying the identity to every component gives

$$\begin{aligned} H_t(x) &= \sum_{i=1}^d \int_{[0,\infty)} x_{t-s,i} \,\mathrm{d}\mu_i(s) \\ &= \sum_{i=1}^d \int_0^\infty x_{t-s,i}\rho_i(s) \,\mathrm{d}s \\ &= \int_0^\infty x_{t-s}^\top \rho(s) \,\mathrm{d}s. \end{aligned}$$

This yields the kernel representation in [LHEL22, p. 16, Eq. (50)].

(iv) Li et al. give the norm identity and the continuity of $t \mapsto H_t(x)$ [LHEL22, p. 16, Eq. (47)]. We give a detailed proof of both properties. We show that the reflection preserves the total variation and that time shifts preserve the norm on $\mathcal{X}$. For continuity, we verify the assumptions of the dominated convergence theorem.

Recall that

$$\|\nu\|_1(A) = \sum_{i=1}^d |\nu_i|(A)$$

denotes the sum of the total variations of the component measures on A.

For each i, let $P_i, N_i \subseteq (-\infty, 0]$ be a Hahn decomposition of $\mu_{0,i}$. Since the reflection $R(r) = -r$ is bijective, $R(P_i)$ and $R(N_i)$ are respectively positive and negative sets for the reflected measure μ_i. Hence, reflection preserves the total masses of the positive and negative parts, and therefore

$$|\mu_i|([0,\infty)) = |\mu_{0,i}|((-\infty, 0]).$$

Summing over i gives

$$\|\mu\|_1([0,\infty)) = \|\mu_0\|_1((-\infty, 0]).$$

By step (i), μ_0 vanishes on $(0,\infty)$. Hence,

$$\|\mu_0\|_1((-\infty, 0]) = \|\mu_0\|_1(\mathbb{R}).$$

Theorem 4.6 gives

$$\|\mu_0\|_1(\mathbb{R}) = \|H_0\|,$$

where $\|H_0\|$ is the operator norm from Definition 3.1. Therefore,

$$\|\mu\|_1([0,\infty)) = \|H_0\|.$$

Finally, time homogeneity gives

$$H_t(x) = H_0\big(x^{(-t)}\big).$$

The shift preserves the norm on $\mathcal{X}$, because

$$\begin{aligned}\|x^{(-t)}\|_{\mathcal{X}} &= \sup_{s\in\mathbb{R}} \|x_{s+t}\|_\infty \\ &= \sup_{r\in\mathbb{R}} \|x_r\|_\infty \\ &= \|x\|_{\mathcal{X}}.\end{aligned}$$

Moreover, the map $x \mapsto x^{(-t)}$ is a bijection of $\mathcal{X}$. It therefore maps the unit ball of $\mathcal{X}$ onto itself. Using the definition of the operator norm in Definition 3.1,

$$\begin{aligned}\|H_t\| &= \sup_{\|x\|_{\mathcal{X}}\le 1} |H_t(x)| \\ &= \sup_{\|x\|_{\mathcal{X}}\le 1} \big|H_0\big(x^{(-t)}\big)\big| \\ &= \sup_{\|z\|_{\mathcal{X}}\le 1} |H_0(z)| \\ &= \|H_0\|.\end{aligned}$$

Therefore,

$$\|H_t\| = \|H_0\|$$

for every $t \in \mathbb{R}$.

For each component, the measure μ_i has density ρ_i with respect to the Lebesgue measure. Its total variation measure therefore has density $|\rho_i|$, so

$$|\mu_i|([0,\infty)) = \int_0^\infty |\rho_i(s)|\,\mathrm{d}s.$$

Summing over all components gives

$$\begin{aligned}\|\mu\|_1([0,\infty)) &= \sum_{i=1}^{d} |\mu_i|([0,\infty)) \\ &= \sum_{i=1}^{d} \int_0^\infty |\rho_i(s)| \,\mathrm{d}s \\ &= \|\rho\|_{L^1([0,\infty))}.\end{aligned}$$

Combining the identities above gives

$$\begin{aligned}\|\rho\|_{L^1([0,\infty))} &= \|\mu\|_1([0,\infty)) \\ &= \|H_0\| \\ &= \|H_t\|\end{aligned}$$

for every $t \in \mathbb{R}$. Therefore,

$$\sup_{t\in\mathbb{R}} \|H_t\| = \|\rho\|_{L^1([0,\infty))}.$$

It remains to verify that $t \mapsto H_t(x)$ is continuous for every fixed $x \in \mathcal{X}$. By (4.4),

$$\begin{aligned}H_{t+\delta}(x) - H_t(x) &= \int_{[0,\infty)} x_{t+\delta-s}^\top \,\mathrm{d}\mu(s) - \int_{[0,\infty)} x_{t-s}^\top \,\mathrm{d}\mu(s) \\ &= \int_{[0,\infty)} (x_{t+\delta-s} - x_{t-s})^\top \,\mathrm{d}\mu(s).\end{aligned}$$

Writing the vector valued integral componentwise gives

$$H_{t+\delta}(x) - H_t(x) = \sum_{i=1}^{d} \int_{[0,\infty)} (x_{t+\delta-s,i} - x_{t-s,i}) \,\mathrm{d}\mu_i(s).$$

Applying (4.1) componentwise and using the triangle inequality, we obtain

$$\begin{aligned}|H_{t+\delta}(x) - H_t(x)| &\le \sum_{i=1}^{d} \int_{[0,\infty)} |x_{t+\delta-s,i} - x_{t-s,i}| \,\mathrm{d}|\mu_i|(s) \\ &\le \sum_{i=1}^{d} \int_{[0,\infty)} \|x_{t+\delta-s} - x_{t-s}\|_\infty \,\mathrm{d}|\mu_i|(s).\end{aligned} \tag{4.5}$$

This corresponds to the continuity argument in the proof of Lemma 15 of Li et al. [LHEL22, p. 16].

Let $(\delta_n)_{n\in\mathbb{N}}$ be an arbitrary sequence satisfying

$$\delta_n \xrightarrow[n\to\infty]{} 0.$$

For every fixed $s \ge 0$, continuity of x gives

$$x_{t+\delta_n-s} \xrightarrow[n\to\infty]{} x_{t-s} \qquad \text{in } \mathbb{R}^d,$$

and therefore

$$\|x_{t+\delta_n-s} - x_{t-s}\|_\infty \xrightarrow[n\to\infty]{} 0.$$

Moreover, the triangle inequality and the definition of $\|x\|_\mathcal{X}$ give

$$\begin{aligned}\|x_{t+\delta_n-s} - x_{t-s}\|_\infty &\leq \|x_{t+\delta_n-s}\|_\infty + \|x_{t-s}\|_\infty \\ &\leq 2\|x\|_\mathcal{X}.\end{aligned}$$

The constant function on the right is integrable with respect to $|\mu_i|$, because

$$\int_{[0,\infty)} 2\|x\|_\mathcal{X} \,\mathrm{d}|\mu_i|(s) = 2\|x\|_\mathcal{X}|\mu_i|([0,\infty)) < \infty.$$

The dominated convergence theorem [Els18, p. 160, Thm. IV.5.2] therefore gives, for every $i \in \{1,\dots,d\}$,

$$\begin{aligned}\lim_{n\to\infty}\int_{[0,\infty)} \|x_{t+\delta_n-s} - x_{t-s}\|_\infty \,\mathrm{d}|\mu_i|(s) &= \int_{[0,\infty)} \lim_{n\to\infty} \|x_{t+\delta_n-s} - x_{t-s}\|_\infty \,\mathrm{d}|\mu_i|(s) \\ &= \int_{[0,\infty)} 0 \,\mathrm{d}|\mu_i|(s) \\ &= 0.\end{aligned}$$

Using (4.5), for every n,

$$\begin{aligned}0 &\leq |H_{t+\delta_n}(x) - H_t(x)| \\ &\leq \sum_{i=1}^{d} \int_{[0,\infty)} \|x_{t+\delta_n-s} - x_{t-s}\|_\infty \,\mathrm{d}|\mu_i|(s).\end{aligned}$$

For every $i \in \{1,\dots,d\}$, the corresponding integral on the right converges to zero. Since the sum contains only finitely many terms,

$$|H_{t+\delta_n}(x) - H_t(x)| \xrightarrow[n\to\infty]{} 0.$$

Therefore,

$$H_{t+\delta_n}(x) \xrightarrow[n\to\infty]{} H_t(x).$$

Since this holds for every sequence $\delta_n \xrightarrow[n\to\infty]{} 0$,

$$\lim_{\delta\to 0} H_{t+\delta}(x) = H_t(x).$$

This shows that the map $t \mapsto H_t(x)$ is continuous.

□

The function ρ is called the memory kernel of the family $\{H_t\}_{t\in\mathbb{R}}$. For $s \geq 0$, the vector $\rho(s)$ describes how strongly the past input x_{t-s} contributes to the output at time t. The component $\rho_i(s)$ describes the contribution of the i-th input component. A kernel concentrated near $s = 0$ mainly depends on recent inputs, while a kernel that remains significant for large s also depends on inputs from the distant past.

4.4 RNN Kernels and Reduction to Kernel Approximation

We now compare the memory kernel ρ of H_t with the kernel generated by a stable linear continuous time RNN. Recall from Section 3.4 that

$$\widehat{H}_t(x) = \int_0^\infty c^\top e^{Ws} U x_{t-s} \, \mathrm{d}s,$$

where

$$W \in \mathcal{W}_m, \qquad U \in \mathbb{R}^{m \times d}, \qquad c \in \mathbb{R}^m.$$

The scalar integrand can be rewritten as

$$\begin{aligned} c^\top e^{Ws} U x_{t-s} &= \left(c^\top e^{Ws} U\right) x_{t-s} \\ &= x_{t-s}^\top U^\top \left(e^{Ws}\right)^\top c \\ &= x_{t-s}^\top U^\top e^{W^\top s} c. \end{aligned}$$

The second equality uses that the scalar $c^\top e^{Ws} U x_{t-s}$ is equal to its transpose. For the last equality, we use

$$\left(e^{Ws}\right)^\top = e^{W^\top s},$$

which follows directly from the power series definition of the matrix exponential.

Definition 4.9 (RNN memory kernel)**.** Let

$$W \in \mathcal{W}_m, \qquad U \in \mathbb{R}^{m \times d}, \qquad c \in \mathbb{R}^m.$$

The memory kernel generated by the RNN is the function

$$\widehat{\rho} : [0, \infty) \to \mathbb{R}^d, \qquad \widehat{\rho}(s) \coloneqq U^\top e^{W^\top s} c.$$

The map $\widehat{\rho}$ is continuous and therefore measurable because $s \mapsto e^{W^\top s}$ is continuous. The RNN functional therefore has the same form as the target functional,

$$\widehat{H}_t(x) = \int_0^\infty x_{t-s}^\top \widehat{\rho}(s) \, \mathrm{d}s$$

[LHEL22, p. 17, Eq. (57)].

We now prove that the RNN kernel is integrable. Our own contribution is the detailed derivation of this bound, including a proof of $\|A^\top\|_2 = \|A\|_2$. Because $W \in \mathcal{W}_m$, Lemma 3.7 gives constants $C > 0$ and $\alpha > 0$ such that

$$\|e^{Ws}\|_2 \leq C e^{-\alpha s}, \qquad s \geq 0.$$

Recall from (3.2) that $\|\cdot\|_2$ denotes the induced Euclidean operator norm for matrices. Let $p \in \mathbb{N}_+$. For every $y \in \mathbb{R}^p$, the Cauchy–Schwarz inequality gives

$$|u^\top y| \leq \|y\|_2$$

for every $u \in \mathbb{R}^p$ with $\|u\|_2 = 1$. If $y \neq 0$, choosing

$$u = \frac{y}{\|y\|_2}$$

gives

$$|u^\top y| = \|y\|_2.$$

For $y = 0$, the equality is immediate. Therefore,

$$\|y\|_2 = \sup_{\|u\|_2=1} |u^\top y|.$$

For every real matrix $A \in \mathbb{R}^{p\times q}$, where $q \in \mathbb{N}_+$,

$$\|A^\top\|_2 = \|A\|_2.$$

Indeed, by (3.2),

$$\begin{aligned}
\|A^\top\|_2 &= \sup_{\substack{v\in\mathbb{R}^p \\ \|v\|_2=1}} \|A^\top v\|_2 \\
&= \sup_{\substack{u\in\mathbb{R}^q,\ \|u\|_2=1 \\ v\in\mathbb{R}^p,\ \|v\|_2=1}} |u^\top A^\top v| \\
&= \sup_{\substack{u\in\mathbb{R}^q,\ \|u\|_2=1 \\ v\in\mathbb{R}^p,\ \|v\|_2=1}} |v^\top A u| \\
&= \|A\|_2.
\end{aligned}$$

The last equality follows from the same identity applied to the vector Au, followed by the definition of the induced matrix norm. Applying this identity to U and e^{Ws} gives

$$\|U^\top\|_2 = \|U\|_2, \qquad \|e^{W^\top s}\|_2 = \|e^{Ws}\|_2.$$

Together with

$$\|v\|_1 \le \sqrt{d}\,\|v\|_2, \qquad v \in \mathbb{R}^d,$$

and the submultiplicativity of the induced operator norm, we get

$$\begin{aligned}
\|\widehat{\rho}(s)\|_1 &\le \sqrt{d}\,\|\widehat{\rho}(s)\|_2 \\
&= \sqrt{d}\,\|U^\top e^{W^\top s} c\|_2 \\
&\le \sqrt{d}\,\|U\|_2 \|e^{Ws}\|_2 \|c\|_2 \\
&\le C\sqrt{d}\,\|U\|_2 \|c\|_2 e^{-\alpha s}.
\end{aligned}$$

Integrating this bound over $[0,\infty)$ gives

$$\begin{aligned}
\|\widehat{\rho}\|_{L^1([0,\infty))} &= \int_0^\infty \|\widehat{\rho}(s)\|_1 \,\mathrm{d}s \\
&\le C\sqrt{d}\,\|U\|_2 \|c\|_2 \int_0^\infty e^{-\alpha s}\,\mathrm{d}s \\
&= \frac{C\sqrt{d}}{\alpha} \|U\|_2 \|c\|_2 \\
&< \infty.
\end{aligned}$$

We get

$$\widehat{\rho} \in L^1([0,\infty), \mathbb{R}^d).$$

The representations of H_t and $\widehat{H}_t$ now differ only in their memory kernels. The following proposition relates the error between the functionals directly to the error between the kernels.

Li et al. use this error bound in their proof of universal approximation [LHEL22, p. 18, Eq. (61)]. We state the bound as a separate proposition for integrable kernels and give a detailed proof.

Proposition 4.10 (Functional error controlled by the kernel error [cf. LHEL22, p. 18, Eq. (61)])**.** *Let*

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}$$

and

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m$$

with representations

$$H_t(x) = \int_0^\infty x_{t-s}^\top \rho(s)\,\mathrm{d}s$$

and

$$\widehat{H}_t(x) = \int_0^\infty x_{t-s}^\top \widehat{\rho}(s)\,\mathrm{d}s,$$

where

$$\rho, \widehat{\rho} \in L^1([0,\infty), \mathbb{R}^d).$$

Then

$$\begin{aligned}\sup_{t\in\mathbb{R}} \|H_t - \widehat{H}_t\| &= \sup_{t\in\mathbb{R}} \sup_{\|x\|_{\mathcal{X}}\le 1} |H_t(x) - \widehat{H}_t(x)| \\ &\le \|\rho - \widehat{\rho}\|_{L^1([0,\infty))},\end{aligned}$$

where $\|H_t - \widehat{H}_t\|$ is the operator norm from Definition 3.1.

Proof. Let $t \in \mathbb{R}$ and $x \in \mathcal{X}$. Subtracting the two representations gives

$$\begin{aligned}H_t(x) - \widehat{H}_t(x) &= \int_0^\infty x_{t-s}^\top \rho(s)\,\mathrm{d}s - \int_0^\infty x_{t-s}^\top \widehat{\rho}(s)\,\mathrm{d}s \\ &= \int_0^\infty x_{t-s}^\top \left(\rho(s) - \widehat{\rho}(s)\right)\,\mathrm{d}s.\end{aligned}$$

Taking absolute values and using the triangle inequality for integrals gives

$$\left|H_t(x) - \widehat{H}_t(x)\right| \le \int_0^\infty \left|x_{t-s}^\top \left(\rho(s) - \widehat{\rho}(s)\right)\right|\,\mathrm{d}s.$$

For vectors $a, b \in \mathbb{R}^d$, we have

$$|a^\top b| = \left|\sum_{i=1}^d a_i b_i\right| \le \sum_{i=1}^d |a_i||b_i| \le \|a\|_\infty \sum_{i=1}^d |b_i| = \|a\|_\infty \|b\|_1.$$

Applying this inequality pointwise gives

$$\left|H_t(x) - \widehat{H}_t(x)\right| \le \int_0^\infty \|x_{t-s}\|_\infty \|\rho(s) - \widehat{\rho}(s)\|_1\,\mathrm{d}s.$$

By the definition of the norm on $\mathcal{X}$,

$$\|x_{t-s}\|_\infty \leq \|x\|_\mathcal{X}$$

for every $s \geq 0$. Therefore,

$$\begin{aligned}\left|H_t(x) - \widehat{H}_t(x)\right| &\leq \|x\|_\mathcal{X} \int_0^\infty \|\rho(s) - \widehat{\rho}(s)\|_1 \,\mathrm{d}s \\ &= \|x\|_\mathcal{X} \|\rho - \widehat{\rho}\|_{L^1([0,\infty))}.\end{aligned}$$

Taking the supremum over all inputs satisfying

$$\|x\|_\mathcal{X} \leq 1$$

and using the operator norm from Definition 3.1 gives

$$\begin{aligned}\|H_t - \widehat{H}_t\| &= \sup_{\|x\|_\mathcal{X} \leq 1} |H_t(x) - \widehat{H}_t(x)| \\ &\leq \|\rho - \widehat{\rho}\|_{L^1([0,\infty))}.\end{aligned}$$

Since the right hand side does not depend on t, taking the supremum over $t \in \mathbb{R}$ gives

$$\sup_{t\in\mathbb{R}} \|H_t - \widehat{H}_t\| \leq \|\rho - \widehat{\rho}\|_{L^1([0,\infty))}.$$

□

Approximation Viewpoint:

We now have two different families of functionals. The target family introduced in Section 3.2 satisfies

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}.$$

In contrast, a stable linear continuous time RNN with hidden dimension m generates a family

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m,$$

as introduced in Section 3.4. The first family is the target that we want to approximate, while the second family is generated by the RNN.

Section 4.3 gives the memory kernel ρ of H_t, and Section 4.4 gives the memory kernel $\widehat{\rho}$ of $\widehat{H}_t$, where

$$\widehat{\rho}(s) = U^\top e^{W^\top s} c.$$

For the approximation analysis, we now ask two questions. Can $\widehat{H}_t$ approximate H_t arbitrarily well, and how fast does the approximation error decrease when the hidden dimension m increases?

Studying these questions directly for H_t and $\widehat{H}_t$ is difficult. Following the kernel approximation viewpoint of Li et al. [LHEL22, p. 11], Proposition 4.10 reduces the problem to approximating ρ by $\widehat{\rho}$ in the L^1 norm. These two functions are easier to analyze directly.

4.5 Uniqueness and kernel induced Extension to Step Inputs

Lemma 4.8 gives the existence of an integrable memory kernel for every family in $\mathcal{H}$. We first prove that this kernel is unique up to Lebesgue almost everywhere equality. We then use the kernel representation to extend the functionals to step inputs.

Corollary 4.11 and Proposition 4.12, including their proofs, are our own additions to the results of Li et al. [LHEL22, Secs. 4–5].

Corollary 4.11 (Uniqueness of the memory kernel)**.** *Let*

$$\rho, \rho' \in L^1([0,\infty), \mathbb{R}^d)$$

and suppose that

$$\int_0^\infty x_{t-s}^\top \rho(s)\,\mathrm{d}s = \int_0^\infty x_{t-s}^\top \rho'(s)\,\mathrm{d}s$$

for every $x \in \mathcal{X}$ and every $t \in \mathbb{R}$. Then

$$\rho = \rho'$$

Lebesgue almost everywhere on $[0,\infty)$.

Proof. The idea is to turn the two kernels into representing measures and to use the uniqueness in Theorem 4.6.

Set

$$\kappa \coloneqq \rho - \rho' \in L^1([0,\infty), \mathbb{R}^d).$$

Subtracting the two representations gives

$$\int_0^\infty x_{t-s}^\top \kappa(s)\,\mathrm{d}s = 0$$

for every $x \in \mathcal{X}$ and every $t \in \mathbb{R}$. Setting $t = 0$ and using the substitution $r = -s$, we get

$$\sum_{i=1}^{d} \int_{-\infty}^{0} x_{r,i}\kappa_i(-r)\,\mathrm{d}r = 0$$

for every $x \in \mathcal{X}$.

For $i = 1, \dots, d$, define

$$g_i \colon \mathbb{R} \to \mathbb{R}$$

by

$$g_i(r) \coloneqq \begin{cases} \kappa_i(-r), & r \le 0, \\ 0, & r > 0. \end{cases}$$

Since

$$\kappa_i \in L^1([0,\infty)),$$

we have

$$\begin{aligned} \int_{\mathbb{R}} |g_i(r)|\,\mathrm{d}r &= \int_{-\infty}^{0} |\kappa_i(-r)|\,\mathrm{d}r \\ &= \int_0^\infty |\kappa_i(s)|\,\mathrm{d}s \\ &< \infty. \end{aligned}$$

Thus,

$$g_i \in L^1(\mathbb{R}).$$

Now define the signed Borel measure

$$\nu_i \colon \mathcal{B}(\mathbb{R}) \to \mathbb{R}$$

by

$$\nu_i(A) \coloneqq \int_A g_i(r)\,\mathrm{d}r, \qquad A \in \mathcal{B}(\mathbb{R}).$$

Its total variation is finite because

$$|\nu_i|(\mathbb{R}) = \int_{\mathbb{R}} |g_i(r)|\,\mathrm{d}r < \infty.$$

The finite positive Borel measures ν_i^+ and ν_i^- on $\mathbb{R}$ are regular by [Bog07, Thm. 1.4.8, pp. 15–16]. Hence, the signed measure $\nu_i = \nu_i^+ - \nu_i^-$ is regular.

Set

$$\nu \coloneqq (\nu_1, \dots, \nu_d)^\top.$$

By the definition of ν_i,

$$\begin{aligned}\sum_{i=1}^d \int_{\mathbb{R}} x_{r,i}\,\mathrm{d}\nu_i(r) &= \sum_{i=1}^d \int_{\mathbb{R}} x_{r,i} g_i(r)\,\mathrm{d}r \\ &= \sum_{i=1}^d \int_{-\infty}^0 x_{r,i}\kappa_i(-r)\,\mathrm{d}r \\ &= 0.\end{aligned}$$

Hence, ν represents the zero functional on $\mathcal{X}$.

The zero measure also represents the zero functional. By the uniqueness of the representing measure in Theorem 4.6,

$$\nu_i = 0, \qquad i = 1, \dots, d.$$

The measure ν_i has Lebesgue density g_i, while the zero measure has Lebesgue density zero. By the uniqueness of the Radon–Nikodym density [Rud87, p. 121, Thm. 6.10(b)],

$$g_i = 0$$

Lebesgue almost everywhere on $\mathbb{R}$. Since

$$g_i(r) = \kappa_i(-r)$$

for $r \leq 0$, we get

$$\kappa_i(s) = 0$$

for Lebesgue almost every $s \geq 0$. This holds for every $i = 1, \dots, d$, so

$$\kappa = 0$$

Lebesgue almost everywhere on $[0, \infty)$. Since

$$\kappa = \rho - \rho',$$

we conclude that

$$\rho = \rho'$$

Lebesgue almost everywhere on $[0, \infty)$. □

The memory kernel is therefore unique as an element of $L^1([0,\infty),\mathbb{R}^d)$.

Li et al. evaluate the functionals directly at step inputs [LHEL22, p. 12, Eq. (24)]. However, these inputs do not belong to $\mathcal{X}$, so the original functionals are not defined on them. We use the unique memory kernel to define an extension to these inputs.

Proposition 4.12 (kernel induced extension to step inputs [cf. LHEL22, p. 18, Eq. (62)])**.** *Let*

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}$$

with memory kernel

$$\rho \in L^1([0,\infty),\mathbb{R}^d).$$

For $i = 1,\dots,d$, let $e_i \in \mathbb{R}^d$ be the i-th standard basis vector and define

$$\mathbf{e}_i \colon \mathbb{R} \to \mathbb{R}^d, \qquad \mathbf{e}_{i,t} \coloneqq e_i \mathbf{1}_{\{t\geq 0\}}.$$

Then the expression

$$\widetilde{H}_t(\mathbf{e}_i) \coloneqq \int_0^\infty \mathbf{e}_{i,t-s}^\top \rho(s)\,\mathrm{d}s$$

is well defined and independent of the representative of the memory kernel.

Define $y_i \colon \mathbb{R} \to \mathbb{R}$ by

$$y_i(t) \coloneqq \widetilde{H}_t(\mathbf{e}_i).$$

Then

$$y_i(t) = \begin{cases} 0, & t < 0, \\ \displaystyle\int_0^t \rho_i(s)\,\mathrm{d}s, & t \geq 0. \end{cases}$$

Proof. The function $\mathbf{e}_i$ is discontinuous at $t = 0$ and does not vanish as $t \to +\infty$. Therefore,

$$\mathbf{e}_i \notin \mathcal{X} = C_0(\mathbb{R},\mathbb{R}^d).$$

The original functional

$$H_t \colon \mathcal{X} \to \mathbb{R}$$

is therefore not defined on $\mathbf{e}_i$.

The kernel representation nevertheless gives a finite value because

$$\begin{aligned} \left|\widetilde{H}_t(\mathbf{e}_i)\right| &\leq \int_0^\infty \left|\mathbf{e}_{i,t-s}^\top \rho(s)\right| \,\mathrm{d}s \\ &\leq \int_0^\infty |\rho_i(s)|\,\mathrm{d}s \\ &< \infty. \end{aligned}$$

To show that the kernel induced extension is well defined, we have to show that its value does not depend on the choice of the memory kernel. Suppose that

$$\rho, \rho' \in L^1([0,\infty),\mathbb{R}^d)$$

are two memory kernels representing the same family

$$\{H_t\}_{t\in\mathbb{R}}.$$

Then, for every $x \in \mathcal{X}$ and every $t \in \mathbb{R}$,

$$H_t(x) = \int_0^\infty x_{t-s}^\top \rho(s)\,\mathrm{d}s = \int_0^\infty x_{t-s}^\top \rho'(s)\,\mathrm{d}s.$$

By Corollary 4.11, this implies

$$\rho(s) = \rho'(s)$$

for Lebesgue almost every $s \geq 0$.

Now fix $i \in \{1, \ldots, d\}$ and $t \in \mathbb{R}$. Multiplying the almost everywhere equality by the same vector $\mathbf{e}_{i,t-s}^\top$ gives

$$\mathbf{e}_{i,t-s}^\top \rho(s) = \mathbf{e}_{i,t-s}^\top \rho'(s)$$

for Lebesgue almost every $s \geq 0$. Hence, the two integrands in the definitions of the corresponding extensions are equal almost everywhere. Since Lebesgue integrals of integrable functions that agree almost everywhere are equal,

$$\int_0^\infty \mathbf{e}_{i,t-s}^\top \rho(s)\,\mathrm{d}s = \int_0^\infty \mathbf{e}_{i,t-s}^\top \rho'(s)\,\mathrm{d}s.$$

Therefore, the value of

$$\widetilde{H}_t(\mathbf{e}_i)$$

does not depend on the chosen memory kernel, so the kernel induced extension is well defined.

For $t \geq 0$,

$$\begin{aligned} y_i(t) &= \int_0^\infty \mathbf{e}_{i,t-s}^\top \rho(s)\,\mathrm{d}s \\ &= \int_0^\infty \mathbf{1}_{\{t-s\geq 0\}} \rho_i(s)\,\mathrm{d}s \\ &= \int_0^t \rho_i(s)\,\mathrm{d}s. \end{aligned}$$

For $t < 0$ and $s \geq 0$, we have

$$t - s < 0.$$

Therefore,

$$\mathbf{1}_{\{t-s\geq 0\}} = 0,$$

and

$$y_i(t) = 0.$$

This proves the stated representation. □

The functions y_i are the step responses used by Li et al. in the approximation-rate result [LHEL22, p. 12, Eqs. (24)–(26)]. We will use them later in Section 5.3 to formulate the smoothness and decay assumptions needed for the approximation-rate bound.

5 Approximation Theory

This chapter studies the approximation of

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}$$

by RNN families

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m.$$

We first discuss a classical universal approximation result for recurrent neural networks. We then study universal approximation and approximation rates for stable linear continuous time RNNs. Finally, we consider inverse approximation results and the curse of memory.

The main reference for this chapter is Li et al. [LHEL22, Secs. 4–5, pp. 9–21]. For the classical universal approximation result, we use Schäfer and Zimmermann [SZ06]. For the polynomial approximation estimate, we use Timan [Tim63]. For the discussion of nonnormal matrices, we use Trefethen and Embree [TE05].

Our own work in this chapter consists of several proofs and numerical experiments. We construct a counterexample to the inverse approximation theorem of Li et al. and prove a modified result for normal recurrent matrices [LHEL22, pp. 12–13, Thm. 11]. We also adapt the decay assumption in the approximation rate theorem and give a detailed proof, including the missing steps. Finally, we implement a numerical experiment to illustrate the approximation rate.

5.1 Classical Universality for Recurrent Neural Networks

Before we study universal approximation in our functional setting, we look at the result of Schäfer and Zimmermann [SZ06]. Their theorem approximates a given dynamical system by an RNN. This gives a useful comparison with the functional approximation problem studied by Li et al. [LHEL22].

For consistency with our setting, we restrict the discussion to the scalar output case $n = 1$.

Definition 5.1 (Discrete time target system [SZ06, p. 633, Eq. (1)])**.** Using notation adapted to this thesis, a discrete-time target system is given by

$$\begin{aligned} z_{k+1} &= g(z_k, x_k), \\ y_k &= r(z_k), \end{aligned} \qquad k = 1, \dots, T,$$

where

$$x_k \in \mathbb{R}^d, \qquad z_k \in \mathbb{R}^q, \qquad y_k \in \mathbb{R},$$

and

$$g\colon \mathbb{R}^q \times \mathbb{R}^d \to \mathbb{R}^q, \qquad r\colon \mathbb{R}^q \to \mathbb{R}.$$

Here, x_k is the input, z_k is the internal state of the target, and y_k is the target output. The function g determines how the target state changes, while r maps the state to the output. We use r for the output map instead of h to avoid confusion with the hidden state notation used elsewhere in this thesis.

The target is approximated by a nonlinear discrete time RNN with its own hidden state.

Definition 5.2 (Approximating recurrent neural network [SZ06, p. 637, Def. 8]). In the scalar output case, the approximating RNN is of the form

$$\begin{aligned} \widehat{z}_{k+1} &= \sigma(A\widehat{z}_k + Bx_k - b), \\ \widehat{y}_k &= c^\top \widehat{z}_k, \end{aligned}$$

where

$$\widehat{z}_k \in \mathbb{R}^m, \qquad A \in \mathbb{R}^{m\times m}, \qquad B \in \mathbb{R}^{m\times d}, \qquad b, c \in \mathbb{R}^m.$$

The function

$$\sigma \colon \mathbb{R} \to \mathbb{R}$$

is applied componentwise. In Theorem 5.3, we assume that σ is a continuous sigmoid function. A sigmoid function is bounded and monotonically increasing. It has finite limits as its argument tends to $-\infty$ and $+\infty$ [SZ06, p. 635, Def. 3].

The states z_k and $\widehat{z}_k$ are different objects. The first belongs to the target system, while the second belongs to the approximating RNN.

In the formulation of Schäfer and Zimmermann, the main result is the following.

Theorem 5.3 (Universal Approximation for Recurrent Neural Networks [SZ06, p. 637, Thm. 2]). *Let*

$$g \colon \mathbb{R}^q \times \mathbb{R}^d \to \mathbb{R}^q$$

be measurable and let

$$r \colon \mathbb{R}^q \to \mathbb{R}$$

be continuous. Then, on every finite time horizon $k = 1, \dots, T$, the target system

$$\begin{aligned} z_{k+1} &= g(z_k, x_k), \\ y_k &= r(z_k) \end{aligned}$$

can be approximated with arbitrary accuracy by a recurrent neural network with a sufficiently large hidden dimension m and a continuous sigmoid activation function.

The proof first approximates the state transition g by a feedforward network and rewrites this approximation as a recurrent state equation. It then approximates the output map r and combines both constructions by enlarging the hidden state [SZ06, pp. 637–639].

The approximation problem considered in this thesis has a different form. Schäfer and Zimmermann start from a given dynamical system with an internal state z_k and approximate this system by an RNN. In our setting, the target is instead given directly by

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H},$$

and we approximate it by

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m.$$

No internal state representation of the target H_t is assumed [LHEL22, p. 9, Eq. (21)].

There is also a difference in the approximation horizon. The result of Schäfer and Zimmermann is formulated for a fixed finite number of time steps $k = 1, \dots, T$. In contrast, Theorem 5.5 controls the approximation error uniformly over all $t \in \mathbb{R}$ [LHEL22, p. 10, Thm. 7].

Li et al. point out that their functional formulation does not require the target relationship to be generated by an underlying dynamical system [LHEL22, p. 11, Rem. 9]. This difference is important in the linear case. If a finite dimensional linear dynamical system generates the target functionals, Li et al. observe that an RNN with a large enough hidden state can represent the target exactly [LHEL22, p. 10]. The more general functional setting lets us study approximation errors and how they depend on memory.

Also, the theorem of Schäfer and Zimmermann is an existence result [SZ06, p. 637, Thm. 2]. It does not give an approximation rate in terms of the hidden dimension. The approximation rates we study later in this thesis address this question.

5.2 Universal Approximation

As shown in Section 4.4, the approximation problem can be reduced to approximating the target memory kernel ρ by an RNN kernel $\widehat{\rho}$. The following lemma gives the approximation result for the kernels that we need.

Lemma 5.4 (Density of Exponential Sums [LHEL22, p. 16, Lem. 16])**.** *Let*

$$\rho \in L^1([0,\infty)).$$

Then, for every $\varepsilon > 0$, there exists a polynomial p with $p(0) = 0$ such that

$$\left\|\rho - p(e^{-\cdot})\right\|_{L^1([0,\infty))} = \int_0^\infty \left|\rho(t) - p(e^{-t})\right| \,\mathrm{d}t \leq \varepsilon.$$

The condition $p(0) = 0$ means that p has no constant term. A nonzero constant term would give a non decaying mode in $p(e^{-t})$. This mode is not integrable on $[0,\infty)$. For some $m \in \mathbb{N}$, we can write

$$p(u) = \sum_{j=1}^{m} \alpha_j u^j,$$

and

$$p(e^{-t}) = \sum_{j=1}^{m} \alpha_j e^{-jt}.$$

Lemma 5.4 therefore shows that finite sums of decaying exponential functions can approximate every integrable kernel arbitrarily well in the L^1 norm.

Proof. We follow the proof of Lemma 16 of Li et al. [LHEL22, pp. 16–17], replacing the Müntz–Szász step with the Weierstrass approximation theorem. We change the approximation problem on $[0,\infty)$ into a polynomial approximation problem on $[0,1]$. We use the substitution $u = e^{-t}$. Then the exponential terms e^{-jt} become the monomials u^j. The factor $1/u$ below comes from the change of variables in the L^1 integral.

Set

$$u = e^{-t}, \qquad t = -\log u,$$

and define $R\colon [0,1] \to \mathbb{R}$ by

$$R(u) \coloneqq \begin{cases} \dfrac{\rho(-\log u)}{u}, & u \in (0,1], \\ 0, & u = 0. \end{cases}$$

We justify this substitution for the Lebesgue integral. We consider the map

$$\phi\colon (0,\infty) \to (0,1), \qquad \phi(t) = e^{-t}.$$

This map is a C^1-diffeomorphism. It is bijective. Both ϕ and its inverse

$$\phi^{-1}(u) = -\log u$$

are continuously differentiable. Also,

$$|\phi'(t)| = e^{-t}.$$

The change-of-variables formula for Lebesgue integrals gives [Els18, Thm. V.4.2, pp. 221–222, Cor. V.4.4, p. 225]

$$\begin{aligned}
\|R\|_{L^1([0,1])} &= \int_0^1 \frac{|\rho(-\log u)|}{u}\,\mathrm{d}u \\
&= \int_0^\infty \frac{|\rho(t)|}{e^{-t}}\,|\phi'(t)|\,\mathrm{d}t \\
&= \int_0^\infty |\rho(t)|\,\mathrm{d}t \\
&= \|\rho\|_{L^1([0,\infty))}.
\end{aligned}$$

We get

$$R \in L^1([0,1])$$

[LHEL22, p. 17, Eqs. (52)–(53)].

Since R does not have to be continuous, we first approximate it by a continuous function $\widetilde{R}$. The density of continuous functions in $L^1([0,1])$ gives

$$\widetilde{R} \in C([0,1])$$

such that

$$\left\|R - \widetilde{R}\right\|_{L^1([0,1])} \leq \frac{\varepsilon}{2}$$

[Els18, p. 373, Thm. VIII.2.27].

We next approximate $\widetilde{R}$ by a polynomial. Here, the Weierstrass approximation theorem is enough. The definition

$$p(u) = uq(u)$$

below ensures $p(0) = 0$.

For a bounded function $g\colon [0,1] \to \mathbb{R}$, define

$$\|g\|_{\infty,[0,1]} \coloneqq \sup_{u\in[0,1]} |g(u)|.$$

By the Weierstrass approximation theorem [FL23, p. 410, Thm. 22.9], there exists a polynomial q such that

$$\left\|\widetilde{R} - q\right\|_{\infty,[0,1]} \leq \frac{\varepsilon}{2}.$$

Since $[0,1]$ has Lebesgue measure one,

$$\begin{aligned}\left\|\widetilde{R}-q\right\|_{L^1([0,1])} &= \int_0^1 |\widetilde{R}(u)-q(u)|\,\mathrm{d}u \\ &\leq \left\|\widetilde{R}-q\right\|_{\infty,[0,1]} \\ &\leq \frac{\varepsilon}{2}.\end{aligned}$$

Define

$$p(u) \coloneqq uq(u).$$

Then p is a polynomial with $p(0)=0$, and for $u>0$,

$$q(u) = \frac{p(u)}{u}.$$

The value at $u=0$ does not affect the Lebesgue integral because the set $\{0\}$ has Lebesgue measure zero.

We now transform the approximation error back to the original variable. Applying the same change-of-variables formula to $\phi(t)=e^{-t}$ gives

$$\begin{aligned}\left\|\rho - p(e^{-\cdot})\right\|_{L^1([0,\infty))} &= \int_0^\infty \left|\rho(t)-p(e^{-t})\right|\,\mathrm{d}t \\ &= \int_0^1 \left|R(u)-\frac{p(u)}{u}\right|\,\mathrm{d}u \\ &= \int_0^1 |R(u)-q(u)|\,\mathrm{d}u \\ &\leq \left\|R-\widetilde{R}\right\|_{L^1([0,1])} + \left\|\widetilde{R}-q\right\|_{L^1([0,1])} \\ &\leq \varepsilon.\end{aligned}$$

This proves the claim. □

With this density result, we can now state the universal approximation theorem for stable linear continuous time RNNs.

Theorem 5.5 (Universal Approximation for Linear RNNs [LHEL22, p. 10, Thm. 7]). *Let*

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}.$$

Then, for every $\varepsilon>0$, there exist $m\in\mathbb{N}_+$ and a family

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m$$

such that

$$\begin{aligned}\sup_{t\in\mathbb{R}}\|H_t-\widehat{H}_t\| &= \sup_{t\in\mathbb{R}}\sup_{\|x\|_{\mathcal{X}}\leq 1}\left|H_t(x)-\widehat{H}_t(x)\right| \\ &\leq \varepsilon.\end{aligned}$$

Here, $\|\cdot\|$ is the operator norm from Definition 3.1.

Proof. The proof follows Li et al. [LHEL22, pp. 17–18]. By Lemma 4.8, there exists a common memory kernel

$$\rho = (\rho_1, \dots, \rho_d)^\top \in L^1([0,\infty), \mathbb{R}^d)$$

such that

$$H_t(x) = \int_0^\infty x_{t-s}^\top \rho(s) \, \mathrm{d}s.$$

A stable linear continuous time RNN generates a functional of the form

$$\widehat{H}_t(x) = \int_0^\infty x_{t-s}^\top U^\top e^{W^\top s} c \, \mathrm{d}s$$

[LHEL22, p. 17, Eqs. (57)–(58)]. We therefore need to construct W, U, and c such that the RNN kernel approximates ρ.

Since

$$\rho = (\rho_1, \dots, \rho_d)^\top \in L^1([0,\infty), \mathbb{R}^d),$$

each component satisfies

$$\rho_i \in L^1([0,\infty)), \qquad i = 1, \dots, d.$$

We can therefore apply Lemma 5.4 separately to each component ρ_i, with accuracy ε/d. For every $i = 1, \dots, d$, there exists a polynomial p_i with $p_i(0) = 0$ such that

$$\left\| \rho_i - p_i(e^{-\cdot}) \right\|_{L^1([0,\infty))} \leq \frac{\varepsilon}{d}.$$

Summing these inequalities over $i = 1, \dots, d$ gives

$$\begin{aligned} \sum_{i=1}^d \left\| \rho_i - p_i(e^{-\cdot}) \right\|_{L^1([0,\infty))} &\leq \sum_{i=1}^d \frac{\varepsilon}{d} \\ &= \varepsilon. \end{aligned}$$

Choose $m \in \mathbb{N}_+$ such that

$$m \geq \deg p_i$$

for every nonzero polynomial p_i. By adding zero coefficients where necessary, we can write

$$p_i(u) = \sum_{j=1}^m \alpha_{ij} u^j, \qquad i = 1, \dots, d.$$

Here, $\alpha_{ij} \in \mathbb{R}$ for $i = 1, \dots, d$ and $j = 1, \dots, m$. We then have

$$p_i(e^{-s}) = \sum_{j=1}^m \alpha_{ij} e^{-js}.$$

We now choose an RNN with hidden dimension m and set

$$W \coloneqq \operatorname{diag}(-1, -2, \dots, -m) \in \mathbb{R}^{m \times m}, \qquad c \coloneqq (1, \dots, 1)^\top \in \mathbb{R}^m,$$

and define $U \in \mathbb{R}^{m \times d}$ by

$$U_{ji} \coloneqq \alpha_{ij}.$$

Since all eigenvalues of W are negative, we have $W \in \mathcal{W}_m$. The constructed RNN defines a family

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m,$$

and its kernel $\widehat{\rho}$ belongs to $L^1([0,\infty),\mathbb{R}^d)$ by Section 4.4. And we get

$$e^{W^\top s} = \operatorname{diag}\left(e^{-s}, e^{-2s}, \dots, e^{-ms}\right).$$

For each $i = 1, \dots, d$, we get

$$\begin{aligned}\left(U^\top e^{W^\top s} c\right)_i &= \sum_{j=1}^{m} U_{ji} e^{-js} \\ &= \sum_{j=1}^{m} \alpha_{ij} e^{-js} \\ &= p_i(e^{-s}).\end{aligned}$$

The RNN kernel is therefore exactly the exponential sum constructed above [LHEL22, p. 18].

By construction, the RNN kernel is

$$\widehat{\rho}(s) = U^\top e^{W^\top s} c = \left(p_1(e^{-s}), \dots, p_d(e^{-s})\right)^\top.$$

By Proposition 4.10 and the definition of the vector valued L^1 norm in Definition 4.7, we get

$$\begin{aligned}\sup_{t\in\mathbb{R}} \|H_t - \widehat{H}_t\| &= \sup_{t\in\mathbb{R}} \sup_{\|x\|_{\mathcal{X}}\le 1} \left|H_t(x) - \widehat{H}_t(x)\right| \\ &\le \|\rho - \widehat{\rho}\|_{L^1([0,\infty))} \\ &= \int_0^\infty \|\rho(s) - \widehat{\rho}(s)\|_1 \,\mathrm{d}s \\ &= \sum_{i=1}^{d} \int_0^\infty \left|\rho_i(s) - p_i(e^{-s})\right| \,\mathrm{d}s \\ &= \sum_{i=1}^{d} \left\|\rho_i - p_i(e^{-\cdot})\right\|_{L^1([0,\infty))} \\ &\le \varepsilon.\end{aligned}$$

The third equality uses

$$\widehat{\rho}_i(s) = p_i(e^{-s}), \qquad i = 1, \dots, d.$$

□

The theorem shows that, in this linear functional setting, long memory does not prevent approximation by stable linear continuous time RNNs if we allow the hidden dimension to grow. But the theorem does not say how large the hidden dimension needs to be.

5.3 Approximation Rates

The goal of this subsection is to get a bound for the approximation error that depends on the hidden dimension m. Li et al. derive such a bound under additional smoothness and decay assumptions on the target [LHEL22, pp. 11–12].

5.3.1 Jackson Approximation

Their proof uses a classical result from polynomial approximation. Jackson's work provides the historical background for this estimate [Jac30]. Here, we use Timan's formulation on a general interval $[a, b]$ [Tim63, p. 261, Sec. 5.1.5, Eq. (22)]. This formulation can be applied directly to $[0, 1]$, avoiding an additional affine change of variables. We first introduce the approximation error and the modulus of continuity used in the Jackson estimate.

Definition 5.6 (Best Uniform Polynomial Approximation [Tim63, p. 26, Sec. 2.1])**.** Let $a, b \in \mathbb{R}$ with $a < b$, and let $\mathcal{P}_n$ denote the space of real polynomials of degree at most n. For

$$f \in C([a, b]),$$

the error of the best uniform polynomial approximation of degree at most n is defined by

$$E_n(f; a, b) \coloneqq \inf_{p \in \mathcal{P}_n} \|f - p\|_{\infty,[a,b]},$$

where

$$\|f\|_{\infty,[a,b]} \coloneqq \sup_{u \in [a,b]} |f(u)|.$$

Definition 5.7 (Modulus of Continuity [Tim63, p. 93, Sec. 3.1.1])**.** For a continuous function

$$g \colon [a, b] \to \mathbb{R},$$

the modulus of continuity is defined by

$$\omega(g; \delta) \coloneqq \sup_{\substack{u,v \in [a,b] \\ |u-v| \le \delta}} |g(u) - g(v)|, \qquad \delta \ge 0.$$

The value $\omega(g; \delta)$ measures the largest possible change in the function value between two points whose distance is at most δ.

For any $u, v \in [a, b]$, the triangle inequality gives

$$\begin{aligned} |g(u) - g(v)| &\le |g(u)| + |g(v)| \\ &\le 2\|g\|_{\infty,[a,b]}. \end{aligned}$$

Taking the supremum over all $u, v \in [a, b]$ satisfying $|u - v| \le \delta$ gives

$$\omega(g; \delta) \le 2\|g\|_{\infty,[a,b]}.$$

Theorem 5.8 (Jackson Approximation Theorem [Tim63, p. 261, Sec. 5.1.5, Eq. (22)])**.** *Let $r \in \mathbb{N}_+$ and $f \in C^r([a, b])$. Then, for $n > r$,*

$$E_n(f; a, b) \le M_r \left(\frac{b-a}{n}\right)^r \omega\left(f^{(r)}; \frac{b-a}{n}\right),$$

where $M_r > 0$ depends only on r.

For the interval $[0,1]$, Theorem 5.8 gives

$$E_n(f;0,1) \leq \frac{M_r}{n^r}\omega\left(f^{(r)};\frac{1}{n}\right).$$

Using

$$\omega\left(f^{(r)};\frac{1}{n}\right) \leq 2\|f^{(r)}\|_{\infty,[0,1]},$$

we get

$$\inf_{p\in\mathcal{P}_n}\|f-p\|_{\infty,[0,1]} \leq \frac{2M_r}{n^r}\|f^{(r)}\|_{\infty,[0,1]} = \frac{C_J(r)}{n^r}\|f^{(r)}\|_{\infty,[0,1]}, \tag{5.1}$$

where $C_J(r) > 0$ depends only on r. The smoothness of f therefore gives a rate for the polynomial approximation. Li et al. apply this estimate to a transformed version of the memory kernel [LHEL22, p. 19, Eq. (68)].

5.3.2 Approximation Rate for Linear Continuous Time RNNs

The approximation rate result is formulated in terms of the step responses introduced in Proposition 4.12. For $i = 1,\dots,d$, let

$$y_i\colon \mathbb{R}\to\mathbb{R}, \qquad y_i(t) \coloneqq \widetilde{H}_t(\mathbf{e}_i),$$

where $\widetilde{H}_t$ denotes the kernel induced extension from Proposition 4.12. By Proposition 4.12,

$$y_i(t) = \begin{cases} 0, & t<0,\\ \displaystyle\int_0^t \rho_i(s)\,\mathrm{d}s, & t\geq 0. \end{cases}$$

The derivatives of y_i are directly related to the smoothness and decay of the memory kernel.

Theorem 5.9 (Approximation Rates of Linear RNNs [cf. LHEL22, p. 12, Thm. 10, Eqs. (24)–(27)])**.** *Let*

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}.$$

For $i = 1,\dots,d$, let

$$y_i(t) \coloneqq \widetilde{H}_t(\mathbf{e}_i)$$

be the step responses introduced in Proposition 4.12.

Suppose there exist constants

$$\alpha\in\mathbb{N}_+, \qquad \beta,\gamma>0,$$

such that, for $i = 1,\dots,d$,

$$y_i \in C^{\alpha+1}(\mathbb{R})$$

and, for $k = 1,\dots,\alpha+1$,

$$e^{\beta t}y_i^{(k)}(t) \longrightarrow 0 \qquad \text{as } t\to+\infty,$$

and

$$\sup_{t\geq 0}\frac{\left|e^{\beta t}y_i^{(k)}(t)\right|}{\beta^{k-1}} \leq \gamma.$$

Then there exists a constant $C(\alpha) > 0$, depending only on α, such that, for every $m \in \mathbb{N}_+$, there exists

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m$$

satisfying

$$\sup_{t\in\mathbb{R}} \sup_{\|x\|_{\mathcal{X}}\le 1} \left| H_t(x) - \widehat{H}_t(x) \right| \le \frac{C(\alpha)\gamma d}{\beta m^{\alpha}}.$$

The first condition describes how the derivatives behave as $t \to \infty$, while the second condition gives a uniform bound for all $t \ge 0$. Here, β describes the exponential decay rate and γ controls the size of the derivatives.

Li et al. use β^k in the denominator of the second condition [LHEL22, p. 12, Eq. (26)]. We use β^{k-1} instead. This change is our own. For the exponential function $e^{-\beta t}$, each derivative introduces another factor of β. Because $y_i^{(1)} = \rho_i$, we have $y_i^{(k)} = \rho_i^{(k-1)}$. Setting $j = k - 1$, the second condition gives

$$\left| \rho_i^{(j)}(t) \right| \le \gamma \beta^j e^{-\beta t}, \qquad t \ge 0, \quad j = 0, \dots, \alpha.$$

This is the bound used in our proof and also in the proof of Li et al. [LHEL22, p. 18, Eq. (66)].

Proof. We follow the proof of Li et al. [LHEL22, pp. 18–19] and give several details that are not shown there. In step (i), we choose a continuous representative of ρ_i. In step (iii), we prove the derivative formula by induction. In step (iv), we show that the derivatives extend continuously to $s = 0$. In step (v), we also consider small widths separately. These detailed arguments are our own.

We prove the result in six steps.

(i) We choose a continuous representative of ρ_i and translate the assumptions on y_i into smoothness and decay properties of ρ_i.

(ii) We transform $[0, \infty)$ to $[0, 1]$ and define the transformed kernel q_i.

(iii) We derive a formula for the derivatives of q_i on $(0, 1]$.

(iv) We bound these derivatives and extend them continuously to $s = 0$, which gives $q_i \in C^{\alpha}([0, 1])$.

(v) We approximate q_i by a polynomial using the Jackson estimate.

(vi) We transform the polynomial approximation back to the memory kernel, realize the resulting exponential sums by a stable RNN, and estimate the functional approximation error.

(i) *From the step response to the memory kernel.*

The assumptions of the theorem are stated for the step responses y_i, while the approximation argument below is formulated in terms of the memory kernel. We first transfer the smoothness and decay assumptions from y_i to ρ_i.

Fix $i \in \{1, \dots, d\}$. For $t \ge 0$,

$$y_i(t) = \int_0^t \rho_i(s)\, \mathrm{d}s.$$

The assumption

$$y_i \in C^{\alpha+1}(\mathbb{R})$$

implies that $y_i^{(1)}$ exists and is continuous.

The version of the fundamental theorem of calculus used here is stated on compact intervals. Since

$$\rho_i \in L^1([0,\infty)),$$

we have

$$\rho_i \in L^1([0,n])$$

for every $n \in \mathbb{N}_+$. The fundamental theorem of calculus for the Lebesgue integral gives

$$y_i^{(1)}(t) = \rho_i(t)$$

for Lebesgue almost every $t \in [0,n]$ [Els18, p. 326, Thm. VII.4.14(a)]. Since

$$[0,\infty) = \bigcup_{n=1}^{\infty} [0,n]$$

and a countable union of Lebesgue null sets is again a Lebesgue null set,

$$y_i^{(1)}(t) = \rho_i(t)$$

for Lebesgue almost every $t \geq 0$.

We choose $y_i^{(1)}$ as the continuous representative of ρ_i on $[0,\infty)$. This changes ρ_i only on a Lebesgue null set, so the kernel representation remains unchanged. With this representative,

$$\rho_i \in C^{\alpha}([0,\infty))$$

and, for $j = 0, \ldots, \alpha$,

$$\rho_i^{(j)} = y_i^{(j+1)}.$$

The smoothness and decay relation agrees with the corresponding step in Li et al. [LHEL22, p. 18, Eqs. (62)–(63)].

The first assumption of the theorem gives

$$e^{\beta t}\rho_i^{(j)}(t) \longrightarrow 0 \qquad \text{as } t \to +\infty.$$

The second assumption of the theorem gives

$$|\rho_i^{(j)}(t)| \leq \gamma \beta^j e^{-\beta t}, \qquad t \geq 0.$$

(ii) *Transformation to the interval* $[0,1]$.

The aim is to approximate ρ_i by a sum of m decaying exponential functions. To apply the Jackson approximation theorem, we first transform $[0,\infty)$ to $[0,1]$. We use

$$s = e^{-\frac{\beta}{\alpha+1}t}.$$

The exponent $\beta/(\alpha+1)$ is chosen so that the factors $s^{-(k+1)}$ appearing after k derivatives can be controlled by the decay $e^{-\beta t}$ for every $k \leq \alpha$. We verify this in step (iv).

The value $t = 0$ corresponds to $s = 1$, while $t \to \infty$ corresponds to $s \to 0$. Define

$$\psi : (0, 1] \to [0, \infty), \qquad \psi(s) \coloneqq -\frac{\alpha + 1}{\beta} \log s,$$

and

$$q_i : [0, 1] \to \mathbb{R}, \qquad q_i(s) \coloneqq \begin{cases} 0, & s = 0, \\ \dfrac{\rho_i(\psi(s))}{s}, & s \in (0, 1]. \end{cases}$$

This is the transformation used by Li et al. [LHEL22, p. 18, Eq. (64)].

The factor $1/s$ is chosen so that the polynomial approximation of q_i gives only decaying exponential terms after transforming back. It also cancels the factor $1/s$ in the change of variables for the L^1 error estimate in step (vi).

For

$$s = e^{-\frac{\beta}{\alpha+1}t},$$

we have

$$\psi(s) = t.$$

The definition of q_i gives

$$q_i\left(e^{-\frac{\beta}{\alpha+1}t}\right) = e^{\frac{\beta}{\alpha+1}t}\rho_i(t).$$

Equivalently,

$$\rho_i(t) = e^{-\frac{\beta}{\alpha+1}t} q_i\left(e^{-\frac{\beta}{\alpha+1}t}\right).$$

Approximating q_i by a polynomial gives an approximation of ρ_i by a finite sum of decaying exponential functions.

(iii) *Derivatives of the transformed kernel.*

Li et al. state the derivative representation

$$q_i^{(k)}(s) = (-1)^k \sum_{j=0}^{k} c(j, k) \left(\frac{\alpha + 1}{\beta}\right)^j \frac{\rho_i^{(j)}(\psi(s))}{s^{k+1}}, \qquad k = 0, \ldots, \alpha, \tag{5.2}$$

and note that it follows by induction [LHEL22, p. 18, Eq. (65)]. We carry out the induction on $(0, 1]$.

For $k = 0$, the definition of q_i gives

$$q_i(s) = \frac{\rho_i(\psi(s))}{s}.$$

The formula holds with

$$c(0, 0) = 1.$$

Assume that the formula holds for some $k < \alpha$. For $j = 0, \ldots, k$, we have $j < \alpha$, so $\rho_i^{(j)}$ is continuously differentiable because

$$\rho_i \in C^\alpha([0, \infty)).$$

The function ψ is differentiable on $(0, 1]$ with

$$\psi'(s) = -\frac{\alpha + 1}{\beta s}.$$

Each term in the induction hypothesis is therefore differentiable on $(0, 1]$. Its derivative is

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{\rho_i^{(j)}(\psi(s))}{s^{k+1}}\right) &= \frac{\rho_i^{(j+1)}(\psi(s))\psi'(s)}{s^{k+1}} - (k+1)\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}} \\ &= -\frac{\alpha+1}{\beta}\frac{\rho_i^{(j+1)}(\psi(s))}{s^{k+2}} - (k+1)\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}}.\end{aligned}$$

Differentiating the induction hypothesis gives

$$\begin{aligned}q_i^{(k+1)}(s) = (-1)^{k+1}\sum_{j=0}^{k} c(j,k)\left(\frac{\alpha+1}{\beta}\right)^{j+1}\frac{\rho_i^{(j+1)}(\psi(s))}{s^{k+2}} \\ + (-1)^{k+1}(k+1)\sum_{j=0}^{k} c(j,k)\left(\frac{\alpha+1}{\beta}\right)^{j}\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}}.\end{aligned}$$

In the first sum, we replace $j+1$ by j. This gives

$$\sum_{j=0}^{k} c(j,k)\left(\frac{\alpha+1}{\beta}\right)^{j+1}\frac{\rho_i^{(j+1)}(\psi(s))}{s^{k+2}} = \sum_{j=1}^{k+1} c(j-1,k)\left(\frac{\alpha+1}{\beta}\right)^{j}\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}}.$$

We separate the term with $j = k+1$:

$$\begin{aligned}&\sum_{j=1}^{k+1} c(j-1,k)\left(\frac{\alpha+1}{\beta}\right)^{j}\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}} \\ &\quad= \sum_{j=1}^{k} c(j-1,k)\left(\frac{\alpha+1}{\beta}\right)^{j}\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}} \\ &\qquad + c(k,k)\left(\frac{\alpha+1}{\beta}\right)^{k+1}\frac{\rho_i^{(k+1)}(\psi(s))}{s^{k+2}}.\end{aligned}$$

For the second sum, we separate the term with $j = 0$:

$$\begin{aligned}&(k+1)\sum_{j=0}^{k} c(j,k)\left(\frac{\alpha+1}{\beta}\right)^{j}\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}} \\ &\quad= (k+1)c(0,k)\frac{\rho_i(\psi(s))}{s^{k+2}} \\ &\qquad + (k+1)\sum_{j=1}^{k} c(j,k)\left(\frac{\alpha+1}{\beta}\right)^{j}\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}}.\end{aligned}$$

The terms for $j = 1, \ldots, k$ now have the same form and can be combined. We get

$$\begin{aligned}q_i^{(k+1)}(s) = (-1)^{k+1}\Bigg[&(k+1)c(0,k)\frac{\rho_i(\psi(s))}{s^{k+2}} \\ &+ \sum_{j=1}^{k}\left(c(j-1,k) + (k+1)c(j,k)\right)\left(\frac{\alpha+1}{\beta}\right)^{j}\frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}} \\ &+ c(k,k)\left(\frac{\alpha+1}{\beta}\right)^{k+1}\frac{\rho_i^{(k+1)}(\psi(s))}{s^{k+2}}\Bigg].\end{aligned}$$

We define

$$\begin{aligned} c(0,k+1) &:= (k+1)c(0,k), \\ c(j,k+1) &:= c(j-1,k) + (k+1)c(j,k), \qquad j = 1,\dots,k, \\ c(k+1,k+1) &:= c(k,k). \end{aligned}$$

Since $c(0,0) = 1$, these relations show that all coefficients $c(j,k)$ are positive integers depending only on j and k.

With these coefficients,

$$q_i^{(k+1)}(s) = (-1)^{k+1} \sum_{j=0}^{k+1} c(j,k+1) \left(\frac{\alpha+1}{\beta}\right)^j \frac{\rho_i^{(j)}(\psi(s))}{s^{k+2}}.$$

The representation also holds for $k+1$, which completes the induction.

(iv) *Bounds and smoothness at $s = 0$.*

We next bound these derivatives on $(0,1]$. Fix $k \in \{0,\dots,\alpha\}$. Set

$$s = e^{-\frac{\beta}{\alpha+1}t}, \qquad t \geq 0.$$

The map

$$t \longmapsto e^{-\frac{\beta}{\alpha+1}t}$$

is surjective from $[0,\infty)$ onto $(0,1]$. We have

$$\psi(s) = t, \qquad s^{-(k+1)} = e^{\frac{(k+1)\beta}{\alpha+1}t}.$$

Using Equation (5.2) and the triangle inequality gives

$$\left|q_i^{(k)}\left(e^{-\frac{\beta}{\alpha+1}t}\right)\right| \leq \sum_{j=0}^{k} |c(j,k)| \left(\frac{\alpha+1}{\beta}\right)^j |\rho_i^{(j)}(t)| e^{\frac{(k+1)\beta}{\alpha+1}t}.$$

The quantitative decay assumption gives

$$|\rho_i^{(j)}(t)| \leq \gamma \beta^j e^{-\beta t}.$$

Substituting this bound gives

$$\begin{aligned} \left|q_i^{(k)}\left(e^{-\frac{\beta}{\alpha+1}t}\right)\right| &\leq \gamma \sum_{j=0}^{k} |c(j,k)| (\alpha+1)^j e^{-\beta t} e^{\frac{(k+1)\beta}{\alpha+1}t} \\ &= \gamma \sum_{j=0}^{k} |c(j,k)| (\alpha+1)^j e^{-\frac{(\alpha-k)\beta}{\alpha+1}t} \\ &\leq \gamma \sum_{j=0}^{k} |c(j,k)| (\alpha+1)^j. \end{aligned}$$

Because $k \leq \alpha$,

$$e^{-\frac{(\alpha-k)\beta}{\alpha+1}t} \leq 1.$$

This calculation explains the choice of the exponent $\beta/(\alpha+1)$ in step (ii): it keeps all derivatives up to order α bounded.

Define

$$C_1(\alpha) \coloneqq \max_{0\le k\le\alpha} \sum_{j=0}^{k} |c(j,k)|(\alpha+1)^j.$$

We then have

$$\sup_{s\in(0,1]} \left|q_i^{(k)}(s)\right| \le C_1(\alpha)\gamma, \qquad k=0,\dots,\alpha. \tag{5.3}$$

This is the uniform estimate used by Li et al. [LHEL22, p. 18, Eq. (66)].

We now study the behavior at $s=0$. In Equation (5.2), each summand contains the factor

$$\frac{\rho_i^{(j)}(\psi(s))}{s^{k+1}}.$$

Recall that

$$\psi(s) = -\frac{\alpha+1}{\beta}\log s, \qquad s\in(0,1].$$

For $j=0,\dots,k$ and $k\le\alpha$, using $t=\psi(s)$ gives

$$\begin{aligned}\lim_{s\to 0+} \frac{\rho_i^{(j)}(\psi(s))}{s^{k+1}} &= \lim_{t\to\infty} \frac{\rho_i^{(j)}(t)}{e^{-\frac{(k+1)\beta}{\alpha+1}t}} \\ &= \lim_{t\to\infty} \left(e^{\beta t}\rho_i^{(j)}(t)\right) e^{-\frac{(\alpha-k)\beta}{\alpha+1}t} \qquad (1) \\ &= 0.\end{aligned}$$

To show that (1) converges to zero, we distinguish the cases $k<\alpha$ and $k=\alpha$.

For $k<\alpha$, the quantitative decay bound from step (i) shows that

$$\left|e^{\beta t}\rho_i^{(j)}(t)\right| \le \gamma\beta^j,$$

while

$$e^{-\frac{(\alpha-k)\beta}{\alpha+1}t} \xrightarrow[t\to\infty]{} 0.$$

For $k=\alpha$, the second factor is equal to one, and the qualitative decay assumption gives

$$e^{\beta t}\rho_i^{(j)}(t) \xrightarrow[t\to\infty]{} 0.$$

The two assumptions of the theorem therefore give the limit for every $k=0,\dots,\alpha$. In both cases (1) converges to zero as $t\to\infty$.

Using Equation (5.2), the derivative representation is a finite sum, so we can take the limit term by term. We get

$$\lim_{s\to 0+} q_i^{(k)}(s) = 0, \qquad k=0,\dots,\alpha.$$

The limit above allows us to extend each $q_i^{(k)}$, $k=0,\dots,\alpha$, continuously to $[0,1]$ by setting

$$q_i^{(k)}(0) \coloneqq 0.$$

For $k=0$, this agrees with the definition $q_i(0)=0$. For $k=1,\dots,\alpha$, we use induction to show that the continuous extensions are the derivatives of q_i at $s=0$.

Induction start: For $k = 0$, we have already shown that

$$\lim_{s \to 0+} q_i(s) = 0.$$

Together with the definition $q_i(0) = 0$, this shows that q_i is continuous at 0.

Induction hypothesis: Let $k \in \{1, \dots, \alpha\}$. Assume that $q_i^{(k-1)}$ is continuous on $[0, 1]$ and that

$$q_i^{(k-1)}(0) = 0.$$

Induction step: For $s > 0$, the function $q_i^{(k-1)}$ is continuous on $[0, s]$ by the induction hypothesis and differentiable on $(0, s)$ by step (iii). The mean value theorem gives a point $\xi_s \in (0, s)$ such that

$$\frac{q_i^{(k-1)}(s) - q_i^{(k-1)}(0)}{s} = q_i^{(k)}(\xi_s).$$

Because $0 < \xi_s < s$, we have

$$\xi_s \xrightarrow[s \to 0+]{} 0.$$

The limit derived above gives

$$q_i^{(k)}(\xi_s) \xrightarrow[s \to 0+]{} 0.$$

It follows that

$$\lim_{s \to 0+} \frac{q_i^{(k-1)}(s) - q_i^{(k-1)}(0)}{s} = 0.$$

This is the derivative of $q_i^{(k-1)}$ at 0, so

$$q_i^{(k)}(0) = 0.$$

Together with

$$\lim_{s \to 0+} q_i^{(k)}(s) = 0,$$

this gives

$$\lim_{s \to 0+} q_i^{(k)}(s) = q_i^{(k)}(0),$$

so $q_i^{(k)}$ is continuous at 0.

Repeating the induction step for $k = 1, \dots, \alpha$ gives

$$q_i \in C^\alpha([0, 1])$$

and

$$q_i(0) = q_i^{(1)}(0) = \cdots = q_i^{(\alpha)}(0) = 0$$

[LHEL22, p. 18, Eq. (67)].

Using Equation (5.3) and $q_i^{(k)}(0) = 0$ for $k = 0, \dots, \alpha$, we get

$$\max_{0 \le k \le \alpha} \|q_i^{(k)}\|_{\infty,[0,1]} \le C_1(\alpha)\gamma.$$

(v) *Polynomial approximation.*

We now approximate q_i uniformly on $[0,1]$ by a polynomial of degree at most $m-1$. The choice of degree $m-1$ gives m polynomial terms. In step (vi), these terms will become m exponential terms in the memory kernel.

Let

$$\mathcal{P}_{m-1} = \{p : [0,1] \to \mathbb{R} \ : \ p \text{ is a polynomial with } \deg(p) \leq m-1\}.$$

The quantity

$$\inf_{p \in \mathcal{P}_{m-1}} \|q_i - p\|_{\infty,[0,1]}$$

is the smallest possible uniform approximation error among all polynomials of degree at most $m-1$.

By step (iv),

$$q_i \in C^\alpha([0,1])$$

and

$$\|q_i^{(\alpha)}\|_{\infty,[0,1]} \leq C_1(\alpha)\gamma.$$

We apply Equation (5.1) with

$$r = \alpha, \qquad n = m-1.$$

The estimate requires $n > r$, so we first consider

$$m > \alpha + 1.$$

For these values of m, Equation (5.1) gives

$$\inf_{p \in \mathcal{P}_{m-1}} \|q_i - p\|_{\infty,[0,1]} \leq \frac{C_J(\alpha)}{(m-1)^\alpha} \|q_i^{(\alpha)}\|_{\infty,[0,1]}.$$

We now choose a polynomial that attains this infimum. Let $(p_\ell)_{\ell \in \mathbb{N}} \subset \mathcal{P}_{m-1}$ be a sequence such that

$$\|q_i - p_\ell\|_{\infty,[0,1]} \xrightarrow[\ell \to \infty]{} \inf_{p \in \mathcal{P}_{m-1}} \|q_i - p\|_{\infty,[0,1]}.$$

The approximation errors are bounded, and the triangle inequality gives

$$\|p_\ell\|_{\infty,[0,1]} \leq \|q_i - p_\ell\|_{\infty,[0,1]} + \|q_i\|_{\infty,[0,1]}.$$

The sequence $(p_\ell)_{\ell \in \mathbb{N}}$ is bounded in $\mathcal{P}_{m-1}$. Since $\mathcal{P}_{m-1}$ is finite dimensional, it has a convergent subsequence with a limit $Q_{i,m} \in \mathcal{P}_{m-1}$. By continuity of the norm,

$$\|q_i - Q_{i,m}\|_{\infty,[0,1]} = \inf_{p \in \mathcal{P}_{m-1}} \|q_i - p\|_{\infty,[0,1]}.$$

Using Equation (5.1) and the bound from step (iv), we get

$$\begin{aligned} \|q_i - Q_{i,m}\|_{\infty,[0,1]} &\leq \frac{C_J(\alpha)}{(m-1)^\alpha} \|q_i^{(\alpha)}\|_{\infty,[0,1]} \\ &\leq \frac{C_J(\alpha) C_1(\alpha) \gamma}{(m-1)^\alpha}. \end{aligned}$$

Because $\alpha \geq 1$ and $m > \alpha + 1$, we have $m \geq 3$. In particular, $m \geq 2$, so

$$m - 1 \geq \frac{m}{2},$$

which gives

$$\frac{1}{(m-1)^\alpha} \leq \frac{2^\alpha}{m^\alpha}.$$

Define

$$C_2(\alpha) \coloneqq 2^\alpha C_J(\alpha) C_1(\alpha).$$

We get

$$\|q_i - Q_{i,m}\|_{\infty,[0,1]} \leq \frac{C_2(\alpha)\gamma}{m^\alpha}.$$

It remains to consider

$$1 \leq m \leq \alpha + 1.$$

For these values, choose

$$Q_{i,m} = 0.$$

This polynomial belongs to $\mathcal{P}_{m-1}$, and step (iv) gives

$$\|q_i - Q_{i,m}\|_{\infty,[0,1]} = \|q_i\|_{\infty,[0,1]} \leq C_1(\alpha)\gamma.$$

Because $m \leq \alpha + 1$,

$$1 \leq \frac{(\alpha+1)^\alpha}{m^\alpha}.$$

We get

$$\|q_i - Q_{i,m}\|_{\infty,[0,1]} \leq \frac{C_1(\alpha)(\alpha+1)^\alpha \gamma}{m^\alpha}.$$

Define

$$C_3(\alpha) \coloneqq C_1(\alpha)(\alpha+1)^\alpha.$$

Then

$$\|q_i - Q_{i,m}\|_{\infty,[0,1]} \leq \frac{C_3(\alpha)\gamma}{m^\alpha}.$$

Set

$$C_4(\alpha) \coloneqq \max\left\{C_2(\alpha), C_3(\alpha)\right\}.$$

For every $m \in \mathbb{N}_+$, there exists a polynomial $Q_{i,m} \in \mathcal{P}_{m-1}$ satisfying

$$\|q_i - Q_{i,m}\|_{\infty,[0,1]} \leq \frac{C_4(\alpha)\gamma}{m^\alpha}. \tag{5.4}$$

We write

$$Q_{i,m}(s) = \sum_{j=0}^{m-1} a_{i,j} s^j.$$

Here, $a_{i,j} \in \mathbb{R}$ for $i = 1, \ldots, d$ and $j = 0, \ldots, m-1$.

This is the polynomial approximation used by Li et al. [LHEL22, p. 19, Eqs. (68)–(69)].

(vi) *Exponential sums and RNN approximation.*

We now transform the polynomial approximation back to the original time variable. We first construct an exponential approximation of each component ρ_i, then realize all components with one RNN, estimate the L^1 kernel error, and use Proposition 4.10 to bound the functional error.

For $i = 1, \dots, d$, define $\phi_{i,m} \colon [0, \infty) \to \mathbb{R}$ by

$$\phi_{i,m}(t) \coloneqq e^{-\frac{\beta}{\alpha+1}t} Q_{i,m}\left(e^{-\frac{\beta}{\alpha+1}t}\right), \qquad t \geq 0. \tag{5.5}$$

Using

$$Q_{i,m}(s) = \sum_{j=0}^{m-1} a_{i,j} s^j,$$

we get

$$\begin{aligned}\phi_{i,m}(t) &= e^{-\frac{\beta}{\alpha+1}t} \sum_{j=0}^{m-1} a_{i,j} e^{-\frac{j\beta}{\alpha+1}t} \\ &= \sum_{j=0}^{m-1} a_{i,j} e^{-\frac{(j+1)\beta}{\alpha+1}t}.\end{aligned}$$

After transforming back to the time variable t, the m monomials of $Q_{i,m}$ become the m decaying exponential terms in $\phi_{i,m}$ [LHEL22, p. 19, Eq. (70)].

Because $\beta > 0$, every exponential term is integrable on $[0, \infty)$. The sum contains finitely many terms, so

$$\begin{aligned}\|\phi_{i,m}\|_{L^1([0,\infty))} &\leq \sum_{j=0}^{m-1} |a_{i,j}| \int_0^\infty e^{-\frac{(j+1)\beta}{\alpha+1}t} \, \mathrm{d}t \\ &= \frac{\alpha+1}{\beta} \sum_{j=0}^{m-1} \frac{|a_{i,j}|}{j+1} < \infty.\end{aligned}$$

We have

$$\phi_{i,m} \in L^1([0, \infty)).$$

Together with

$$\rho_i \in L^1([0, \infty)),$$

this gives

$$\rho_i - \phi_{i,m} \in L^1([0, \infty)).$$

We next realize these exponential sums with one linear continuous time RNN. The exponential terms

$$e^{-\frac{\beta}{\alpha+1}t}, e^{-\frac{2\beta}{\alpha+1}t}, \dots, e^{-\frac{m\beta}{\alpha+1}t}$$

are the same for every component i. Only the coefficients $a_{i,j}$ depend on i. We can use the same recurrent matrix for all components and store the component dependent coefficients in the columns of the input matrix.

Define

$$\begin{aligned}W &\coloneqq \operatorname{diag}\left(-\frac{\beta}{\alpha+1}, -\frac{2\beta}{\alpha+1}, \dots, -\frac{m\beta}{\alpha+1}\right) \in \mathbb{R}^{m \times m}, \\ c &\coloneqq (1, \dots, 1)^\top \in \mathbb{R}^m, \\ u_i &\coloneqq (a_{i,0}, \dots, a_{i,m-1})^\top \in \mathbb{R}^m.\end{aligned}$$

For every $i = 1, \dots, d$,

$$\phi_{i,m}(t) = c^\top e^{Wt} u_i.$$

All eigenvalues of W are negative because $\beta > 0$, so $W \in \mathcal{W}_m$.

Set

$$U \coloneqq [u_1, \dots, u_d] \in \mathbb{R}^{m \times d}.$$

By Definition 4.9, the memory kernel generated by this RNN is

$$\widehat{\rho}_m(t) \coloneqq U^\top e^{W^\top t} c = \big(\phi_{1,m}(t), \dots, \phi_{d,m}(t)\big)^\top$$

[LHEL22, p. 19, Eqs. (70)–(74)].

Let

$$\{\widehat{H}_t\}_{t \in \mathbb{R}} \in \widehat{\mathcal{H}}_m$$

be the functional family generated by the RNN with parameters (W, U, c).

We now estimate the kernel error. Recall from step (ii) that

$$\rho_i(t) = e^{-\frac{\beta}{\alpha+1}t} q_i\left(e^{-\frac{\beta}{\alpha+1}t}\right).$$

Set

$$s = e^{-\frac{\beta}{\alpha+1}t}.$$

Then, by step (ii) and Equation (5.5),

$$\begin{aligned} \rho_i(t) &= s q_i(s), \\ \phi_{i,m}(t) &= s Q_{i,m}(s). \end{aligned}$$

We now use the change of variables from step (ii). Define

$$\varphi : (0, \infty) \to (0, 1), \qquad \varphi(t) \coloneqq e^{-\frac{\beta}{\alpha+1}t}.$$

The map φ is a C^1 diffeomorphism with inverse

$$\psi : (0, 1) \to (0, \infty), \qquad \psi(s) = -\frac{\alpha+1}{\beta} \log s,$$

and

$$|\psi'(s)| = \frac{\alpha+1}{\beta s}.$$

The change of variables formula for Lebesgue integrals gives [Els18, Thm. V.4.2, pp. 221–222, Cor. V.4.4, p. 225]

$$\|\rho_i - \phi_{i,m}\|_{L^1([0,\infty))} = \int_0^1 |\rho_i(\psi(s)) - \phi_{i,m}(\psi(s))| \, |\psi'(s)| \, \mathrm{d}s.$$

By step (ii) and Equation (5.5),

$$\rho_i(\psi(s)) = s q_i(s)$$

and

$$\phi_{i,m}(\psi(s)) = s Q_{i,m}(s).$$

Using these identities and the formula for $|\psi'(s)|$, we get

$$\begin{aligned}\|\rho_i - \phi_{i,m}\|_{L^1([0,\infty))} &= \int_0^1 s\,|q_i(s) - Q_{i,m}(s)|\,\frac{\alpha+1}{\beta s}\,\mathrm{d}s \\ &= \frac{\alpha+1}{\beta}\int_0^1 |q_i(s) - Q_{i,m}(s)|\,\mathrm{d}s \\ &\leq \frac{\alpha+1}{\beta}\|q_i - Q_{i,m}\|_{\infty,[0,1]} \\ &\leq \frac{(\alpha+1)C_4(\alpha)\gamma}{\beta m^\alpha}.\end{aligned}$$

The last inequality follows from Equation (5.4). Define

$$C_5(\alpha) \coloneqq (\alpha+1)C_4(\alpha).$$

Then

$$\|\rho_i - \phi_{i,m}\|_{L^1([0,\infty))} \leq \frac{C_5(\alpha)\gamma}{\beta m^\alpha}$$

[LHEL22, p. 19, Eq. (75)].

Using the definition of the vector valued L^1 norm,

$$\begin{aligned}\|\rho - \widehat{\rho}_m\|_{L^1([0,\infty))} &= \int_0^\infty \sum_{i=1}^d |\rho_i(t) - \phi_{i,m}(t)|\,\mathrm{d}t \\ &= \sum_{i=1}^d \|\rho_i - \phi_{i,m}\|_{L^1([0,\infty))} \\ &\leq \frac{C_5(\alpha)\gamma d}{\beta m^\alpha}.\end{aligned}$$

Proposition 4.10 gives

$$\sup_{t\in\mathbb{R}}\sup_{\|x\|_{\mathcal{X}}\leq 1}\left|H_t(x) - \widehat{H}_t(x)\right| \leq \|\rho - \widehat{\rho}_m\|_{L^1([0,\infty))} \leq \frac{C_5(\alpha)\gamma d}{\beta m^\alpha}.$$

Because $C_5(\alpha)$ depends only on α, set

$$C(\alpha) \coloneqq C_5(\alpha).$$

We get

$$\sup_{t\in\mathbb{R}}\sup_{\|x\|_{\mathcal{X}}\leq 1}\left|H_t(x) - \widehat{H}_t(x)\right| \leq \frac{C(\alpha)\gamma d}{\beta m^\alpha}.$$

The factor $m^{-\alpha}$ comes from the Jackson approximation in step (v), the factor $1/\beta$ from the change of variables above, the factor γ from the derivative bounds, and the factor d from summing the errors of the d kernel components. This proves the result [LHEL22, p. 19, Eqs. (76)–(77)]. □

The error decreases polynomially in the hidden dimension m with rate $m^{-\alpha}$. A larger smoothness parameter α gives a faster dependence on the RNN width.

The parameter β connects the approximation rate to the memory of the target. A smaller β means slower exponential decay of the memory kernel and gives a larger upper bound. This already indicates that slowly decaying memory can make approximation less efficient. We study this effect more directly in Section 5.5.

5.3.3 Numerical Illustration

We give our own numerical illustration of the approximation rate. We choose two target kernels with different smoothness, fit exponential sums by least squares, and compare the approximation errors for different RNN widths.

We consider the scalar case $d = 1$ and choose

$$\alpha = 1, \qquad \beta = 1.$$

For these parameters, Theorem 5.9 requires the corresponding step response to belong to $C^2(\mathbb{R})$ and its first two derivatives to satisfy the weighted decay conditions with the factor e^t. The resulting approximation rate is m^{-1}.

For each target memory kernel ρ, we construct approximations of the form

$$\widehat{\rho}_m(t) = \sum_{j=1}^{m} a_{j,m} e^{-\frac{j}{2}t}.$$

The exponential functions are fixed by the construction in the proof of Theorem 5.9. For each width m, we determine only the coefficients $a_{j,m}$ numerically by least squares.

The first target kernel is analytic on $[0, \infty)$. The second target kernel is only continuously differentiable, so its step response has exactly the smoothness required by Theorem 5.9 for $\alpha = 1$. This allows us to compare targets with different smoothness.

Proposition 5.10 (Target kernels for the numerical illustration)**.** *Let*

$$\rho_{\text{analytic}}(t) = t^2 e^{-2t}, \qquad t \geq 0,$$

and let

$$\rho_{\text{limited}}(t) = e^{-3t/2} \left(1 - e^{-t/2}\right)^2 \left| e^{-t/2} - \frac{1}{2} \right|^{1+\delta}, \qquad t \geq 0, \qquad \delta = 0.1.$$

For either choice

$$\rho \in \left\{ \rho_{\text{analytic}}, \rho_{\text{limited}} \right\},$$

define

$$H_t(x) = \int_0^\infty x_{t-s} \rho(s) \, \mathrm{d}s.$$

Then $\{H_t\}_{t \in \mathbb{R}} \in \mathcal{H}$, and its step response satisfies the assumptions of Theorem 5.9 for $\alpha = \beta = 1$. Moreover,

$$\rho_{\text{limited}} \in C^1([0, \infty)), \qquad \rho_{\text{limited}} \notin C^2([0, \infty)).$$

Proof. For $t \geq 0$,

$$\rho_{\text{analytic}}(t) = t^2 e^{-2t} \leq 4e^{-2} e^{-t},$$

and

$$|\rho_{\text{limited}}(t)| \leq e^{-3t/2} \leq e^{-t}.$$

Both kernels therefore belong to $L^1([0, \infty))$. Linearity is immediate, and

$$\|H_t\| = \sup_{\|x\|_{\mathcal{X}} \leq 1} |H_t(x)| \leq \|\rho\|_{L^1([0,\infty))}$$

gives continuity. Causality and time homogeneity follow directly from the integral representation, while regularity follows by dominated convergence. Therefore, both kernels define families in $\mathcal{H}$.

For the analytic kernel,

$$\rho_{\text{analytic}}(0) = \rho'_{\text{analytic}}(0) = 0.$$

The corresponding step response is equal to zero for $t < 0$ and satisfies

$$y'_{\text{analytic}}(t) = \rho_{\text{analytic}}(t), \qquad y''_{\text{analytic}}(t) = \rho'_{\text{analytic}}(t)$$

for $t > 0$. The two boundary conditions at $t = 0$ give

$$y_{\text{analytic}} \in C^2(\mathbb{R}).$$

Furthermore,

$$e^t \rho_{\text{analytic}}(t) = t^2 e^{-t} \xrightarrow[t\to\infty]{} 0$$

and

$$e^t \rho'_{\text{analytic}}(t) = 2t(1-t)e^{-t} \xrightarrow[t\to\infty]{} 0.$$

For the second kernel, the only loss of smoothness occurs when

$$e^{-t/2} = \frac{1}{2},$$

which is equivalent to

$$t = 2\log 2.$$

At this point, the factor

$$\left| e^{-t/2} - \frac{1}{2} \right|^{1+\delta}$$

is continuously differentiable but not twice continuously differentiable. The other factors are smooth and nonzero at $t = 2\log 2$. Therefore,

$$\rho_{\text{limited}} \in C^1([0,\infty)), \qquad \rho_{\text{limited}} \notin C^2([0,\infty)).$$

At $t = 0$, the factor

$$\left(1 - e^{-t/2}\right)^2$$

gives

$$\rho_{\text{limited}}(0) = \rho'_{\text{limited}}(0) = 0.$$

Together with $\rho_{\text{limited}} \in C^1([0,\infty))$, this gives

$$y_{\text{limited}} \in C^2(\mathbb{R}).$$

For the first weighted decay condition,

$$e^t \rho_{\text{limited}}(t) = e^{-t/2} \left(1 - e^{-t/2}\right)^2 \left| e^{-t/2} - \frac{1}{2} \right|^{1+\delta}$$
$$\xrightarrow[t\to\infty]{} 0.$$

Differentiating the kernel gives

$$\begin{aligned}\rho'_{\text{limited}}(t) = &-\frac{3}{2}e^{-3t/2}\left(1-e^{-t/2}\right)^2\left|e^{-t/2}-\frac{1}{2}\right|^{1+\delta}\\ &+e^{-2t}\left(1-e^{-t/2}\right)\left|e^{-t/2}-\frac{1}{2}\right|^{1+\delta}\\ &-\frac{1+\delta}{2}e^{-2t}\left(1-e^{-t/2}\right)^2\operatorname{sgn}\left(e^{-t/2}-\frac{1}{2}\right)\left|e^{-t/2}-\frac{1}{2}\right|^{\delta}.\end{aligned}$$

Multiplication by e^t gives

$$\begin{aligned}e^t\rho'_{\text{limited}}(t) = &-\frac{3}{2}e^{-t/2}\left(1-e^{-t/2}\right)^2\left|e^{-t/2}-\frac{1}{2}\right|^{1+\delta}\\ &+e^{-t}\left(1-e^{-t/2}\right)\left|e^{-t/2}-\frac{1}{2}\right|^{1+\delta}\\ &-\frac{1+\delta}{2}e^{-t}\left(1-e^{-t/2}\right)^2\operatorname{sgn}\left(e^{-t/2}-\frac{1}{2}\right)\left|e^{-t/2}-\frac{1}{2}\right|^{\delta}\\ &\xrightarrow[t\to\infty]{} 0.\end{aligned}$$

The weighted functions are continuous and converge to zero, so they are bounded on $[0,\infty)$. Because $\beta = 1$, these bounds are exactly the quantitative decay condition of Theorem 5.9 for some finite $\gamma > 0$. □

For the analytic target, the quantitative decay constant is

$$\begin{aligned}\gamma_{\text{analytic}} &= \max\left\{\sup_{t\geq 0} e^t|\rho_{\text{analytic}}(t)|, \sup_{t\geq 0} e^t|\rho'_{\text{analytic}}(t)|\right\}\\ &= (4+2\sqrt{5})e^{-(3+\sqrt{5})/2} \approx 0.6180.\end{aligned}$$

For the limited smoothness target, we evaluate the same two suprema numerically and get

$$\gamma_{\text{limited}} \approx 0.03819.$$

We perform the numerical fit for

$$m = 2, \ldots, 16$$

using 4000 equally spaced points in $[0, 20]$.

For example, for the analytic target and $m = 4$, the fitted coefficients give

$$\widehat{\rho}_4(t) = -0.1657e^{-t/2} + 1.8046e^{-t} - 2.4036e^{-3t/2} + 0.7313e^{-2t}.$$

The corresponding RNN parameters can be chosen as

$$W = \operatorname{diag}\left(-\frac{1}{2}, -1, -\frac{3}{2}, -2\right), \qquad c = \begin{pmatrix}1\\1\\1\\1\end{pmatrix},$$

and

$$U = \begin{pmatrix} -0.1657 \\ 1.8046 \\ -2.4036 \\ 0.7313 \end{pmatrix}.$$

We measure the approximation error by

$$E(m) := \|\rho - \widehat{\rho}_m\|_{L^1([0,\infty))},$$

where the integral is evaluated numerically by adaptive quadrature on $[0, \infty)$.

We use SciPy for the numerical implementation [VGO+20]. We add some remarks on the numerical computation. The exponential basis is ill conditioned. The condition number of the least squares matrix grows from about 6 for $m = 2$ to about $3 \cdot 10^{11}$ for $m = 16$, and the fitted coefficients reach magnitudes of up to about $8 \cdot 10^6$. For $m \geq 17$, the basis is treated as numerically rank deficient by the least squares solver in double precision, and the fit is no longer reliable, so we restrict the experiment to $m \leq 16$.

The fit minimizes a discrete L^2 error on $[0, 20]$, while $E(m)$ is an L^1 error on $[0, \infty)$. The error $E(m)$ does not necessarily decrease when m increases, even though larger models can represent everything that smaller models can. The numerical quadrature is split at $t = 2 \log 2$ for the limited smoothness target and at the sign changes of $\rho - \widehat{\rho}_m$. Two different tolerance settings give relative differences below 10^{-6}.

Figure 2 shows the analytic and limited smoothness target kernels.

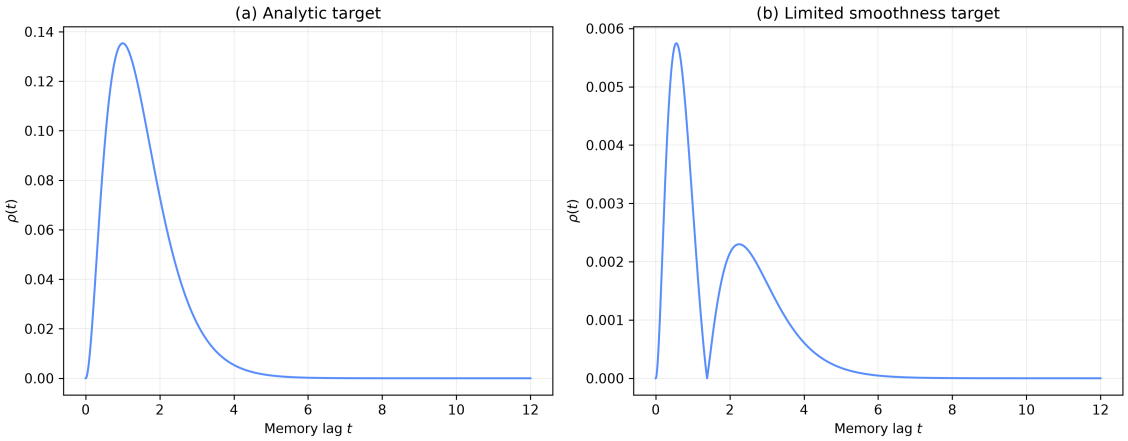


Figure 2: Target kernels used in the numerical illustration of Theorem 5.9.

Figure 3 shows the observed L^1 kernel error for increasing RNN width.

The proof of Theorem 5.9 gives, for $d = 1$, an L^1 kernel approximation rate of the form

$$\frac{C(\alpha)\gamma}{\beta m^\alpha}.$$

We compare the errors of our numerically fitted exponential sums with this scaling. The proof does not provide an explicit value for $C(\alpha)$, so we cannot plot the theoretical upper bound directly. We define

$$C_{\mathrm{req}}(m) := \frac{\beta m^\alpha E(m)}{\gamma}.$$

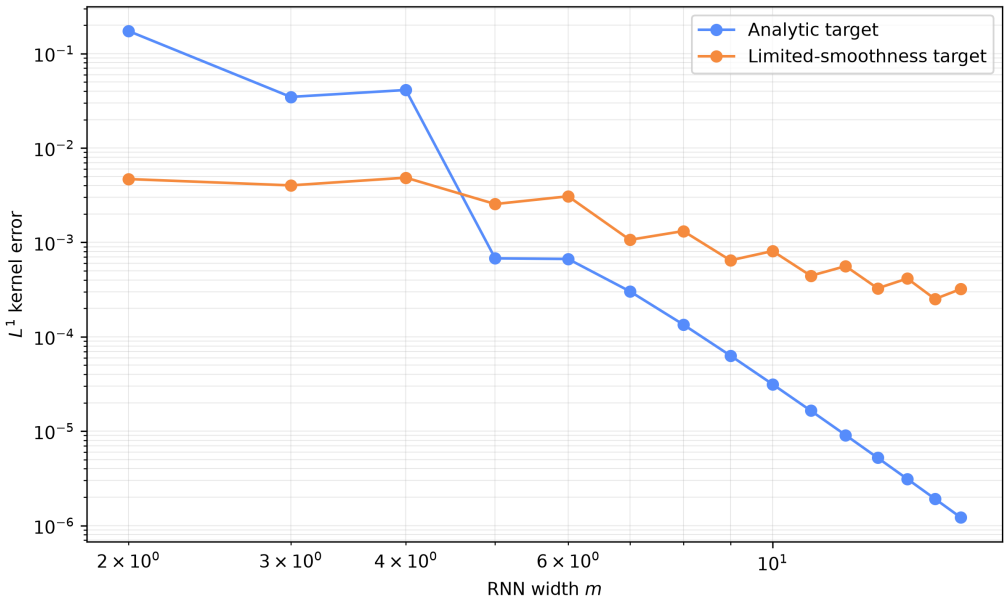


Figure 3: Numerical L^1 kernel approximation error as a function of the RNN width m for the analytic and limited smoothness targets.

This is the value of the multiplicative constant for which

$$\frac{C_{\mathrm{req}}(m)\gamma}{\beta m^{\alpha}} = E(m).$$

We use $C_{\mathrm{req}}(m)$ to compare the observed errors with the $m^{-\alpha}$ dependence in the theorem.

Figure 4 shows $C_{\mathrm{req}}(m)$ for $m = 2, \ldots, 16$. The largest observed value for the analytic target is approximately

$$\max_m C_{\mathrm{req}}(m) \approx 0.563$$

and occurs at $m = 2$. For the limited smoothness target, the largest observed value is approximately

$$\max_m C_{\mathrm{req}}(m) \approx 0.508$$

and occurs at $m = 4$. Using the slightly larger rounded values 0.563 and 0.509, respectively, gives, for all $m = 2, \ldots, 16$,

$$E_{\mathrm{analytic}}(m) \leq \frac{0.563\,\gamma_{\mathrm{analytic}}}{m} < \frac{0.348}{m},$$

and

$$E_{\mathrm{limited}}(m) \leq \frac{0.509\,\gamma_{\mathrm{limited}}}{m} < \frac{0.0195}{m}.$$

The same constants are already determined by the widths $m = 2, \ldots, 10$. Keeping them fixed, both inequalities also hold for $m = 11, \ldots, 16$.

For both examples, the observed errors are consistent with an empirical upper bound having the m^{-1} dependence predicted for $\alpha = 1$. For larger tested widths, $C_{\mathrm{req}}(m)$ is substantially smaller than its maximum value, so the observed errors decrease faster than m^{-1} over much of the tested range.

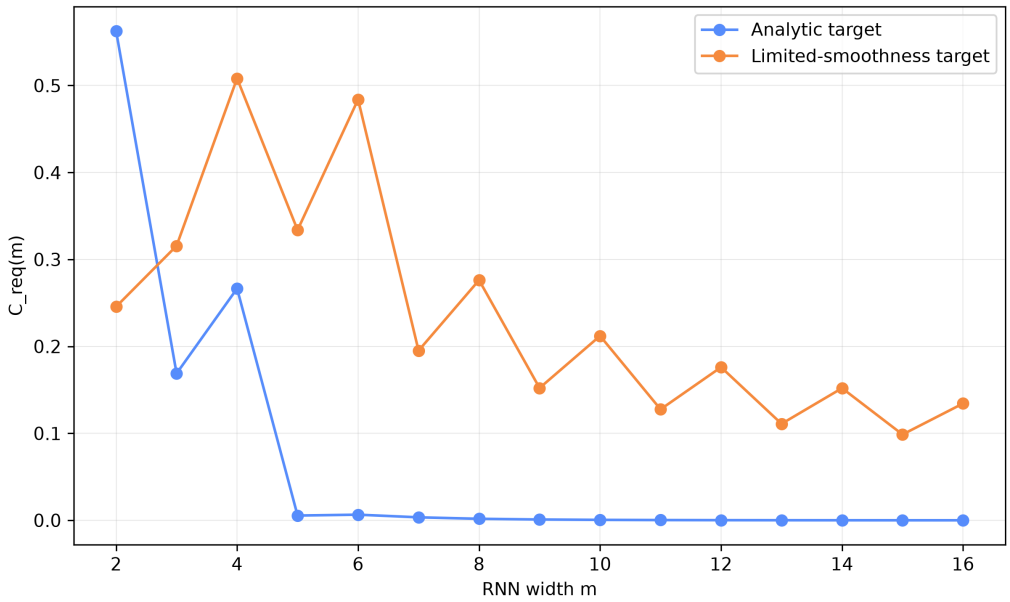


Figure 4: Normalized kernel approximation errors $C_{\mathrm{req}}(m) = \beta m^{\alpha} E(m)/\gamma$ for the analytic and limited smoothness targets.

These experiments only illustrate the theoretical estimate numerically. The coefficients are fitted using discrete least squares and not by the Jackson construction from the proof. We only consider the finite range $m = 2, \ldots, 16$. The values of $C_{\mathrm{req}}(m)$ are computed from the observed errors themselves and are not estimates of the universal constant $C(\alpha)$ in Theorem 5.9.

The code for this experiment is provided in `experiments/04_theorem10_approximation_rates.py`. The corresponding notebook is available at `notebooks/03_theorem10_approximation_rates.ipynb`.

5.4 Inverse Approximation

Theorem 5.9 showed one direction. If the derivatives of the step responses decay sufficiently fast, in particular faster than $e^{-\beta t}$, then we get a polynomial approximation rate in the hidden dimension.

We now consider the opposite direction. We ask whether such exponential decay is also necessary if the target derivatives can be approximated uniformly by linear continuous time RNNs whose recurrent matrices remain uniformly stable. Li et al. state the following inverse approximation result under additional assumptions [LHEL22, pp. 12–13, Thm. 11]. We show below that these assumptions are not sufficient for general nonnormal recurrent matrices.

5.4.1 Original Inverse Approximation Result

Theorem 5.11 (Inverse Approximation Theorem of Li et al. [LHEL22, pp. 12–13, Thm. 11, Eqs. (28)–(31)]). *Assume the conditions of Theorem 5.5. For*

$$y_i(t) := \widetilde{H}_t(\mathbf{e}_i), \qquad i = 1, \ldots, d,$$

suppose that

$$y_i \in C^{\alpha+1}(\mathbb{R}), \qquad \alpha \in \mathbb{N}_+.$$

For every $m \in \mathbb{N}_+$, let

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m$$

be generated by

$$W_m \in \mathbb{R}^{m\times m}, \qquad U_m \in \mathbb{R}^{m\times d}, \qquad c_m \in \mathbb{R}^m.$$

Let

$$\widehat{\rho}_m(s) = \big(\widehat{\rho}_{m,1}(s), \dots, \widehat{\rho}_{m,d}(s)\big)^\top \coloneqq U_m^\top e^{W_m^\top s} c_m \in \mathbb{R}^d$$

be the corresponding RNN memory kernel. For $i = 1, \dots, d$, define $\widehat{y}_{i,m} \colon \mathbb{R} \to \mathbb{R}$ by

$$\widehat{y}_{i,m}(t) \coloneqq \begin{cases} 0, & t < 0, \\ \displaystyle\int_0^t \widehat{\rho}_{m,i}(s)\,\mathrm{d}s, & t \geq 0, \end{cases} \qquad i = 1, \dots, d.$$

Suppose that, for $i = 1, \dots, d$ and $k = 1, \dots, \alpha + 1$,

$$\lim_{m\to\infty} \sup_{t\geq 0} \left| \widehat{y}_{i,m}^{(k)}(t) - y_i^{(k)}(t) \right| = 0.$$

Define

$$w_m \coloneqq \max_{\lambda\in\sigma(W_m)} \operatorname{Re}\lambda.$$

We interpret the uniform parameter bound as

$$\sup_{m\in\mathbb{N}_+} \max\{\|W_m\|_2, \|U_m\|_2, \|c_m\|_2\} < \infty,$$

where $\|\cdot\|_2$ is defined in Equation (3.2). Assume further that there exists $\beta > 0$ such that

$$\limsup_{m\to\infty} w_m < -\beta.$$

Then

$$e^{\beta t} y_i^{(k)}(t) = o(1) \qquad \text{as } t \to +\infty$$

for all $i = 1, \dots, d$ and $k = 1, \dots, \alpha + 1$.

5.4.2 The Nonnormal Case

A central step in the proof of Li et al. requires an exponentially weighted tail estimate with a constant that is independent of the hidden dimension m [LHEL22, p. 20, Eq. (83)]. For every fixed m, Lemma 3.7 applied to W_m gives constants $C_m > 0$ and $\eta_m > 0$ such that

$$\|e^{W_m t}\|_2 \leq C_m e^{-\eta_m t}, \qquad t \geq 0.$$

The lemma does not give constants that are uniform in m. For general nonnormal matrices, these assumptions do not control $\|e^{W_m t}\|_2$ uniformly in m. Classical examples based on Jordan blocks of increasing dimension show that the matrix exponential can be much larger than the eigenvalues alone suggest [TE05, Sec. 19, pp. 185–186]. We construct our own counterexample to show that the inverse approximation result does not hold for general nonnormal recurrent matrices.

Proposition 5.12 (A nonnormal counterexample)**.** *Let $d = 1$, $\alpha = 1$, and $\beta = 1$. There exist a family*

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}$$

and, for every $m \in \mathbb{N}_+$, a family

$$\{\widehat{H}_t\}_{t\in\mathbb{R}} \in \widehat{\mathcal{H}}_m$$

generated by

$$W_m \in \mathcal{W}_m, \qquad U_m \in \mathbb{R}^{m\times 1}, \qquad c_m \in \mathbb{R}^m,$$

that satisfy all assumptions of Theorem 5.11, while the step response y of $\{H_t\}_{t\in\mathbb{R}}$ satisfies

$$e^t y^{(1)}(t) \xrightarrow[t\to\infty]{} \infty.$$

Therefore, the conclusion of Theorem 5.11 does not hold in general.

Proof. The proof has six steps:

(i) construction of the target functional and its step response,
(ii) construction of the RNN sequence,
(iii) verification of the parameter bounds and the spectral condition,
(iv) uniform approximation of the first derivative,
(v) uniform approximation of the second derivative,
(vi) verification that the claimed decay of the target fails.

(i) *Target functional and step response.*
We consider the scalar case

$$d = 1, \qquad \alpha = 1, \qquad \beta = 1.$$

Consider the memory kernel

$$\rho : [0,\infty) \to \mathbb{R}, \qquad \rho(t) \coloneqq t^2 e^{-t}.$$

For each $t \in \mathbb{R}$, define

$$H_t : \mathcal{X} \to \mathbb{R}, \qquad H_t(x) \coloneqq \int_0^\infty x_{t-s}\rho(s)\,\mathrm{d}s.$$

Since

$$\int_0^\infty t^2 e^{-t}\,\mathrm{d}t = 2,$$

we have

$$\rho \in L^1([0,\infty)).$$

This gives

$$\sup_{\|x\|_{\mathcal{X}}\le 1} |H_t(x)| \le \sup_{\|x\|_{\mathcal{X}}\le 1} 2\|x\|_{\mathcal{X}} \le 2.$$

Therefore, H_t is a continuous linear functional. Causality and time homogeneity follow directly from the functional representation.

To verify regularity, let

$$(x^{(n)})_{n\in\mathbb{N}} \subset \mathcal{X}$$

satisfy

$$\sup_n \|x^{(n)}\|_{\mathcal{X}} \leq 1$$

and

$$x_t^{(n)} \xrightarrow[n\to\infty]{} 0$$

for Lebesgue almost every $t \in \mathbb{R}$. For fixed t, the map $s \mapsto t - s$ maps Lebesgue null sets to Lebesgue null sets, so

$$x_{t-s}^{(n)} \xrightarrow[n\to\infty]{} 0$$

for Lebesgue almost every $s \geq 0$. Moreover,

$$\left|x_{t-s}^{(n)}\rho(s)\right| \leq \left|x_{t-s}^{(n)}\right| |\rho(s)| \leq |\rho(s)|$$

for almost every $s \geq 0$. Since

$$\rho \in L^1([0,\infty)),$$

dominated convergence gives

$$H_t(x^{(n)}) \xrightarrow[n\to\infty]{} 0.$$

Therefore,

$$\{H_t\}_{t\in\mathbb{R}} \in \mathcal{H}.$$

By the kernel induced extension from Section 4.5, the step response $y\colon \mathbb{R} \to \mathbb{R}$ is

$$y(t) \coloneqq \widetilde{H}_t(\mathbf{e}_1) = \begin{cases} 0, & t < 0, \\ \displaystyle\int_0^t s^2 e^{-s}\,\mathrm{d}s, & t \geq 0. \end{cases}$$

For $t > 0$, we get

$$y^{(1)}(t) = t^2 e^{-t}, \qquad y^{(2)}(t) = (2t - t^2)e^{-t}.$$

Moreover, $\rho(0) = 0$ and $\rho^{(1)}(0) = 0$.

For $t < 0$, the step response is identically zero, so its left derivative at the origin is zero. For $t \geq 0$,

$$y(t) = \int_0^t \rho(s)\,\mathrm{d}s.$$

Because ρ is continuous, the fundamental theorem of calculus gives the right derivative $y^{(1)}(0) = \rho(0) = 0$. The left and right derivatives therefore agree.

For $t > 0$, we have $y^{(1)}(t) = \rho(t)$. Using $y^{(1)}(0) = \rho(0) = 0$ and $\rho^{(1)}(0) = 0$, the right derivative of $y^{(1)}$ at the origin is zero. For $t < 0$, $y^{(1)}(t) = 0$, so its left derivative at the origin is also zero. Therefore,

$$y^{(1)}(0) = y^{(2)}(0) = 0.$$

Since

$$\lim_{t\to 0+} y^{(1)}(t) = \lim_{t\to 0+} y^{(2)}(t) = 0$$

and both derivatives vanish for $t < 0$, they are continuous at the origin. Hence,

$$y \in C^2(\mathbb{R}).$$

(ii) *Construction of the RNN sequence.*

We now construct a sequence of RNNs that approximates these two derivatives. For $t \geq 0$,

$$y^{(1)}(t) = t^2 e^{-t}.$$

We rewrite it as

$$y^{(1)}(t) = t^2 e^{-3t/2} e^{t/2}.$$

Using the Taylor expansion of the last exponential factor gives

$$y^{(1)}(t) = t^2 e^{-3t/2} \sum_{n=0}^{\infty} \frac{(t/2)^n}{n!}.$$

We construct width m RNNs that reproduce more and more terms of this Taylor expansion while all recurrent eigenvalues remain equal to $-3/2$. The construction uses the classical Jordan block mechanism from the example of Zabczyk [TE05, Sec. 19, pp. 185–186]. In this example, Jordan blocks of increasing dimension produce strong nonnormal effects although the real parts of their eigenvalues remain fixed [TE05, Sec. 19, pp. 185–186].

For $m \in \mathbb{N}_+$, let

$$N_m \in \mathbb{R}^{m \times m}$$

be the upper shift matrix

$$N_m \coloneqq \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & 0 \end{pmatrix},$$

where $N_1 = (0)$, and let

$$e_1, \ldots, e_m$$

denote the standard basis vectors of $\mathbb{R}^m$. We define

$$W_m \coloneqq -\frac{3}{2} I_m + N_m \in \mathbb{R}^{m \times m}.$$

Since W_m is upper triangular with diagonal entries $-3/2$, all its eigenvalues are equal to $-3/2$, so $W_m \in \mathcal{W}_m$. For $m \geq 2$,

$$N_m N_m^\top \neq N_m^\top N_m,$$

so N_m is not normal. The same holds for W_m.

Since $N_m^m = 0$, the matrix exponential satisfies

$$e^{W_m t} = e^{-3t/2} \sum_{r=0}^{m-1} \frac{t^r}{r!} N_m^r$$

by the standard formula for the exponential of a Jordan block [Tes12, pp. 62–63, Eqs. (3.18)–(3.19), p. 70, Eq. (3.42)].

We choose

$$c_m \coloneqq e_1 \in \mathbb{R}^m$$

and let $u_m \in \mathbb{R}^m$ be specified below. Since $d = 1$, we set

$$U_m \coloneqq u_m \in \mathbb{R}^{m\times 1}.$$

By Definition 4.9, the RNN memory kernel is

$$\widehat{\rho}_m(t) = U_m^\top e^{W_m^\top t} c_m.$$

Let $\widehat{y}_m$ denote the corresponding RNN step response. By the kernel induced extension from Section 4.5,

$$\widehat{y}_m(t) = \begin{cases} 0, & t < 0, \\ \displaystyle\int_0^t \widehat{\rho}_m(s)\,\mathrm{d}s, & t \geq 0. \end{cases}$$

For $t > 0$,

$$\widehat{y}_m^{(1)}(t) = \widehat{\rho}_m(t).$$

Since the expression is scalar,

$$U_m^\top e^{W_m^\top t} c_m = c_m^\top e^{W_m t} U_m.$$

Therefore,

$$\begin{aligned} \widehat{y}_m^{(1)}(t) &= c_m^\top e^{W_m t} u_m \\ &= e^{-3t/2} e_1^\top \sum_{r=0}^{m-1} \frac{t^r}{r!} N_m^r u_m. \end{aligned}$$

Since

$$e_1^\top N_m^r = e_{r+1}^\top, \qquad r = 0, \ldots, m-1,$$

we get

$$\widehat{y}_m^{(1)}(t) = e^{-3t/2} \sum_{r=0}^{m-1} \frac{t^r}{r!} (u_m)_{r+1}.$$

We choose u_m by matching the coefficients with the Taylor expansion of $y^{(1)}$. The factor t^2 shifts all powers by two, so the target has no terms of degree zero or one.

For $m \geq 3$, let

$$r = 2, \ldots, m-1.$$

The term of degree r in the target is obtained by setting $n = r - 2$, because

$$t^2 (t/2)^n$$

has degree $n + 2$. The corresponding target term is

$$t^2 \frac{(t/2)^{r-2}}{(r-2)!}.$$

The RNN gives the term

$$\frac{t^r}{r!} (u_m)_{r+1}.$$

Matching the two coefficients requires

$$\frac{t^r}{r!}(u_m)_{r+1} = t^2 \frac{(t/2)^{r-2}}{(r-2)!}.$$

Solving for the coefficient gives

$$\begin{aligned}(u_m)_{r+1} &= \frac{r!}{(r-2)!} 2^{-(r-2)} \\ &= r(r-1)2^{-(r-2)}.\end{aligned}$$

Setting

$$j = r + 1$$

gives

$$(u_m)_j = (j-1)(j-2)2^{-(j-3)}, \qquad j = 3, \dots, m.$$

The target expansion starts with degree two. In the RNN expansion, the coefficients of degrees zero and one are $(u_m)_1$ and $(u_m)_2$. We therefore set

$$(u_m)_1 = (u_m)_2 = 0.$$

The same formula gives these two values, so we define

$$(u_m)_j \coloneqq (j-1)(j-2)2^{-(j-3)}, \qquad j = 1, \dots, m.$$

For $m \geq 3$, the definition of u_m gives, for $t \geq 0$,

$$\begin{aligned}\widehat{\rho}_m(t) &= e^{-3t/2} \sum_{r=2}^{m-1} \frac{r(r-1)t^r}{r!2^{r-2}} \\ &= t^2 e^{-3t/2} \sum_{n=0}^{m-3} \frac{(t/2)^n}{n!}.\end{aligned} \tag{5.6}$$

For $t > 0$, the RNN kernel contains exactly the first $m-2$ terms of the Taylor expansion of $e^{t/2}$ in

$$y^{(1)}(t) = t^2 e^{-3t/2} e^{t/2}.$$

For $m \geq 3$, Equation (5.6) shows that $\widehat{\rho}_m$ contains the factor t^2, so $\widehat{\rho}_m(0) = \widehat{\rho}_m^{(1)}(0) = 0$. For $m = 1, 2$, $\widehat{\rho}_m$ vanishes identically. By the same argument as in step (i), with $\widehat{\rho}_m$ in place of ρ, we get

$$\widehat{y}_m^{(1)}(0) = \widehat{y}_m^{(2)}(0) = 0.$$

(iii) *Parameter bounds and spectral condition.*

We next verify the remaining assumptions of Theorem 5.11, before showing that the truncated expansion converges uniformly on $[0, \infty)$. We first show that the parameters are uniformly bounded.

Since N_m shifts a vector by one coordinate,

$$N_m \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} x_2 \\ x_3 \\ \vdots \\ 0 \end{pmatrix},$$

we have

$$\|N_m x\|_2 \leq \|x\|_2$$

for every $x \in \mathbb{R}^m$. We get

$$\|N_m\|_2 \leq 1.$$

Using the triangle inequality for the induced matrix norm gives

$$\begin{aligned}\|W_m\|_2 &= \left\| -\frac{3}{2} I_m + N_m \right\|_2 \\ &\leq \frac{3}{2}\|I_m\|_2 + \|N_m\|_2 \\ &\leq \frac{5}{2}.\end{aligned}$$

We also have

$$\|c_m\|_2 = 1.$$

It remains to bound the input vector. For $m = 1, 2$, we have $u_m = 0$. For $m \geq 3$, the definition of u_m gives

$$\|u_m\|_2^2 = \sum_{j=3}^{m} (j-1)^2 (j-2)^2 4^{-(j-3)}.$$

Setting

$$n = j - 3$$

gives

$$\|u_m\|_2^2 = \sum_{n=0}^{m-3} (n+2)^2 (n+1)^2 4^{-n}.$$

Since all summands are nonnegative,

$$\|u_m\|_2^2 \leq \sum_{n=0}^{\infty} (n+2)^2 (n+1)^2 4^{-n}.$$

To see that the series on the right converges, define

$$a_n \coloneqq (n+2)^2 (n+1)^2 4^{-n}.$$

Then

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{(n+3)^2 (n+2)^2 4^{-(n+1)}}{(n+2)^2 (n+1)^2 4^{-n}} \\ &= \frac{1}{4}\left(\frac{n+3}{n+1}\right)^2 \xrightarrow[n\to\infty]{} \frac{1}{4}.\end{aligned}$$

Because

$$\frac{1}{4} < 1,$$

the ratio test shows that the series converges. With

$$C_u \coloneqq \left(\sum_{n=0}^{\infty} (n+2)^2 (n+1)^2 4^{-n} \right)^{1/2} < \infty,$$

we have $\|u_m\|_2 \leq C_u$ for every $m \in \mathbb{N}_+$. Since $d = 1$ and $U_m = u_m$, we get

$$\sup_{m \in \mathbb{N}_+} \max\{\|W_m\|_2, \|U_m\|_2, \|c_m\|_2\} < \infty.$$

The spectral condition is also satisfied. As shown in step (ii), all eigenvalues of W_m are equal to $-3/2$. Therefore, $w_m = -3/2$ for every $m \in \mathbb{N}_+$, and

$$\limsup_{m \to \infty} w_m = -\frac{3}{2} < -1 = -\beta.$$

(iv) *Approximation of the first derivative.*

It remains to show that the constructed RNNs satisfy the uniform approximation condition in Theorem 5.11. We first consider the first derivative. Let $m \geq 3$ and $t \geq 0$. By construction,

$$y^{(1)}(t) = t^2 e^{-3t/2} e^{t/2}$$

and

$$\widehat{y}_m^{(1)}(t) = t^2 e^{-3t/2} \sum_{n=0}^{m-3} \frac{(t/2)^n}{n!}.$$

We get

$$y^{(1)}(t) - \widehat{y}_m^{(1)}(t) = t^2 e^{-3t/2} \left(e^{t/2} - \sum_{n=0}^{m-3} \frac{(t/2)^n}{n!} \right).$$

To estimate the truncation error, we use Taylor's theorem with the Lagrange remainder. Let f be $N+1$ times continuously differentiable on an interval containing a and x. Then there exists a point ξ between a and x such that

$$f(x) = \sum_{k=0}^{N} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(N+1)}(\xi)}{(N+1)!}(x-a)^{N+1}$$

[FL23, p. 388, Thm. 22.2].

We apply this result with $f(x) = e^x$, $a = 0$, and $N = m - 3$. Let $x \geq 0$. Since every derivative of e^x is again e^x, there exists $\xi \in [0, x]$ such that

$$e^x - \sum_{n=0}^{m-3} \frac{x^n}{n!} = e^\xi \frac{x^{m-2}}{(m-2)!}.$$

Since $0 \leq \xi \leq x$,

$$e^\xi \leq e^x.$$

We get

$$0 \leq e^x - \sum_{n=0}^{m-3} \frac{x^n}{n!} \leq e^x \frac{x^{m-2}}{(m-2)!}.$$

Setting

$$x = \frac{t}{2}$$

gives

$$0 \leq e^{t/2} - \sum_{n=0}^{m-3} \frac{(t/2)^n}{n!} \leq e^{t/2} \frac{(t/2)^{m-2}}{(m-2)!}.$$

Substituting this estimate into the approximation error gives

$$\left|y^{(1)}(t) - \widehat{y}_m^{(1)}(t)\right| \leq \frac{t^m e^{-t}}{2^{m-2}(m-2)!}.$$

To get a bound that is uniform in t, we maximize $t^m e^{-t}$ over $t \geq 0$. Its derivative is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(t^m e^{-t}\right) = t^{m-1}e^{-t}(m-t),$$

so the maximum is attained at $t = m$. We get

$$\sup_{t\geq 0}\left|y^{(1)}(t) - \widehat{y}_m^{(1)}(t)\right| \leq \frac{m^m e^{-m}}{2^{m-2}(m-2)!}.$$

By the lower bound in Stirling's formula,

$$n! > \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \geq \left(\frac{n}{e}\right)^n$$

for $n \geq 1$ [FL23, pp. 348–349, Thm. 20.7]. Applying this estimate with $n = m-2$ gives

$$(m-2)! \geq \left(\frac{m-2}{e}\right)^{m-2}.$$

Using this bound,

$$\begin{aligned}\frac{m^m e^{-m}}{2^{m-2}(m-2)!} &\leq \frac{e^{-2}m^2}{2^{m-2}}\left(\frac{m}{m-2}\right)^{m-2} \\ &= \frac{e^{-2}m^2}{2^{m-2}}\left(1 + \frac{2}{m-2}\right)^{m-2} \\ &\leq \frac{m^2}{2^{m-2}}.\end{aligned}$$

The last inequality follows from $1 + x \leq e^x$ for $x \geq 0$, which gives

$$\left(1 + \frac{2}{m-2}\right)^{m-2} \leq e^2.$$

Since exponential growth dominates polynomial growth,

$$\frac{m^2}{2^{m-2}} \xrightarrow[m\to\infty]{} 0.$$

We get

$$\sup_{t\geq 0}\left|\widehat{y}_m^{(1)}(t) - y^{(1)}(t)\right| \xrightarrow[m\to\infty]{} 0.$$

(v) *Approximation of the second derivative.*

Since $\alpha = 1$, it remains to verify the uniform approximation condition for the second derivatives. Let $m \geq 4$. For $t > 0$, linearity of differentiation gives

$$\begin{aligned}y^{(2)}(t) - \widehat{y}_m^{(2)}(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\left(y^{(1)}(t) - \widehat{y}_m^{(1)}(t)\right) \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left[t^2 e^{-3t/2}\left(e^{t/2} - \sum_{n=0}^{m-3}\frac{(t/2)^n}{n!}\right)\right].\end{aligned}$$

We differentiate the two factors separately. First,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(t^2 e^{-3t/2}\right) = \left(2t - \frac{3}{2}t^2\right) e^{-3t/2}.$$

For the second factor,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{t/2} - \sum_{n=0}^{m-3} \frac{(t/2)^n}{n!}\right) &= \frac{1}{2}e^{t/2} - \frac{1}{2}\sum_{n=0}^{m-4} \frac{(t/2)^n}{n!} \\ &= \frac{1}{2}\left(e^{t/2} - \sum_{n=0}^{m-4} \frac{(t/2)^n}{n!}\right).\end{aligned}$$

The product rule gives

$$\begin{aligned}y^{(2)}(t) - \widehat{y}_m^{(2)}(t) = &\left(2t - \frac{3}{2}t^2\right) e^{-3t/2}\left(e^{t/2} - \sum_{n=0}^{m-3} \frac{(t/2)^n}{n!}\right) \\ &+ \frac{1}{2}t^2 e^{-3t/2}\left(e^{t/2} - \sum_{n=0}^{m-4} \frac{(t/2)^n}{n!}\right).\end{aligned}$$

We use the same Taylor remainder estimate as in step (iv). For $t \geq 0$,

$$0 \leq e^{t/2} - \sum_{n=0}^{m-3} \frac{(t/2)^n}{n!} \leq e^{t/2}\frac{(t/2)^{m-2}}{(m-2)!},$$

and the same argument with $N = m - 4$ gives

$$0 \leq e^{t/2} - \sum_{n=0}^{m-4} \frac{(t/2)^n}{n!} \leq e^{t/2}\frac{(t/2)^{m-3}}{(m-3)!}.$$

Using

$$\left|2t - \frac{3}{2}t^2\right| \leq 2t + \frac{3}{2}t^2,$$

we get

$$\begin{aligned}\left|y^{(2)}(t) - \widehat{y}_m^{(2)}(t)\right| \leq &\, 2\frac{t^{m-1}e^{-t}}{2^{m-2}(m-2)!} + \frac{3}{2}\frac{t^m e^{-t}}{2^{m-2}(m-2)!} \\ &+ \frac{t^{m-1}e^{-t}}{2^{m-2}(m-3)!}.\end{aligned}$$

At $t = 0$, we have $y^{(2)}(0) = \widehat{y}_m^{(2)}(0) = 0$ by steps (i) and (ii), so the same bound also holds at the origin.

As above, for every $q \geq 1$, the function

$$t \longmapsto t^q e^{-t}$$

attains its maximum at $t = q$. We have

$$\sup_{t \geq 0} t^{m-1}e^{-t} = (m-1)^{m-1}e^{-(m-1)}$$

and

$$\sup_{t\geq 0} t^m e^{-t} = m^m e^{-m}.$$

Taking the supremum gives

$$\begin{aligned}
&\sup_{t\geq 0} \left|y^{(2)}(t) - \widehat{y}_m^{(2)}(t)\right| \\
&\leq 2\frac{(m-1)^{m-1}e^{-(m-1)}}{2^{m-2}(m-2)!} + \frac{3}{2}\frac{m^m e^{-m}}{2^{m-2}(m-2)!} \\
&\quad + \frac{(m-1)^{m-1}e^{-(m-1)}}{2^{m-2}(m-3)!}.
\end{aligned}$$

Using

$$n! \geq \left(\frac{n}{e}\right)^n, \qquad n \geq 1,$$

together with $1 + x \leq e^x$ for $x \geq 0$, as in step (iv), we get

$$\frac{(m-1)^{m-1}e^{-(m-1)}}{(m-2)!} \leq m-1,$$

$$\frac{m^m e^{-m}}{(m-2)!} \leq m^2,$$

and

$$\frac{(m-1)^{m-1}e^{-(m-1)}}{(m-3)!} \leq (m-1)^2.$$

Since

$$2(m-1) + (m-1)^2 = m^2 - 1,$$

we have

$$\begin{aligned}
\sup_{t\geq 0} \left|y^{(2)}(t) - \widehat{y}_m^{(2)}(t)\right| &\leq \frac{2(m-1) + \frac{3}{2}m^2 + (m-1)^2}{2^{m-2}} \\
&= \frac{\frac{5}{2}m^2 - 1}{2^{m-2}} \\
&\leq \frac{10m^2}{2^m} \xrightarrow[m\to\infty]{} 0.
\end{aligned}$$

(vi) *Failure of the claimed conclusion.*

We have now verified all assumptions of Theorem 5.11. However,

$$e^t y^{(1)}(t) = e^t t^2 e^{-t} = t^2 \xrightarrow[t\to\infty]{} \infty.$$

We also have

$$e^t y^{(2)}(t) = e^t(2t - t^2)e^{-t} = 2t - t^2,$$

which does not converge to zero as $t \to \infty$. The conclusion of Theorem 5.11 therefore fails for this sequence of nonnormal recurrent matrices. □

5.4.3 Inverse Approximation for Normal Recurrent Matrices

A related result by Wang and Li does not face this problem because they restrict the recurrent matrices to the real diagonal case, which is sufficient for their approximation setting [WL24, p. 2, Sec. 2.1]. For real diagonal matrices, each component of the matrix exponential is a scalar exponential, so eigenvalue bounds directly control its decay.

For the linear RNN setting considered here, we use the additional assumption that the recurrent matrices W_m are normal. Under this assumption, the matrix exponential can also be controlled directly by the eigenvalues. This gives the uniform estimate that is missing for general nonnormal matrices and allows us to prove a modified inverse approximation result.

Lemma 5.13 (Matrix Exponential of a Normal Matrix)**.** *Let*

$$W \in \mathbb{R}^{m\times m}$$

be normal, that is,

$$W^\top W = WW^\top,$$

and define

$$w \coloneqq \max_{\lambda\in\sigma(W)} \operatorname{Re}\lambda.$$

Then, for every $t \geq 0$,

$$\|e^{Wt}\|_2 = e^{wt}.$$

Proof. By the discussion following Equation (3.2), the induced Euclidean operator norm of a real matrix has the same value over $\mathbb{R}^m$ and $\mathbb{C}^m$. We can therefore regard W as a matrix in $\mathbb{C}^{m\times m}$ without changing its norm.

Since W is real,

$$W^* = W^\top.$$

The assumption $W^\top W = WW^\top$ therefore shows that W is also normal as a complex matrix.

By the spectral theorem for normal matrices, there exist a unitary matrix $Q \in \mathbb{C}^{m\times m}$ and a diagonal matrix $D \in \mathbb{C}^{m\times m}$ with eigenvalues $\lambda_1, \ldots, \lambda_m \in \mathbb{C}$ such that

$$D = \operatorname{diag}(\lambda_1, \ldots, \lambda_m)$$

$$W = QDQ^*$$

[FS25, Sec. 6.7.7, pp. 370–371].

Since $Q^* = Q^{-1}$ and similarity transformations commute with the matrix exponential,

$$e^{Wt} = Qe^{Dt}Q^*$$

[Tes12, p. 61, Eq. (3.14)].

We next use that unitary matrices preserve the Euclidean norm. Hence,

$$\begin{aligned}
\|e^{Wt}\|_2 &= \sup_{\|x\|_2=1} \|Qe^{Dt}Q^*x\|_2 \\
&= \sup_{\|x\|_2=1} \|e^{Dt}Q^*x\|_2 \\
&= \sup_{\|y\|_2=1} \|e^{Dt}y\|_2 \\
&= \|e^{Dt}\|_2.
\end{aligned}$$

Here, we used $y = Q^*x$, which again satisfies $\|y\|_2 = \|x\|_2$ because Q^* is unitary.

Since D is diagonal,

$$e^{Dt} = \operatorname{diag}\left(e^{\lambda_1 t}, \dots, e^{\lambda_m t}\right).$$

Therefore, its operator 2 norm is the largest absolute value of its diagonal entries, and thus

$$\begin{aligned}\|e^{Wt}\|_2 &= \|e^{Dt}\|_2 \\ &= \max_{j=1,\dots,m} \left|e^{\lambda_j t}\right| \\ &= \max_{j=1,\dots,m} e^{\operatorname{Re}(\lambda_j)t} \\ &= e^{wt}.\end{aligned}$$

The last equality uses $t \geq 0$.

□

For normal recurrent matrices, Lemma 5.13 gives the uniform estimate that is missing in the general case. Assume

$$\limsup_{m\to\infty} w_m < -\beta.$$

Choose $\delta > 0$ such that

$$\limsup_{m\to\infty} w_m < -(\beta + \delta).$$

By the definition of the limit superior, there exists $m_0 \in \mathbb{N}$ such that

$$w_m \leq -(\beta + \delta), \qquad m \geq m_0.$$

If W_m is normal, Lemma 5.13 gives

$$\|e^{W_m t}\|_2 \leq e^{-(\beta+\delta)t}, \qquad m \geq m_0, \quad t \geq 0.$$

The bound is independent of m. We can therefore prove the inverse approximation result under the additional assumption that the recurrent matrices are normal.

The following theorem is adapted from the inverse approximation theorem of Li et al. [LHEL22, pp. 12–13, Thm. 11]. We add the assumption that every recurrent matrix W_m is normal. This adaptation and its proof are our own. We use a bound on the matrix exponential that is uniform in m. This allows us to pass to the limit directly, without the transformation used by Li et al. [LHEL22, p. 20, Eqs. (80)–(84)].

Theorem 5.14 (Inverse Approximation for Normal Recurrent Matrices [cf. LHEL22, pp. 12–13, Thm. 11])**.** *Assume the hypotheses stated in Theorem 5.11 and suppose, in addition, that every recurrent matrix W_m is normal. Then, for $i = 1, \dots, d$ and $k = 1, \dots, \alpha + 1$,*

$$e^{\beta t} y_i^{(k)}(t) = o(1) \qquad \text{as } t \to +\infty.$$

Proof. Fix $i \in \{1, \dots, d\}$ and $k \in \{1, \dots, \alpha + 1\}$, and set

$$u_{i,m} \coloneqq U_m e_i \in \mathbb{R}^m.$$

By the definition of the RNN memory kernel,

$$\begin{aligned}\widehat{\rho}_{m,i}(t) &= e_i^\top \widehat{\rho}_m(t) \\ &= e_i^\top U_m^\top e^{W_m^\top t} c_m \\ &= c_m^\top e^{W_m t} U_m e_i \\ &= c_m^\top e^{W_m t} u_{i,m}.\end{aligned}$$

For $t > 0$, the definition of the RNN step response and the fundamental theorem of calculus give

$$\widehat{y}_{i,m}^{(1)}(t) = \widehat{\rho}_{m,i}(t) = c_m^\top e^{W_m t} u_{i,m}.$$

Repeated application of Equation (3.5) gives

$$\widehat{y}_{i,m}^{(k)}(t) = c_m^\top W_m^{k-1} e^{W_m t} u_{i,m}, \qquad t > 0,$$

which agrees with the derivative representation of Li et al. [LHEL22, p. 20, Eq. (79)].

By the uniform parameter bound in Theorem 5.11, there exists $M \geq 1$, independent of m, such that

$$\max\{\|c_m\|_2, \|W_m\|_2, \|U_m\|_2\} \leq M$$

for every $m \in \mathbb{N}_+$. Because $u_{i,m} = U_m e_i$ and $\|e_i\|_2 = 1$,

$$\|u_{i,m}\|_2 \leq \|U_m\|_2 \leq M.$$

The spectral gap assumption gives

$$\limsup_{m\to\infty} w_m < -\beta.$$

We can therefore choose $\delta > 0$ such that

$$\limsup_{m\to\infty} w_m < -(\beta + \delta).$$

By the definition of the limit superior, there exists $m_0 \in \mathbb{N}_+$ such that

$$w_m \leq -(\beta + \delta), \qquad m \geq m_0.$$

Because W_m is normal, Lemma 5.13 gives

$$\|e^{W_m t}\|_2 = e^{w_m t} \leq e^{-(\beta+\delta)t}, \qquad m \geq m_0, \quad t \geq 0.$$

Cauchy Schwarz and submultiplicativity of the induced matrix norm give

$$\begin{aligned}\left|\widehat{y}_{i,m}^{(k)}(t)\right| &\leq \|c_m\|_2 \|W_m\|_2^{k-1} \|e^{W_m t}\|_2 \|u_{i,m}\|_2 \\ &\leq M^{k+1} e^{-(\beta+\delta)t}\end{aligned}$$

for every $m \geq m_0$ and $t > 0$.

Fix $t > 0$. The approximation assumption in Theorem 5.11 gives

$$\widehat{y}_{i,m}^{(k)}(t) \xrightarrow[m\to\infty]{} y_i^{(k)}(t).$$

Continuity of the absolute value gives

$$\left|y_i^{(k)}(t)\right| = \lim_{m\to\infty} \left|\widehat{y}_{i,m}^{(k)}(t)\right| \leq M^{k+1} e^{-(\beta+\delta)t}.$$

Multiplying by $e^{\beta t}$ gives

$$e^{\beta t}\left|y_i^{(k)}(t)\right| \leq M^{k+1} e^{-\delta t} \xrightarrow[t\to\infty]{} 0.$$

Because i and k were arbitrary, the claim follows. □

A similar argument based on uniform exponential decay of the approximating models is used by Wang et al. [WLL24, Appendix A.10, Lem. A.10]. Under the normality assumption considered here, the required uniform bound follows directly from Lemma 5.13.

The inverse result gives a more direct connection between approximation and long memory. Since

$$y_i^{(1)}(t) = \rho_i(t),$$

the theorem requires the memory kernel itself to decay faster than $e^{-\beta t}$ if the target derivatives can be approximated uniformly by normal RNNs whose recurrent matrices remain uniformly stable.

Targets with slower memory decay, such as polynomially decaying kernels, do not satisfy this decay condition.

5.5 The Curse of Memory in Approximation

Theorem 5.5 shows that stable linear continuous time RNNs can approximate every target in $\mathcal{H}$ arbitrarily well if the hidden dimension is allowed to increase. The theorem does not, however, give a bound on how large this dimension has to be.

The approximation rate in Theorem 5.9 already shows a connection between memory decay and the required RNN width. Li et al. show that this effect can become much stronger for polynomially decaying memory [LHEL22, Sec. 4.3, pp. 13–14]. For a fixed approximation accuracy, the required width can grow exponentially as the memory becomes longer. They refer to this effect as the *curse of memory in approximation* [LHEL22, Sec. 4.3, pp. 13–14].

This gives an answer to the approximation question considered in this chapter. Long memory does not prevent approximation in principle, as Theorem 5.5 shows, but it can make the required RNN much larger. This difficulty is different from the vanishing and exploding gradients discussed in Section 2.2.4. The curse of memory in approximation concerns the size of the model needed to represent the target, while vanishing and exploding gradients concern the training procedure.

The two effects therefore describe different problems. Approximation theory asks whether suitable parameters exist and how large the model has to be. Optimization theory asks whether these parameters can be found by training. Chapter 6 studies this second question.

6 Optimization Theory

As discussed in Section 2.2.4, discrete time RNNs with differentiable activation functions can have vanishing or exploding gradients. In particular, information from the distant past can have only a very small effect on the gradient and make training difficult.

We study this problem in the linear continuous time setting. In the previous chapter, we showed that the target functionals can be approximated by stable linear continuous time RNNs. In this chapter, we study optimization difficulties in a simplified setting with diagonal recurrent matrices and gradient flow in the reduced parameters.

Li et al. show that increasing the memory of the target can lead to arbitrarily long plateaus in gradient flow [LHEL22, pp. 27–28, Thm. 20]. In contrast to Chapter 5, we do not give a complete proof of the plateau time bound. Our own work in this chapter is mainly to explain the main steps of the optimization analysis in detail. We also show that the reduced loss J is continuously differentiable and that J_ω and $\overline{J}$ are twice continuously differentiable. For the more technical estimates and the plateau time bound, we refer to Li et al. [LHEL22, Secs. 6.2.1 and 7.1]. Finally, we illustrate the plateau mechanism using our own numerical implementation.

The main reference for this chapter is Li et al. [LHEL22, Secs. 6 and 7.1, pp. 21–45].

6.1 Reduction to a Kernel Optimization Problem

We first describe how the optimization problem arises. Recall from Lemma 4.8 and Definition 4.9 that the target family $\{H_t\}_{t\in\mathbb{R}}$ has memory kernel

$$\rho\colon [0,\infty) \to \mathbb{R}^d,$$

while the RNN family $\{\widehat{H}_t\}_{t\in\mathbb{R}}$ has memory kernel

$$\widehat{\rho}\colon [0,\infty) \to \mathbb{R}^d, \qquad \widehat{\rho}(t) = U^\top e^{W^\top t} c.$$

and

$$c^\top e^{Wt} U - \rho(t)^\top = \big(\widehat{\rho}(t) - \rho(t)\big)^\top.$$

Li et al. measure the training error at a fixed terminal time $T > 0$ by the expected squared difference between the target output and the RNN output [LHEL22, p. 21, Eq. (87)],

$$\mathbb{E}_x\left[\left|\widehat{H}_T(x) - H_T(x)\right|^2\right] = \mathbb{E}_x\left[\left|\int_0^T \left[c^\top e^{Wt} U - \rho(t)^\top\right] x_{T-t}\,\mathrm{d}t\right|^2\right].$$

Here, the input is assumed to satisfy $x_t = 0$ for $t \le 0$.

To simplify this expression, Li et al. model the input as white noise and write formally

$$x_{T-t}\,\mathrm{d}t \overset{\mathrm{d}}{=} \mathrm{d}B_t,$$

where B is a standard d-dimensional Brownian motion [LHEL22, p. 23, Eq. (90)]. This relation is understood only under integration against suitable deterministic test functions. A rigorous stochastic interpretation of this step in our setting is beyond

the scope of this thesis. The general foundations of Itô integration, including the construction of the integral and the Itô isometry, are in Øksendal [Øks98, pp. 21–29, Sec. 3.1]. Using the Itô isometry, Li et al. reduce the expected loss to the kernel loss for $c \in \mathbb{R}^m$, $W \in \mathcal{W}_m$, and $U \in \mathbb{R}^{m\times d}$,

$$J_T(c, W, U) \coloneqq \int_0^T \|\widehat{\rho}(t) - \rho(t)\|_2^2 \, \mathrm{d}t = \int_0^T \left\|c^\top e^{Wt} U - \rho(t)^\top\right\|_2^2 \, \mathrm{d}t \tag{6.1}$$

[LHEL22, p. 23, Eq. (91)]. We take this kernel loss as the starting point of the optimization analysis. For finite T, this loss is finite whenever

$$\rho \in L^2([0, T], \mathbb{R}^d).$$

Li et al. introduce further simplifications to isolate the effect of temporal memory [LHEL22, p. 24, Eqs. (92)–(94)]. First, they restrict the input dimension to $d = 1$, so that the target kernel ρ is scalar valued.

Second, the time horizon is extended to infinity in order to study arbitrarily large memory lags. For the optimization analysis, we assume

$$\rho \in L^2([0, \infty)).$$

Unlike in the previous chapters, L^1-integrability is not required for the kernel loss considered here. Let $b \in \mathbb{R}^m$ denote the single column of U, and let $W \in \mathcal{W}_m$. By Lemma 3.7, there exist constants $C, \alpha > 0$ such that

$$\left|c^\top e^{Wt} b\right| \le C\|c\|_2\|b\|_2 e^{-\alpha t}, \qquad t \ge 0.$$

Since $e^{-\alpha t} \in L^2([0, \infty))$, the RNN kernel belongs to $L^2([0, \infty))$.

Define $h\colon [0, \infty) \to [0, \infty)$ by

$$h(t) \coloneqq \left(c^\top e^{Wt} b - \rho(t)\right)^2.$$

Since $h \ge 0$, the functions $h\mathbf{1}_{[0,T]}$ increase pointwise to h as $T \to \infty$. For $c, b \in \mathbb{R}^m$ and $W \in \mathcal{W}_m$, the monotone convergence theorem [Els18, p. 139, Thm. IV.2.7] gives

$$J_\infty(c, W, b) \coloneqq \lim_{T\to\infty} J_T(c, W, b) = \int_0^\infty \left(c^\top e^{Wt} b - \rho(t)\right)^2 \, \mathrm{d}t \tag{6.2}$$

[LHEL22, p. 24, Eq. (92)]. Since both the RNN kernel and ρ belong to $L^2([0, \infty))$, we have

$$J_\infty(c, W, b) < \infty.$$

Third, the dependence of e^{Wt} on a general recurrent matrix W makes derivatives with respect to W difficult to handle. Li et al. therefore restrict the recurrent matrix to the diagonal form

$$W = \operatorname{diag}(-w_1, \ldots, -w_m), \qquad w \in (0, \infty)^m$$

[LHEL22, p. 24, Eq. (93)]. Li et al. also observe plateauing numerically for full recurrent matrices, nonlinear activations, and the Adam optimizer [LHEL22, p. 32, Fig. 4].

Because $w_i > 0$ for all i, all eigenvalues of W are negative, so

$$W \in \mathcal{W}_m.$$

For this diagonal matrix,

$$e^{Wt} = \operatorname{diag}\left(e^{-w_1 t}, \dots, e^{-w_m t}\right),$$

so that

$$c^\top e^{Wt} b = \sum_{i=1}^{m} c_i b_i e^{-w_i t}.$$

Define $a = (a_1, \dots, a_m)^\top \in \mathbb{R}^m$ by

$$a_i \coloneqq b_i c_i, \qquad i = 1, \dots, m.$$

The RNN kernel then has the exponential sum representation $\widehat{\rho}_{a,w} \colon [0,\infty) \to \mathbb{R}$,

$$\widehat{\rho}_{a,w}(t) \coloneqq \sum_{i=1}^{m} a_i e^{-w_i t}.$$

A smaller value of w_i gives a more slowly decaying exponential, so the corresponding term acts on larger memory lags.

We therefore define the reduced objective $J \colon \mathbb{R}^m \times (0,\infty)^m \to [0,\infty)$ by

$$J(a,w) \coloneqq \int_0^\infty \left(\widehat{\rho}_{a,w}(t) - \rho(t)\right)^2 \, \mathrm{d}t, \qquad (a,w) \in \mathbb{R}^m \times (0,\infty)^m. \tag{6.3}$$

Whenever $a_i = b_i c_i$ for $i = 1, \dots, m$, this objective satisfies

$$J(a,w) = J_\infty\big(c, \operatorname{diag}(-w_1, \dots, -w_m), b\big).$$

This is the reduced optimization objective considered by Li et al. [LHEL22, p. 24, Eq. (94)]. It depends on b and c only through a, and training reduces to approximating the scalar target kernel ρ by a finite sum of decaying exponentials in $L^2([0,\infty))$.

For the subsequent optimization analysis, we additionally assume

$$\rho \in C^2([0,\infty)),$$

as in Li et al. [LHEL22, p. 24]. The target kernel therefore satisfies

$$\rho \in L^2([0,\infty)) \cap C^2([0,\infty)).$$

The reduced objective in (6.3) is the starting point for the optimization analysis. In the next subsection, we derive its gradient with respect to the reduced parameters (a,w).

6.2 Gradient of the Kernel Loss

Unless stated otherwise, all integrals in this chapter are understood as Lebesgue integrals with respect to the Lebesgue measure.

We next compute the gradient of the reduced loss. Li et al. give the gradient formulas without a detailed derivation [LHEL22, p. 25, Eqs. (96)–(97)]. Our own contribution is to justify differentiation under the integral sign and to prove that the gradient is continuous.

Define $F\colon [0,\infty) \times \mathbb{R}^m \times (0,\infty)^m \to [0,\infty)$ by

$$F(t,a,w) \coloneqq \left(\sum_{i=1}^{m} a_i e^{-w_i t} - \rho(t) \right)^2,$$

so that

$$J(a,w) = \int_0^\infty F(t,a,w)\,\mathrm{d}t.$$

For every $a \in \mathbb{R}^m$ and $w \in (0,\infty)^m$,

$$\sum_{i=1}^{m} a_i e^{-w_i t} - \rho(t) \in L^2([0,\infty)),$$

because each exponential $e^{-w_i \cdot}$ belongs to $L^2([0,\infty))$ and $\rho \in L^2([0,\infty))$. The function F is the square of this function, so

$$F(\cdot,a,w) \in L^1([0,\infty)).$$

For every fixed $t \geq 0$, the function $F(t,\cdot,\cdot)$ is continuously differentiable on $\mathbb{R}^m \times (0,\infty)^m$.

Fix $k \in \{1,\dots,m\}$, and let $e_k \in \mathbb{R}^m$ denote the k-th standard basis vector. For $|h| < 1$, we have

$$|a_k + h| \leq |a_k| + 1,$$

and therefore

$$\begin{aligned}
\left| \frac{\partial F}{\partial a_k}(t, a + h e_k, w) \right| &= 2e^{-w_k t} \left| \sum_{i \neq k} a_i e^{-w_i t} + (a_k + h) e^{-w_k t} - \rho(t) \right| \\
&\leq 2e^{-w_k t} \left(\sum_{i \neq k} |a_i| e^{-w_i t} + (|a_k| + 1) e^{-w_k t} + |\rho(t)| \right) \\
&=: g_{a_k}(t).
\end{aligned}$$

The terms that contain only exponentials are integrable because

$$\int_0^\infty e^{-\alpha t}\,\mathrm{d}t < \infty \qquad \text{for every } \alpha > 0.$$

For the remaining term, the Cauchy–Schwarz inequality gives

$$\begin{aligned}
\int_0^\infty e^{-w_k t} |\rho(t)|\,\mathrm{d}t &\leq \left(\int_0^\infty e^{-2w_k t}\,\mathrm{d}t \right)^{1/2} \left(\int_0^\infty |\rho(t)|^2\,\mathrm{d}t \right)^{1/2} \\
&< \infty.
\end{aligned}$$

We therefore have

$$g_{a_k} \in L^1([0,\infty)).$$

For the derivative with respect to w_k, let

$$|h| < \frac{w_k}{2}.$$

Then

$$w_k + h > \frac{w_k}{2},$$

which gives

$$e^{-(w_k+h)t} \leq e^{-w_k t/2}.$$

We get

$$\begin{aligned}\left|\frac{\partial F}{\partial w_k}(t,a,w+he_k)\right| &= 2|a_k|te^{-(w_k+h)t}\left|\sum_{i\neq k} a_i e^{-w_i t} + a_k e^{-(w_k+h)t} - \rho(t)\right| \\ &\leq 2|a_k|te^{-w_k t/2}\left(\sum_{i\neq k}|a_i|e^{-w_i t} + |a_k|e^{-w_k t/2} + |\rho(t)|\right) \\ &=: g_{w_k}(t).\end{aligned}$$

The terms that contain only exponentials are again integrable because

$$\int_0^\infty te^{-\alpha t}\,\mathrm{d}t < \infty \qquad \text{for every } \alpha > 0.$$

For the remaining term, the Cauchy–Schwarz inequality gives

$$\begin{aligned}\int_0^\infty te^{-w_k t/2}|\rho(t)|\,\mathrm{d}t &\leq \left(\int_0^\infty t^2 e^{-w_k t}\,\mathrm{d}t\right)^{1/2}\left(\int_0^\infty |\rho(t)|^2\,\mathrm{d}t\right)^{1/2} \\ &< \infty.\end{aligned}$$

We therefore have

$$g_{w_k} \in L^1([0,\infty)).$$

The theorem on differentiation under the integral sign then allows us to interchange differentiation and integration [Els18, pp. 162–163, Thm. IV.5.7 and its addendum].

For $k = 1, \ldots, m$, this gives

$$\begin{aligned}\frac{\partial J}{\partial a_k}(a,w) &= \frac{\partial}{\partial a_k}\int_0^\infty \left(\sum_{i=1}^m a_i e^{-w_i t} - \rho(t)\right)^2 \mathrm{d}t \\ &= 2\int_0^\infty \left(\sum_{i=1}^m a_i e^{-w_i t} - \rho(t)\right) e^{-w_k t}\,\mathrm{d}t \qquad (6.4) \\ &= 2\int_0^\infty e^{-w_k t}\left(\sum_{i=1}^m a_i e^{-w_i t} - \rho(t)\right)\mathrm{d}t.\end{aligned}$$

$$\begin{aligned}\frac{\partial J}{\partial w_k}(a,w) &= \frac{\partial}{\partial w_k}\int_0^\infty \left(\sum_{i=1}^m a_i e^{-w_i t} - \rho(t)\right)^2 \mathrm{d}t \\ &= 2\int_0^\infty \left(\sum_{i=1}^m a_i e^{-w_i t} - \rho(t)\right)\left(-a_k t e^{-w_k t}\right)\mathrm{d}t \\ &= -2a_k \int_0^\infty t e^{-w_k t}\left(\sum_{i=1}^m a_i e^{-w_i t} - \rho(t)\right)\mathrm{d}t.\end{aligned} \tag{6.5}$$

These are the gradient expressions used by Li et al. [LHEL22, p. 25, Eqs. (96)–(97)].

We now show that the partial derivatives of J are continuous. For this, we apply the theorem on continuity under the integral sign [Els18, p. 162, Theorem IV.5.6]. The first two assumptions are already satisfied: We have shown that the partial derivatives of F are integrable for every fixed (a,w), and they are continuous in (a,w) for every fixed t. It remains to find an integrable bound that holds for all parameters in a neighbourhood of a fixed point.

Fix

$$(a,w) \in \mathbb{R}^m \times (0,\infty)^m$$

and choose

$$0 < \varepsilon < \min\left\{1, \frac{1}{2}\min_{1\le i\le m} w_i\right\}.$$

For every (a',w') satisfying

$$\|(a',w') - (a,w)\|_2 < \varepsilon,$$

we have

$$|a_i' - a_i| < 1, \qquad |w_i' - w_i| < \frac{w_i}{2}, \qquad i = 1,\dots,m.$$

The triangle inequality gives

$$|a_i'| \le |a_i| + 1, \qquad w_i' > \frac{w_i}{2} > 0.$$

For $t \ge 0$, the second inequality implies

$$e^{-w_i' t} \le e^{-w_i t/2}.$$

Fix $k \in \{1,\dots,m\}$. Using these bounds and the triangle inequality, we get

$$\begin{aligned}\left|\frac{\partial F}{\partial a_k}(t,a',w')\right| &= 2e^{-w_k' t}\left|\sum_{i=1}^m a_i' e^{-w_i' t} - \rho(t)\right| \\ &\le 2e^{-w_k t/2}\left(\sum_{i=1}^m (|a_i|+1)e^{-w_i t/2} + |\rho(t)|\right)\end{aligned}$$

and

$$\begin{aligned}\left|\frac{\partial F}{\partial w_k}(t,a',w')\right| &= 2|a_k'| t e^{-w_k' t}\left|\sum_{i=1}^m a_i' e^{-w_i' t} - \rho(t)\right| \\ &\le 2(|a_k|+1) t e^{-w_k t/2}\left(\sum_{i=1}^m (|a_i|+1)e^{-w_i t/2} + |\rho(t)|\right).\end{aligned}$$

The right hand sides do not depend on (a', w'). We now show that they are integrable. The terms containing only exponentials are integrable because $w_i > 0$ for all i. For the terms containing ρ, the Cauchy–Schwarz inequality gives

$$\begin{aligned}\int_0^\infty t^j e^{-w_k t/2} |\rho(t)| \,\mathrm{d}t &\leq \left(\int_0^\infty t^{2j} e^{-w_k t} \,\mathrm{d}t\right)^{1/2} \left(\int_0^\infty |\rho(t)|^2 \,\mathrm{d}t\right)^{1/2} \\ &< \infty, \qquad j = 0, 1.\end{aligned}$$

The first factor is finite because $w_k > 0$, and the second because $\rho \in L^2([0, \infty))$.

We have therefore found an integrable bound for each partial derivative of F that holds for all parameters in the chosen neighbourhood. By Theorem IV.5.6 of Elstrodt, the partial derivatives of J are continuous [Els18, p. 162]. Since all partial derivatives exist and are continuous, we get

$$J \in C^1\big(\mathbb{R}^m \times (0, \infty)^m\big).$$

Gradient descent was introduced in Section 2.2.3. For the theoretical analysis, Li et al. use a continuous time version of the discrete parameter updates and consider the gradient flow [LHEL22, p. 24, Eq. (95)]. We use τ for the optimization time to distinguish it from the memory lag t. For some $\tau^* \in (0, \infty]$, consider a differentiable curve

$$\vartheta \colon [0, \tau^*) \to \mathbb{R}^m \times (0, \infty)^m, \qquad \vartheta(\tau) = \big(a(\tau), w(\tau)\big),$$

satisfying the gradient flow equation

$$\vartheta'(\tau) = -\nabla J\big(\vartheta(\tau)\big), \qquad \vartheta(0) = \vartheta_0. \tag{6.6}$$

Here, $\nabla J(\vartheta) \in \mathbb{R}^{2m}$ is the gradient with respect to the reduced parameters $\vartheta = (a, w)$. Equivalently,

$$a'(\tau) = -\nabla_a J\big(a(\tau), w(\tau)\big), \qquad w'(\tau) = -\nabla_w J\big(a(\tau), w(\tau)\big).$$

This gradient flow is defined for the reduced parameters (a, w). Since $a_i = b_i c_i$, it does not in general coincide with gradient flow in the original parameters (c, W, b).

Since $J \in C^1$, the chain rule gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\tau} J\big(\vartheta(\tau)\big) &= \nabla J\big(\vartheta(\tau)\big)^\top \vartheta'(\tau) \\ &= -\left\|\nabla J\big(\vartheta(\tau)\big)\right\|_2^2 \leq 0.\end{aligned} \tag{6.7}$$

The loss is nonincreasing along the gradient flow.

Equations (6.4) and (6.5) show how long memory can lead to a small gradient. The kernel error at memory lag t is multiplied by

$$e^{-w_k t} \qquad \text{or} \qquad t e^{-w_k t}.$$

Both factors converge to zero as $t \to \infty$. This suggests that errors at large memory lags can contribute substantially to the loss while having only a small influence on the gradient. Li et al. use this heuristic as the starting point of their analysis [LHEL22, p. 25, Sec. 6.1.3].

6.3 Long Memory and the Plateau Mechanism

We now study how moving part of the target kernel to larger memory lags affects optimization. This is similar to the copy-first memory task in Example 2.3, where the relevant input lies farther in the past as the sequence length increases.

We follow the target-kernel construction of Li et al. [LHEL22, pp. 25–26, Eqs. (98), (101)]. For $\omega > 0$, we consider target kernels $\rho_\omega \colon [0,\infty) \to \mathbb{R}$ of the form

$$\rho_\omega(t) \coloneqq \bar{\rho}(t) + \rho_{0,\omega}(t), \qquad \rho_{0,\omega}(t) \coloneqq \rho_0\left(t - \frac{1}{\omega}\right), \qquad \omega > 0. \tag{6.8}$$

Here, $\bar{\rho}$ describes the short term memory component, while $\rho_{0,\omega}$ describes the long term memory component. As $\omega \to 0^+$, the shift $1/\omega$ increases and the long term component moves to larger memory lags.

For the following analysis, we take the short term component $\bar{\rho} \colon [0,\infty) \to \mathbb{R}$ to be the exponential sum

$$\bar{\rho}(t) \coloneqq \sum_{j=1}^{m^*} a_j^* e^{-w_j^* t}, \qquad m^* < m, \tag{6.9}$$

where

$$a_j^* \neq 0, \qquad w_j^* > 0, \qquad w_i^* \neq w_j^* \quad \text{for } i \neq j.$$

These are the assumptions on the short term component used by Li et al. in their plateau analysis [LHEL22, pp. 27, 34]. The condition $m^* < m$ means that the RNN has more exponential modes than are needed to represent $\bar{\rho}$.

For the function

$$\rho_0 \colon \mathbb{R} \to \mathbb{R},$$

we assume

$$\rho_0 \in L^2(\mathbb{R}) \cap L^\infty(\mathbb{R}) \cap C^2(\mathbb{R}).$$

We also assume that $\rho_0 \not\equiv 0$ and

$$\lim_{t \to -\infty} \rho_0(t) = 0.$$

These assumptions follow Li et al. [LHEL22, p. 27]. Since $\bar{\rho}$ decays exponentially and translations preserve the L^2 norm on $\mathbb{R}$, each target ρ_ω belongs to

$$L^2([0,\infty)) \cap C^2([0,\infty)).$$

For the target ρ_ω, the reduced loss from (6.3) becomes

$$J_\omega(a,w) \coloneqq \int_0^\infty \left(\widehat{\rho}_{a,w}(t) - \rho_\omega(t)\right)^2 \mathrm{d}t, \qquad (a,w) \in \mathbb{R}^m \times (0,\infty)^m \tag{6.10}$$

[LHEL22, p. 26, Eq. (100)].

We consider an initial parameter vector

$$\vartheta_0 \coloneqq (a_0, w_0) \in \mathbb{R}^m \times (0,\infty)^m$$

for which the RNN represents the short term component exactly,

$$\widehat{\rho}_{a_0,w_0}(t) = \bar{\rho}(t), \qquad t \geq 0. \tag{6.11}$$

In the following, we also write $\widehat{\rho}_{\vartheta_0} = \widehat{\rho}_{a_0,w_0}$. We fix ϑ_0 independently of ω throughout the analysis.

Such a parameter vector can be constructed by choosing m^* model modes equal to the modes of $\bar{\rho}$ and setting the amplitudes of the remaining $m - m^*$ modes to zero. More generally, Lemma 30 of Li et al. characterizes all parameter vectors that represent $\bar{\rho}$ exactly [LHEL22, p. 37, Lemma 30].

We first study the loss at this parameter vector.

Proposition 6.1 (Loss Remains Bounded Away from Zero [LHEL22, p. 38, Prop. 32])**.** *Let $\vartheta_0 = (a_0, w_0)$ satisfy (6.11). Then there exist constants $C(\rho_0), C'(\rho_0) > 0$, depending only on ρ_0, such that*

$$J_\omega(\vartheta_0) \geq C(\rho_0) > 0 \tag{6.12}$$

for every $\omega \in (0, C'(\rho_0))$.

Proof. By (6.8) and (6.11),

$$\widehat{\rho}_{\vartheta_0}(t) - \rho_\omega(t) = -\rho_0\left(t - \frac{1}{\omega}\right).$$

Inserting this into (6.10) and using the substitution $s = t - 1/\omega$ gives

$$\begin{aligned} J_\omega(\vartheta_0) &= \int_0^\infty \rho_0\left(t - \frac{1}{\omega}\right)^2 \mathrm{d}t \\ &= \int_{-1/\omega}^\infty \rho_0(s)^2 \,\mathrm{d}s. \end{aligned}$$

Because $\rho_0 \in L^2(\mathbb{R})$, the function ρ_0^2 belongs to $L^1(\mathbb{R})$. Furthermore, the lower integration limit tends to $-\infty$ as $\omega \to 0^+$. Since the integrand is nonnegative, the monotone convergence theorem applies [Els18, p. 139, Thm. IV.2.7], and we get

$$\lim_{\omega \to 0^+} J_\omega(\vartheta_0) = \int_{-\infty}^\infty \rho_0(s)^2 \,\mathrm{d}s = \|\rho_0\|^2_{L^2(\mathbb{R})} > 0. \tag{6.13}$$

The limit is positive because ρ_0 is continuous and not identically zero. By the definition of the limit, there exists $C'(\rho_0) > 0$ such that

$$J_\omega(\vartheta_0) \geq \frac{1}{2}\|\rho_0\|^2_{L^2(\mathbb{R})} =: C(\rho_0)$$

for every $\omega \in (0, C'(\rho_0))$. This proves the assertion.

□

Increasing the memory length therefore moves the remaining error farther into the past, but its contribution to the loss does not disappear.

We next study the gradient. Recall from Equations (6.4) and (6.5) that its components contain exponentially decaying weights. To describe the interaction between these weights and the shifted target, we introduce the overlap terms

$$\Delta_{n,\omega}(w) \coloneqq \int_0^\infty t^n e^{-wt} \rho_0\left(t - \frac{1}{\omega}\right) \mathrm{d}t, \qquad n = 0, 1, 2, \quad w, \omega > 0. \tag{6.14}$$

These are the overlap integrals considered by Li et al. [LHEL22, p. 34, Eq. (118)].

The following lemma shows that these terms vanish when the long term component moves farther into the past.

Lemma 6.2 (Decay of the Overlap Terms [LHEL22, pp. 34–36, Lemma 27])**.** *For $n = 0, 1, 2$ and every fixed $w > 0$,*

$$\lim_{\omega \to 0^+} \Delta_{n,\omega}(w) = 0. \tag{6.15}$$

Proof. The proof follows Li et al. [LHEL22, p. 35, proof of Lemma 27(ii), Eq. (121)]. Fix $n \in \{0, 1, 2\}$ and $w > 0$. For every fixed $t \geq 0$,

$$t - \frac{1}{\omega} \xrightarrow[\omega \to 0^+]{} -\infty.$$

The assumption on the left tail of ρ_0 gives

$$\rho_0 \left(t - \frac{1}{\omega} \right) \xrightarrow[\omega \to 0^+]{} 0.$$

Therefore,

$$t^n e^{-wt} \rho_0 \left(t - \frac{1}{\omega} \right) \xrightarrow[\omega \to 0^+]{} 0.$$

Since $\rho_0 \in L^\infty(\mathbb{R})$, we also have

$$\left| t^n e^{-wt} \rho_0 \left(t - \frac{1}{\omega} \right) \right| \leq \|\rho_0\|_{L^\infty(\mathbb{R})} t^n e^{-wt}.$$

The right hand side is integrable because $w > 0$ and

$$\int_0^\infty t^n e^{-wt} \, \mathrm{d}t = \frac{n!}{w^{n+1}} < \infty.$$

By the dominated convergence theorem [Els18, p. 160, Thm. IV.5.2], we get

$$\begin{aligned} \lim_{\omega \to 0^+} \Delta_{n,\omega}(w) &= \lim_{\omega \to 0^+} \int_0^\infty t^n e^{-wt} \rho_0 \left(t - \frac{1}{\omega} \right) \mathrm{d}t \\ &= \int_0^\infty t^n e^{-wt} \lim_{\omega \to 0^+} \rho_0 \left(t - \frac{1}{\omega} \right) \mathrm{d}t \\ &= 0. \end{aligned}$$

□

Under the stronger sub-Gaussian tail assumption, Li et al. also establish quantitative decay estimates for the overlap terms [LHEL22, pp. 34–36, Lemma 27]. We use these estimates in the next subsection to study the plateau timescale.

We can now show that the gradient at ϑ_0 vanishes as $\omega \to 0^+$.

Proposition 6.3 (Vanishing Gradient [LHEL22, pp. 38–39, Prop. 33])**.** *Let $\vartheta_0 = (a_0, w_0)$ satisfy (6.11). Then*

$$\lim_{\omega \to 0^+} \|\nabla J_\omega(\vartheta_0)\|_2 = 0. \tag{6.16}$$

Proof. We derive the gradient identity used by Li et al. [LHEL22, p. 39, proof of Prop. 33] directly from the residual at the initialization. By the exact representation of the short term component,

$$\widehat{\rho}_{\vartheta_0}(t) - \rho_\omega(t) = -\rho_0\left(t - \frac{1}{\omega}\right).$$

Inserting this into (6.4) gives, for $k = 1, \dots, m$,

$$\begin{aligned}\frac{\partial J_\omega}{\partial a_k}(\vartheta_0) &= 2\int_0^\infty e^{-w_{0,k}t}\left(\widehat{\rho}_{\vartheta_0}(t) - \rho_\omega(t)\right)\,\mathrm{d}t \\ &= -2\int_0^\infty e^{-w_{0,k}t}\rho_0\left(t - \frac{1}{\omega}\right)\,\mathrm{d}t \\ &= -2\Delta_{0,\omega}(w_{0,k}).\end{aligned}$$

Similarly, (6.5) gives

$$\begin{aligned}\frac{\partial J_\omega}{\partial w_k}(\vartheta_0) &= -2a_{0,k}\int_0^\infty te^{-w_{0,k}t}\left(\widehat{\rho}_{\vartheta_0}(t) - \rho_\omega(t)\right)\,\mathrm{d}t \\ &= 2a_{0,k}\int_0^\infty te^{-w_{0,k}t}\rho_0\left(t - \frac{1}{\omega}\right)\,\mathrm{d}t \\ &= 2a_{0,k}\Delta_{1,\omega}(w_{0,k}).\end{aligned}$$

Consequently,

$$\begin{aligned}\|\nabla J_\omega(\vartheta_0)\|_2^2 &= \sum_{k=1}^m\left[\left(\frac{\partial J_\omega}{\partial a_k}(\vartheta_0)\right)^2 + \left(\frac{\partial J_\omega}{\partial w_k}(\vartheta_0)\right)^2\right] \\ &= 4\sum_{k=1}^m\left(\Delta_{0,\omega}(w_{0,k})^2 + a_{0,k}^2\Delta_{1,\omega}(w_{0,k})^2\right).\end{aligned} \tag{6.17}$$

By Lemma 6.2, both overlap terms converge to zero for every fixed $w_{0,k} > 0$. Since the sum contains only finitely many terms, we get

$$\lim_{\omega\to 0^+}\|\nabla J_\omega(\vartheta_0)\|_2^2 = 0,$$

which proves the assertion.

□

Propositions 6.1 and 6.3 show that

$$J_\omega(\vartheta_0) \geq C(\rho_0) > 0, \qquad \|\nabla J_\omega(\vartheta_0)\|_2 \longrightarrow 0$$

as $\omega \to 0^+$, where the lower bound holds for sufficiently small $\omega > 0$.

This alone does not show a plateau. The gradient is small at ϑ_0, but it may become large after a small change of the parameters. To understand the behavior near ϑ_0, we next consider the Hessian.

We define the short term loss $\overline{J}\colon \mathbb{R}^m \times (0,\infty)^m \to [0,\infty)$ by

$$\overline{J}(a,w) \coloneqq \int_0^\infty\left(\widehat{\rho}_{a,w}(t) - \bar{\rho}(t)\right)^2\,\mathrm{d}t \tag{6.18}$$

as in Li et al. [LHEL22, p. 40, Eq. (140)].

We first show that J_ω is twice continuously differentiable. The same proof applies to $\overline{J}$ by replacing ρ_ω with $\bar{\rho}$, because both target kernels belong to $L^2([0,\infty))$.

For this, we apply the theorem on differentiation under the integral sign a second time. Equations (6.4) and (6.5) already express the first partial derivatives of the loss as integrals. These formulas also apply to J_ω, since their derivation only requires the target kernel to belong to $L^2([0,\infty))$.

To differentiate these integrals again, we need to show that the second partial derivatives of the integrands exist and admit an integrable bound in a neighborhood of each parameter point.

We first compute the required derivatives. Recall that

$$\widehat{\rho}_{a,w}(t) = \sum_{i=1}^{m} a_i e^{-w_i t}.$$

Its first partial derivatives are

$$\frac{\partial \widehat{\rho}_{a,w}(t)}{\partial a_i} = e^{-w_i t}, \qquad \frac{\partial \widehat{\rho}_{a,w}(t)}{\partial w_i} = -a_i t e^{-w_i t}. \tag{6.19}$$

The only potentially nonzero second partial derivatives, up to symmetry, are

$$\frac{\partial^2 \widehat{\rho}_{a,w}(t)}{\partial a_i \partial w_i} = -t e^{-w_i t}, \qquad \frac{\partial^2 \widehat{\rho}_{a,w}(t)}{\partial w_i^2} = a_i t^2 e^{-w_i t}. \tag{6.20}$$

For any two parameter coordinates $p, q \in \{a_1, \dots, a_m, w_1, \dots, w_m\}$, the product rule gives

$$\begin{aligned} &\frac{\partial^2}{\partial p \partial q} \left(\widehat{\rho}_{a,w}(t) - \rho_\omega(t)\right)^2 \\ &\quad = 2 \frac{\partial \widehat{\rho}_{a,w}(t)}{\partial p} \frac{\partial \widehat{\rho}_{a,w}(t)}{\partial q} \\ &\qquad + 2 \left(\widehat{\rho}_{a,w}(t) - \rho_\omega(t)\right) \frac{\partial^2 \widehat{\rho}_{a,w}(t)}{\partial p \partial q}. \end{aligned} \tag{6.21}$$

We now find an integrable bound for these second partial derivatives.

Fix a parameter point $(a, w) \in \mathbb{R}^m \times (0,\infty)^m$ and define

$$K := \frac{1}{2} \min_{j \in \{1,\dots,m\}} w_j > 0.$$

As in the proof of the continuity of the gradient, we consider a sufficiently small neighborhood of (a, w). Let (a', w') be any parameter point in this neighborhood, with components a_i' and w_i'. We choose the neighborhood such that

$$|a_i' - a_i| < 1, \qquad |w_i' - w_i| < \frac{w_i}{2}, \qquad i = 1, \dots, m.$$

By the triangle inequality, we get

$$|a_i'| \le |a_i| + 1, \qquad w_i' > \frac{w_i}{2} \ge K.$$

For $t \ge 0$, this gives

$$e^{-w_i' t} \le e^{-Kt}.$$

We set

$$C_1 \coloneqq \sum_{i=1}^{m} (|a_i| + 1).$$

Using (6.19), (6.20) and the bounds on the amplitudes and decay rates, we get

$$\begin{aligned} |\widehat{\rho}_{a',w'}(t)| &\leq C_1 e^{-Kt}, \\ \left| \frac{\partial \widehat{\rho}_{a',w'}(t)}{\partial p} \right| &\leq C_1 (1+t) e^{-Kt}, \\ \left| \frac{\partial^2 \widehat{\rho}_{a',w'}(t)}{\partial p \partial q} \right| &\leq C_1 (t+t^2) e^{-Kt}. \end{aligned}$$

Here, the derivatives are evaluated at (a', w'). The constant C_1 depends only on the fixed parameter point and is independent of t and (a', w') in the chosen neighborhood.

Inserting these bounds into (6.21) gives

$$\begin{aligned} &\left| \frac{\partial^2}{\partial p \partial q} \left(\widehat{\rho}_{a',w'}(t) - \rho_\omega(t) \right)^2 \right| \\ &\quad \leq 2C_1^2 (1+t)^2 e^{-2Kt} \\ &\qquad + 2C_1 (t+t^2) e^{-Kt} \left(C_1 e^{-Kt} + |\rho_\omega(t)| \right) \\ &\quad \leq 8C_1^2 (1+t^2) e^{-Kt} \\ &\qquad + 4C_1 (1+t^2) e^{-Kt} |\rho_\omega(t)| \\ &\quad \leq C (1+t^2) e^{-Kt} \big(1 + |\rho_\omega(t)| \big), \end{aligned} \tag{6.22}$$

where we define

$$C \coloneqq 8C_1^2.$$

The second inequality follows from

$$(1+t)^2 \leq 2(1+t^2), \qquad t + t^2 \leq 2(1+t^2), \qquad e^{-2Kt} \leq e^{-Kt}$$

for $t \geq 0$. The last inequality holds because $C_1 \geq 1$, which implies

$$4C_1 \leq 8C_1^2 = C.$$

Both constants depend only on the fixed parameter point and are independent of t and (a', w') in the chosen neighborhood.

The term without ρ_ω in (6.22) is integrable because $K > 0$. For the remaining term, the Cauchy–Schwarz inequality gives

$$\begin{aligned} &\int_0^\infty (1+t^2) e^{-Kt} |\rho_\omega(t)| \, \mathrm{d}t \\ &\quad \leq \left(\int_0^\infty (1+t^2)^2 e^{-2Kt} \, \mathrm{d}t \right)^{1/2} \|\rho_\omega\|_{L^2([0,\infty))} < \infty. \end{aligned}$$

The first factor is finite because $K > 0$, and the second because $\rho_\omega \in L^2([0,\infty))$.

We have therefore found an integrable bound for all second partial derivatives of the integrand in a neighborhood of each parameter point.

The second partial derivatives of the integrands are continuous in the parameters and satisfy the integrable bounds established above. By the theorems on differentiation

and continuity under the integral sign [Els18, pp. 162–163, Thms. IV.5.6 and IV.5.7], we get

$$J_\omega, \overline{J} \in C^2\big(\mathbb{R}^m \times (0,\infty)^m\big).$$

By (6.11),

$$\overline{J}(\vartheta_0) = 0.$$

Because $\overline{J} \geq 0$, the point ϑ_0 is a global minimizer of $\overline{J}$. Since $\overline{J}$ is twice continuously differentiable,

$$\nabla \overline{J}(\vartheta_0) = 0,$$

and

$$\nabla^2 \overline{J}(\vartheta_0)$$

is positive semidefinite.

The important point is that this minimum is not isolated in general. Because $m > m^*$, the RNN contains more exponential modes than are needed to represent $\bar{\rho}$. For example, the additional modes can have amplitude zero. Changing their decay rates then leaves the RNN kernel unchanged.

More generally, Lemma 30 of Li et al. describes all exact representations of $\bar{\rho}$ [LHEL22, p. 37, Lemma 30]. Several model modes can have the same decay rate and together represent one target mode. Other modes with the same decay rate can cancel each other. Different parameter vectors can therefore represent the same short term kernel.

This redundancy is reflected in the rank of the Hessian. Using the characterization of exact representations, Li et al. establish the bound

$$\operatorname{rank}\big(\nabla^2 \overline{J}(\vartheta_0)\big) \leq m + m^* \tag{6.23}$$

[LHEL22, p. 41, Proof of Prop. 34].

As in Li et al. [LHEL22, p. 41, proof of Prop. 34], we apply the rank-nullity theorem to the Hessian acting on the $2m$-dimensional parameter space. By the rank-nullity theorem [FS25, p. 136, Sec. 3.2.4], we get

$$\begin{aligned}
\dim\ker\big(\nabla^2 \overline{J}(\vartheta_0)\big) &= 2m - \operatorname{rank}\big(\nabla^2 \overline{J}(\vartheta_0)\big) \\
&\geq 2m - (m + m^*) \\
&= m - m^*.
\end{aligned}$$

Hence, there exist at least $m - m^*$ linearly independent vectors $v \in \mathbb{R}^{2m}$ such that

$$\nabla^2 \overline{J}(\vartheta_0) v = 0.$$

For each of these vectors,

$$v^\top \nabla^2 \overline{J}(\vartheta_0) v = 0.$$

Since this expression is the second directional derivative of $\overline{J}$ at ϑ_0 in direction v, the short term loss has at least $m - m^*$ linearly independent directions with zero curvature. These directions need not be aligned with individual model modes.

We now consider how adding the long term component changes the Hessian. Expanding the square in (6.10) gives

$$\begin{aligned} J_\omega(a,w) &= \int_0^\infty \left(\widehat{\rho}_{a,w}(t) - \bar{\rho}(t) - \rho_{0,\omega}(t)\right)^2 \,\mathrm{d}t \\ &= \int_0^\infty \left(\widehat{\rho}_{a,w}(t) - \bar{\rho}(t)\right)^2 \,\mathrm{d}t \\ &\quad - 2\int_0^\infty \left(\widehat{\rho}_{a,w}(t) - \bar{\rho}(t)\right)\rho_{0,\omega}(t)\,\mathrm{d}t \\ &\quad + \int_0^\infty \rho_{0,\omega}(t)^2\,\mathrm{d}t \\ &= \overline{J}(a,w) - 2\int_0^\infty \left(\widehat{\rho}_{a,w}(t) - \bar{\rho}(t)\right)\rho_{0,\omega}(t)\,\mathrm{d}t \\ &\quad + \|\rho_{0,\omega}\|^2_{L^2([0,\infty))}\,. \end{aligned} \tag{6.24}$$

The last term does not depend on (a, w). Differentiating twice therefore leaves only the short term loss and the cross term. This gives the Hessian decomposition

$$\nabla^2 J_\omega(a,w) = \nabla^2 \overline{J}(a,w) + 2E_\omega(a,w) \in \mathbb{R}^{2m\times 2m}. \tag{6.25}$$

as derived by Li et al. [LHEL22, p. 41, Eq. (141)].

Here, E_ω is the $2m \times 2m$ matrix

$$E_\omega(a,w) \coloneqq \begin{pmatrix} 0_{m\times m} & \operatorname{diag}\left(\Delta_{1,\omega}(w_1),\dots,\Delta_{1,\omega}(w_m)\right) \\ \operatorname{diag}\left(\Delta_{1,\omega}(w_1),\dots,\Delta_{1,\omega}(w_m)\right) & -\operatorname{diag}\left(a_1\Delta_{2,\omega}(w_1),\dots,a_m\Delta_{2,\omega}(w_m)\right) \end{pmatrix}$$

[LHEL22, p. 40].

The matrix $\nabla^2\overline{J}$ describes the curvature of the short term loss, while E_ω contains the additional curvature caused by the long term component.

By Lemma 6.2,

$$\lim_{\omega\to 0^+} \Delta_{1,\omega}(w_{0,k}) = 0, \qquad \lim_{\omega\to 0^+} \Delta_{2,\omega}(w_{0,k}) = 0$$

for every fixed $w_{0,k} > 0$. Since m is finite, every entry of $E_\omega(\vartheta_0)$ converges to zero. Consequently,

$$E_\omega(\vartheta_0) \xrightarrow[\omega\to 0^+]{} 0.$$

By (6.25), the Hessian of the full loss therefore approaches the Hessian of the short term loss at ϑ_0.

Li et al. describe the corresponding behavior of the Hessian eigenvalues in the following proposition [LHEL22, pp. 39–42, Prop. 34]. We state this result without proof.

Proposition 6.4 (Flat Hessian Directions [LHEL22, pp. 39–42, Prop. 34])**.** *Let $\vartheta_0 = (a_0, w_0)$ satisfy* (6.11) *and assume $m > m^*$. Denote the eigenvalues of $\nabla^2 J_\omega(\vartheta_0)$ by*

$$\lambda_1(\omega) \geq \lambda_2(\omega) \geq \cdots \geq \lambda_{2m}(\omega).$$

Then there exists an integer $m' \leq m + m^*$ such that

$$\lambda_k(\omega) > 0, \qquad k = 1,\dots,m',$$

for all sufficiently small $\omega > 0$, whereas

$$\lim_{\omega\to 0^+} \lambda_k(\omega) = 0, \qquad k = m'+1,\dots,2m. \tag{6.26}$$

By Proposition 6.4 and $m' \leq m + m^*$, at least $m - m^*$ eigenvalues of $\nabla^2 J_\omega(\vartheta_0)$ converge to zero as $\omega \to 0^+$. In the corresponding eigendirections, the gradient changes only weakly to first order.

Together with Propositions 6.1 and 6.3, this gives the local picture of the plateau mechanism: the loss remains bounded away from zero, while the gradient at ϑ_0 becomes small and at least $m - m^*$ Hessian eigenvalues approach zero.

However, these results only describe the loss and its derivatives at ϑ_0. They do not show whether the gradient remains small along the gradient flow or how long the parameters stay near ϑ_0. In the next subsection, we study the duration of this plateau.

6.4 Exponentially Long Plateaus

We now study the training dynamics. We want to determine how long the gradient flow stays close to the initial parameters before the loss decreases by a fixed amount.

Li et al. additionally assume that the long term kernel component ρ_0 has sub-Gaussian tails,

$$|\rho_0(t)| \leq c_0 e^{-c_1 t^2}, \qquad |t| \geq t_0,$$

for constants $c_0, c_1, t_0 > 0$ [LHEL22, p. 27, Eq. (102)].

Under this assumption, Li et al. establish the bound

$$|\Delta_{n,\omega}(w)| \leq C\omega^{-n} e^{-w/\omega} \left(c_2^{w^2} + c_3^w \right), \qquad n = 0, 1, 2,$$

for fixed $w > 0$ and sufficiently small $\omega > 0$ [LHEL22, p. 34, Lemma 27, Eq. (120)]. Here $C > 0$ is independent of ω, while $c_2, c_3 > 1$ depend only on the constants in the sub-Gaussian tail assumption [LHEL22, p. 34, Lemma 27, Eq. (120)].

Compared with Lemma 6.2, this gives a quantitative decay rate. For fixed $w > 0$, the overlap terms decrease exponentially in the memory length $1/\omega$, up to a polynomial factor.

Following Li et al. [LHEL22, p. 27, Eq. (103)], consider the gradient flow

$$\vartheta_\omega'(\tau) = -\nabla J_\omega(\vartheta_\omega(\tau)), \qquad \vartheta_\omega(0) = \vartheta_0, \tag{6.27}$$

where ϑ_0 satisfies (6.11) and does not depend on ω.

Since $J_\omega \in C^2$, its gradient is locally Lipschitz continuous. By the Picard–Lindelöf theorem, the initial value problem has a unique local solution [Tes12, p. 38, Thm. 2.2]. This solution extends uniquely to a maximal interval [Tes12, p. 51, Thm. 2.13].

We denote the maximal solution by

$$\vartheta_\omega \colon [0, \tau_\omega^*) \to \mathbb{R}^m \times (0, \infty)^m,$$

where $\tau_\omega^* \in (0, \infty]$.

We fix a tolerance

$$0 < \delta < \min \left\{ 1, \min_{1 \leq i \leq m} w_{0,i} \right\},$$

independently of ω. The closed ball of radius δ around ϑ_0 is then a compact subset of $\mathbb{R}^m \times (0, \infty)^m$. By the continuation criterion for maximal solutions, the gradient flow cannot cease to exist in finite time while remaining in this ball [Tes12, p. 53, Cor. 2.16].

Following Li et al. [LHEL22, pp. 27–28, Thm. 20, Eqs. (104)–(105)], we define the parameter and loss escape times for this fixed tolerance δ by

$$\tau_0 \coloneqq \inf\left\{\tau \geq 0 : \|\vartheta_\omega(\tau) - \vartheta_0\|_2 > \delta\right\},$$

and

$$\tau_0' \coloneqq \inf\left\{\tau \geq 0 : |J_\omega(\vartheta_\omega(\tau)) - J_\omega(\vartheta_0)| > \delta\right\}.$$

Both infima are taken over the maximal interval of existence of the gradient flow, with the convention $\inf \emptyset = +\infty$.

The quantity τ_0 measures how long the parameters stay close to ϑ_0, while τ_0' measures how long the loss stays close to its initial value. We state the following result of Li et al. for the special case in which the initialization represents the short term component exactly [LHEL22, pp. 27–28, Thm. 20].

We follow the formal asymptotic expansion and local linearization used by Li et al. [LHEL22, pp. 42–44, proof of Thm. 20, Eqs. (145)–(149)]. The calculations below illustrate their argument, but do not provide an independent proof of the theorem.

Theorem 6.5 (Exponential Timescale of Plateauing [LHEL22, pp. 27–28, Thm. 20])**.** *Let $m > m^*$ and let $\vartheta_0 = (a_0, w_0)$ satisfy* (6.11)*.*

Then

$$\lim_{\omega \to 0^+} \tau_0 = \lim_{\omega \to 0^+} \tau_0' = \infty.$$

If, in addition, ρ_0 has sub-Gaussian tails, the fixed tolerance δ is sufficiently small, and the initialization satisfies

$$a_{0,i} \in [a_l^0, a_r^0], \qquad w_{0,i} \in [w_l^0, w_r^0], \qquad a_l^0 < a_r^0, \quad 0 < w_l^0 < w_r^0,$$

then there exists a constant $C > 0$, depending only on ρ_0, the constants in the sub-Gaussian tail assumption, and a_l^0, a_r^0, w_r^0, such that for sufficiently small $\omega > 0$,

$$\tau_0' \geq \tau_0 \geq C\omega^2 e^{w_l^0/\omega} \min\left\{\frac{\delta}{\sqrt{m}}, \ln(1+\delta)\right\}.$$

We now explain the main argument behind the estimate of the parameter escape time. By the definition of τ_0,

$$\|\vartheta_\omega(\tau) - \vartheta_0\|_2 \leq \delta, \qquad 0 \leq \tau < \tau_0.$$

During this time, the parameters stay close to ϑ_0. Since $J_\omega \in C^2$, a first order Taylor expansion of the gradient around ϑ_0 gives

$$\nabla J_\omega(\vartheta_\omega(\tau)) = \nabla J_\omega(\vartheta_0) + \nabla^2 J_\omega(\vartheta_0)(\vartheta_\omega(\tau) - \vartheta_0) + o(\|\vartheta_\omega(\tau) - \vartheta_0\|_2).$$

as $\vartheta_\omega(\tau) \to \vartheta_0$, for fixed ω.

For fixed ω, this expansion describes the local behavior near ϑ_0. We consider the displacement $z(\tau) \in \mathbb{R}^{2m}$ from the initial point,

$$z(\tau) \coloneqq \vartheta_\omega(\tau) - \vartheta_0.$$

Since $\vartheta_\omega(0) = \vartheta_0$, we have $z(0) = 0$.

Inserting the Taylor expansion into the gradient flow equation and neglecting the remainder gives a linearized system. We denote its solution by $\widetilde{z}(\tau) \in \mathbb{R}^{2m}$, which approximates the displacement $z(\tau)$:

$$\widetilde{z}'(\tau) = -\nabla J_\omega(\vartheta_0) - \nabla^2 J_\omega(\vartheta_0)\widetilde{z}(\tau), \qquad \widetilde{z}(0) = 0. \tag{6.28}$$

This is a linear differential equation with constant coefficients. Rearranging gives

$$\widetilde{z}'(\tau) + \nabla^2 J_\omega(\vartheta_0)\widetilde{z}(\tau) = -\nabla J_\omega(\vartheta_0).$$

Multiplying by $e^{\nabla^2 J_\omega(\vartheta_0)\tau}$ and applying the product rule gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\tau}\left[e^{\nabla^2 J_\omega(\vartheta_0)\tau}\widetilde{z}(\tau)\right] \\ = -e^{\nabla^2 J_\omega(\vartheta_0)\tau}\nabla J_\omega(\vartheta_0).\end{aligned}$$

Integrating from 0 to τ and using $\widetilde{z}(0) = 0$, we get

$$\widetilde{z}(\tau) = -\left(\int_0^\tau e^{-\nabla^2 J_\omega(\vartheta_0)s}\,\mathrm{d}s\right)\nabla J_\omega(\vartheta_0). \tag{6.29}$$

This equation describes the movement in the linearized system. The initial gradient determines the initial velocity, while the Hessian determines how this movement changes locally.

Since $\nabla^2 J_\omega(\vartheta_0)$ is symmetric, the spectral theorem [FS25, pp. 363–364, Sec. 6.7.2] gives an orthogonal matrix $P \in \mathbb{R}^{2m\times 2m}$ such that

$$\nabla^2 J_\omega(\vartheta_0) = P^\top \operatorname{diag}\left(\lambda_1(\omega), \ldots, \lambda_{2m}(\omega)\right) P.$$

Taking the Euclidean norm in (6.29) and using the induced matrix norm gives

$$\|\widetilde{z}(\tau)\|_2 \le \left\|\int_0^\tau e^{-\nabla^2 J_\omega(\vartheta_0)s}\,\mathrm{d}s\right\|_2 \|\nabla J_\omega(\vartheta_0)\|_2 .$$

Because the Hessian is orthogonally diagonalizable and the spectral norm is invariant under orthogonal transformations, we get

$$\begin{aligned}&\left\|\int_0^\tau e^{-\nabla^2 J_\omega(\vartheta_0)s}\,\mathrm{d}s\right\|_2 \\ &= \left\|P^\top \operatorname{diag}\left(\int_0^\tau e^{-\lambda_1(\omega)s}\,\mathrm{d}s, \ldots, \int_0^\tau e^{-\lambda_{2m}(\omega)s}\,\mathrm{d}s\right) P\right\|_2 \\ &= \max_{k=1,\ldots,2m}\int_0^\tau e^{-\lambda_k(\omega)s}\,\mathrm{d}s.\end{aligned}$$

Therefore,

$$\begin{aligned}\|\widetilde{z}(\tau)\|_2 \le \|\nabla J_\omega(\vartheta_0)\|_2 \\ \cdot \max_{k=1,\ldots,2m}\int_0^\tau e^{-\lambda_k(\omega)s}\,\mathrm{d}s.\end{aligned}$$

For fixed τ, the integral decreases as $\lambda_k(\omega)$ increases. Since the eigenvalues are ordered as in Proposition 6.4, the maximum is attained at the smallest eigenvalue $\lambda_{2m}(\omega)$. We get

$$\|\widetilde{z}(\tau)\|_2 \le \|\nabla J_\omega(\vartheta_0)\|_2 \int_0^\tau e^{-\lambda_{2m}(\omega)s}\,\mathrm{d}s. \tag{6.30}$$

This is the central estimate for the linearized dynamics. It shows that the displacement depends on the initial gradient and the smallest Hessian eigenvalue.

We distinguish two cases.

If

$$\lambda_{2m}(\omega) \geq 0,$$

then all Hessian eigenvalues are nonnegative. In particular,

$$e^{-\lambda_{2m}(\omega)s} \leq 1, \qquad s \geq 0.$$

By (6.30),

$$\|\widetilde{z}(\tau)\|_2 \leq \tau \, \|\nabla J_\omega(\vartheta_0)\|_2 .$$

If the linearized solution reaches a distance δ and the initial gradient is nonzero, we therefore need

$$\tau \geq \frac{\delta}{\|\nabla J_\omega(\vartheta_0)\|_2}.$$

A small initial gradient therefore implies a long escape time in the linearized system.

If

$$\lambda_{2m}(\omega) < 0,$$

then

$$e^{-\lambda_{2m}(\omega)s} = e^{|\lambda_{2m}(\omega)|s}.$$

This exponential factor can amplify the movement in the corresponding eigendirection. From (6.30), we get

$$\begin{aligned} \|\widetilde{z}(\tau)\|_2 &\leq \|\nabla J_\omega(\vartheta_0)\|_2 \int_0^\tau e^{|\lambda_{2m}(\omega)|s} \, \mathrm{d}s \\ &= \|\nabla J_\omega(\vartheta_0)\|_2 \, \frac{e^{|\lambda_{2m}(\omega)|\tau} - 1}{|\lambda_{2m}(\omega)|}. \end{aligned}$$

If the linearized solution reaches a distance δ and the initial gradient is nonzero, we need

$$\delta \leq \|\nabla J_\omega(\vartheta_0)\|_2 \, \frac{e^{|\lambda_{2m}(\omega)|\tau} - 1}{|\lambda_{2m}(\omega)|}.$$

Solving for τ gives

$$\tau \geq \frac{1}{|\lambda_{2m}(\omega)|} \ln\left(1 + \frac{\delta|\lambda_{2m}(\omega)|}{\|\nabla J_\omega(\vartheta_0)\|_2}\right).$$

Since $0 < \delta < 1$, we can simplify this bound. If

$$|\lambda_{2m}(\omega)| \leq \|\nabla J_\omega(\vartheta_0)\|_2,$$

then $\ln(1+x) \geq x/2$ for $0 \leq x \leq 1$ gives

$$\tau \geq \frac{\delta}{2\|\nabla J_\omega(\vartheta_0)\|_2}.$$

Otherwise, the monotonicity of the logarithm gives

$$\tau \geq \frac{\ln(1+\delta)}{|\lambda_{2m}(\omega)|}.$$

Combining both cases with the case $\lambda_{2m}(\omega) \geq 0$, we get

$$\tau \geq \frac{1}{2} \min \left\{ \frac{\delta}{\|\nabla J_\omega(\vartheta_0)\|_2}, \frac{\ln(1+\delta)}{|\lambda_{2m}(\omega)|} \right\}$$

whenever the linearized solution reaches a distance δ from the initial point. A fraction with zero denominator is interpreted as $+\infty$.

If the initial gradient is zero, the solution of the linearized system stays at the initial point.

The estimates above only apply to the linearized system. Li et al. use a formal asymptotic expansion to extend the argument to the nonlinear gradient flow [LHEL22, pp. 42–43, proof of Thm. 20]. This gives

$$\tau_0 \geq C \min \left\{ \frac{\delta}{\|\nabla J_\omega(\vartheta_0)\|_2}, \frac{\ln(1+\delta)}{|\lambda_{2m}(\omega)|} \right\}, \tag{6.31}$$

where $C > 0$ is independent of ω. We use this estimate below, but do not prove that the neglected terms remain small over the escape time.

By Proposition 6.3,

$$\|\nabla J_\omega(\vartheta_0)\|_2 \xrightarrow[\omega \to 0^+]{} 0.$$

Since $m^* < m$, we have

$$m' \leq m + m^* < 2m.$$

Proposition 6.4 therefore gives

$$\lambda_{2m}(\omega) \xrightarrow[\omega \to 0^+]{} 0.$$

Both quantities in the denominators of (6.31) therefore approach zero. The estimate taken from Li et al. then gives the divergence of the parameter escape time stated in Theorem 6.5 [LHEL22, pp. 42–43, proof of Thm. 20].

For a quantitative bound, Li et al. use the sub-Gaussian tail assumption introduced above [LHEL22, p. 27, Eq. (102)]. Li et al. combine the quantitative overlap estimate with the gradient and Hessian bounds from their Lemma 27 and Propositions 33 and 34 [LHEL22, pp. 34–44]. This gives

$$\tau_0 \geq C\omega^2 e^{w_l^0/\omega} \min \left\{ \frac{\delta}{\sqrt{m}}, \ln(1+\delta) \right\} \tag{6.32}$$

under the bounded initialization from Theorem 6.5 [LHEL22, p. 44, Eq. (152)].

Since $w_l^0 > 0$, the exponential term dominates the polynomial factor, so

$$\lim_{\omega \to 0^+} \omega^2 e^{w_l^0/\omega} = \infty.$$

For fixed m and $\delta > 0$ satisfying the assumptions of Theorem 6.5, the lower bound implies that $\tau_0 \to \infty$ as $\omega \to 0^+$.

Finally, we relate the parameter escape time to the loss escape time. Li et al. use a second order Taylor expansion of the loss around ϑ_0 and estimate its linear and quadratic terms using the initial gradient and the smallest Hessian eigenvalue [LHEL22, p. 44, Proof of Thm. 20].

We do not reproduce the technical argument. Under the additional assumptions of the quantitative part of Theorem 6.5, we use the comparison established by Li et al. [LHEL22, p. 44, Proof of Thm. 20],

$$\tau_0' \geq \tau_0.$$

The loss therefore stays near its initial value for at least as long as the parameters remain near ϑ_0. Combined with (6.32), this gives the exponentially long loss plateau stated in Theorem 6.5.

For the reduced model considered here, a significant long term error can remain even when the short term component is represented exactly. The results of Li et al. show that the loss can remain close to its initial value over a timescale with an exponential lower bound [LHEL22, pp. 27–28, Thm. 20]. Li et al. refer to this optimization difficulty as the *curse of memory* [LHEL22, p. 27].

Li et al. relate this optimization difficulty to the problem of vanishing gradients in discrete RNNs [LHEL22, p. 21]. As discussed in Section 2.2.4, products of Jacobians can suppress gradient contributions from distant time steps.

In the present continuous time setting, a related effect appears through the weights $e^{-w_k t}$ and $te^{-w_k t}$ in the gradient formulas.

6.5 Numerical Illustration

We now illustrate this mechanism numerically using the target family introduced in Section 6.3. The experiment follows the numerical investigation of Li et al., who study plateau and parameter escape times for linear continuous time RNNs [LHEL22, p. 31, Sec. 6.2.2, Fig. 3]. We specify the parameters and initialization explicitly and use our own numerical implementation.

We choose

$$\bar{\rho}(t) = e^{-t/2}, \qquad \rho_0(t) = e^{-2t^2},$$

so that

$$\rho_\omega(t) = e^{-t/2} + e^{-2(t-1/\omega)^2}.$$

The first term is the short term component. The Gaussian long term component is centered at the memory lag

$$L := \frac{1}{\omega}.$$

We use

$$L \in \{8, 10, 12, 14, 16, 18\}$$

and set the model width to $m = 10$. The corresponding RNN kernel is

$$\widehat{\rho}_\vartheta(t) = \sum_{i=1}^{10} a_i e^{-w_i t}, \qquad \vartheta = (a, w) \in \mathbb{R}^{10} \times (0, \infty)^{10}.$$

As in the theoretical analysis, we choose the initialization such that the short term component is represented exactly,

$$\widehat{\rho}_{\vartheta_0}(t) = \bar{\rho}(t) = e^{-t/2}.$$

The first exponential mode has amplitude 1, while all remaining amplitudes are initialized at zero. The initial decay rates are geometrically spaced between 1/2

and 4, with $w_{0,1} = 1/2$. The initialization satisfies the boundedness conditions in Theorem 6.5 with

$$a_l^0 = 0, \qquad a_r^0 = 1, \qquad w_l^0 = \frac{1}{2}, \qquad w_r^0 = 4.$$

The target family and initialization satisfy the structural assumptions of Theorem 6.5. The short term component consists of one exponential, so $m^* = 1 < m = 10$, and the initialization represents this component exactly.

Moreover, ρ_0 belongs to $L^2(\mathbb{R}) \cap L^\infty(\mathbb{R}) \cap C^2(\mathbb{R})$, satisfies $\rho_0(t) \to 0$ as $t \to -\infty$, and has sub-Gaussian tails with $c_0 = 1$ and $c_1 = 2$.

Since the short term component is represented exactly, the initial error comes only from the shifted Gaussian component. This allows us to study how moving the remaining error farther into the past affects optimization.

Figure 5 shows the target kernels for three memory scales. Increasing $L = 1/\omega$ only shifts the long term component to larger memory lags. Its shape and amplitude stay unchanged.

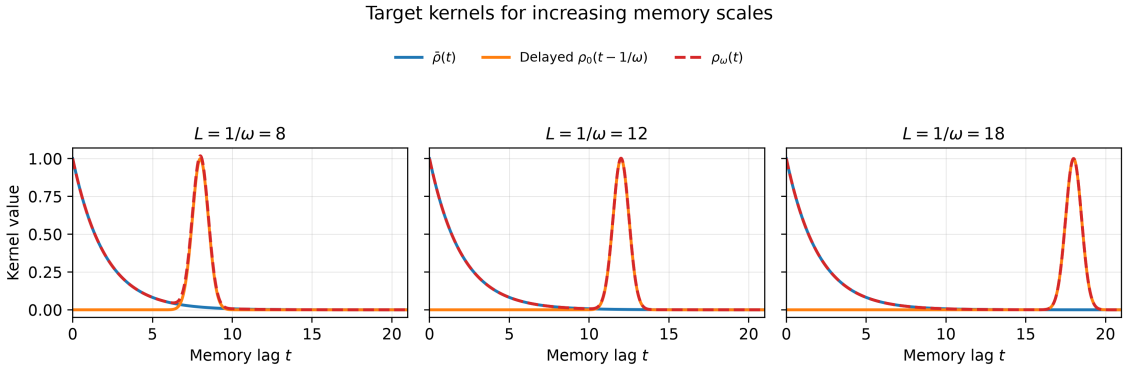


Figure 5: Target kernels for the memory scales $L = 1/\omega \in \{8, 12, 18\}$. The short term component stays fixed, while the Gaussian long term component is shifted to larger memory lags.

We numerically integrate the gradient flow for all six memory scales. For both hitting times, we use the same tolerance

$$\delta = 0.05.$$

This satisfies

$$0 < \delta < \min\{1, \min_i w_{0,i}\}.$$

The numerical loss hitting time is determined from the loss decrease, which agrees with the definition of τ_0' because the loss is non increasing along the gradient flow by (6.7). Table 2 shows the initial loss, the initial gradient norm, and the corresponding hitting times.

Because the initialization represents the short term component exactly, substituting $u = 2(t - L)$ gives

$$\begin{aligned} J_\omega(\vartheta_0) &= \int_0^\infty e^{-4(t-L)^2}\,\mathrm{d}t \\ &= \frac{1}{2}\int_{-2L}^\infty e^{-u^2}\,\mathrm{d}u \xrightarrow[L\to\infty]{} \frac{\sqrt{\pi}}{2} \approx 0.8862. \end{aligned}$$

$L = 1/\omega$	$J_\omega(\vartheta_0)$	$\|\nabla J_\omega(\vartheta_0)\|_2$	τ_0	τ_0'
8	0.8862	0.3764	0.1451	0.5602
10	0.8862	0.1730	0.4091	4.4856
12	0.8862	0.0764	1.6329	17.4300
14	0.8862	0.0328	5.9541	63.3961
16	0.8862	0.0138	22.7790	233.1916
18	0.8862	0.0057	72.3267	881.7643

Table 2: Initial loss, initial gradient norm, and first hitting times for increasing memory scale $L = 1/\omega$.

This limit equals $\|\rho_0\|^2_{L^2(\mathbb{R})}$, as shown in (6.13). For the investigated memory scales, the initial loss is already close to this limit. At the same time,

$$\|\nabla J_\omega(\vartheta_0)\|_2$$

decreases from approximately 0.3764 for $L = 8$ to 0.0057 for $L = 18$. This illustrates the decrease in the gradient norm described in Proposition 6.3.

The last two columns of Table 2 show that both the parameter and loss hitting times increase strongly with the memory scale. Figure 6 shows the corresponding gradient flow trajectories for three values of L. The left panel shows the parameter displacement

$$\|\vartheta_\omega(\tau) - \vartheta_0\|_2 ,$$

and the right panel shows the loss decrease

$$J_\omega(\vartheta_0) - J_\omega\big(\vartheta_\omega(\tau)\big).$$

The dashed horizontal lines show the thresholds used to define τ_0 and τ_0'.

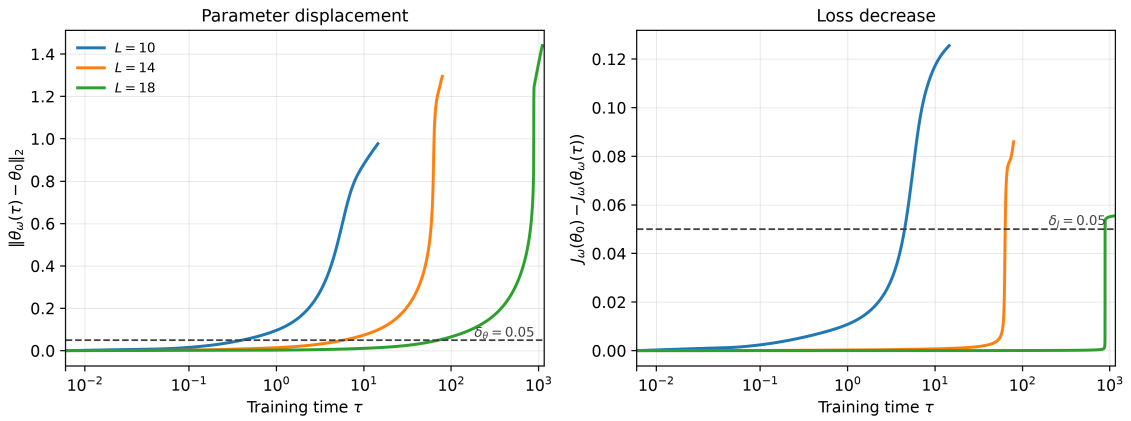


Figure 6: Gradient flow trajectories for $L = 1/\omega \in \{10, 14, 18\}$. The parameter displacement and loss decrease stay below their thresholds for longer training times when the memory scale increases.

Figure 7 compares the learned kernels with their corresponding targets. For each memory scale, we continue the gradient flow beyond the loss hitting time. The final models are therefore evaluated at different training times.

The short term component remains approximately represented, while the shifted Gaussian component is still poorly approximated. This illustrates the difficulty of

learning long term dependencies even after the loss has decreased by the chosen threshold.

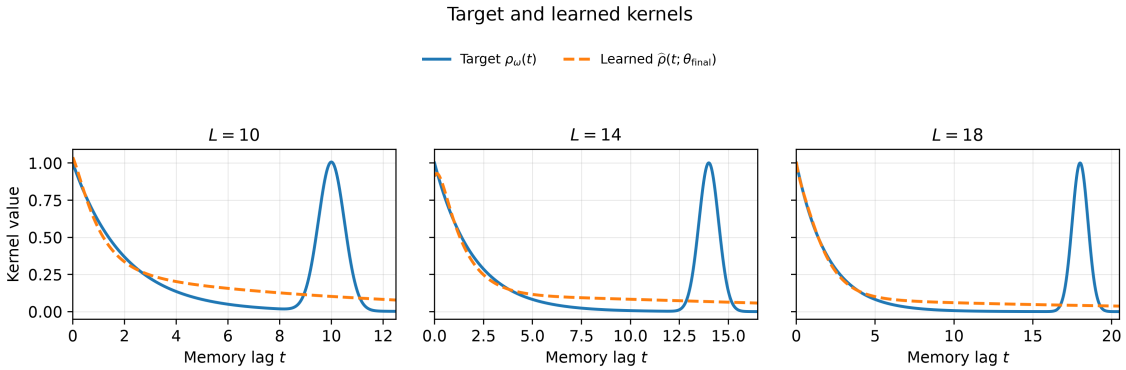


Figure 7: Target kernels ρ_ω and learned kernels $\widehat{\rho}_{\vartheta_{\text{final}}}$ for $L = 1/\omega \in \{10, 14, 18\}$. The final training times are approximately 14.49, 79.25, and 1102.21, respectively. The final parameters are not assumed to be global minimizers.

The results show that increasing the memory scale from $L = 8$ to $L = 18$ increases the parameter hitting time from approximately 0.15 to 72.33 and the loss hitting time from 0.56 to 881.76. The initial loss remains almost unchanged, while the optimization becomes substantially slower. This illustrates the increasing plateau duration described in Theorem 6.5.

For the numerical computation, we evaluate the loss and its gradient using closed form expressions and solve the gradient flow with the adaptive Runge–Kutta method DOP853 [HNW93, Sec. II.5, pp. 181–185] implemented in SciPy's `solve_ivp` [VGO+20]. Event detection gives the parameter and loss hitting times. Further details are given in Appendix 8.2.

The experiment illustrates the theoretical result for finitely many memory scales. The observed increase in the hitting times does not establish an asymptotic growth rate. Theorem 6.5 provides an asymptotic lower bound, not an exact expression for the observed hitting times.

The code for the experiment is available in `notebooks/08_theorem20_plateauing.ipynb` of the accompanying GitHub repository.

7 Conclusion and Outlook

The results of Li et al. show that linear continuous time RNNs already have difficulties with long term memory [LHEL22]. Nonlinear recurrent dynamics and discrete time are not necessary for these difficulties. In approximation, slowly decaying target memory can require a larger RNN width for a fixed approximation accuracy [LHEL22, Sec. 4.3, pp. 13–14]. In optimization, gradient flow can stay on a plateau for a time whose lower bound grows exponentially with the memory length [LHEL22, pp. 27–28, Thm. 20].

The connection between these two mechanisms comes from the considered hypothesis class $\widehat{\mathcal{H}}$. For the time homogeneous target families $\{H_t\}_{t\in\mathbb{R}}$, the input output relation is described by a fixed memory kernel ρ that only depends on the memory lag. To define the RNN representation on the infinite past, we assume that the recurrent matrix is stable, meaning

$$W \in \mathcal{W}_m, \qquad \max_{\lambda\in\sigma(W)} \operatorname{Re}\lambda < 0,$$

as in Proposition 3.8. The stability condition on W restricts which kernels the model can represent. The corresponding RNN kernel

$$\widehat{\rho}(t) = U^\top e^{W^\top t} c$$

decays for large memory lags. In the diagonal setting used in the approximation construction and in the optimization analysis, this gives

$$\widehat{\rho}(t) = \sum_{i=1}^{m} a_i e^{-w_i t}, \qquad w_i > 0.$$

The same exponential family has two different roles. In approximation, finite sums of $e^{-w_i t}$ are used to approximate the target kernel ρ. In optimization, the same exponential functions appear in the overlap terms

$$\Delta_{n,\omega}(w) = \int_0^\infty t^n e^{-wt} \rho_0\left(t - \frac{1}{\omega}\right) \mathrm{d}t,$$

which enter the gradient and Hessian. Because $w > 0$, these functions become small at large memory lags. Slowly decaying target memory can require a larger model for an accurate approximation. At the same time, an error at a large memory lag has only a small overlap with the functions that determine the optimization dynamics. The same exponential structure therefore creates one difficulty in approximation complexity and another in optimization time.

One of our own results deals with the inverse approximation theorem of Li et al. [LHEL22, pp. 12–13, Thm. 11]. We construct a counterexample (Proposition 5.12) showing that the theorem does not hold for general nonnormal recurrent matrices. We then prove a modified version of the theorem (Theorem 5.14) under the additional assumption that the recurrent matrices are normal. Under this assumption, stable approximation implies exponential decay of the target memory, provided the other assumptions of the theorem hold. We also give detailed proofs of several results from Li et al. and fill in steps that are only briefly explained in their paper [LHEL22].

Similar restrictions on memory decay also appear for nonlinear RNNs. Wang et al. show that, under assumptions on stable approximation, activation functions, uniform memory decay of the perturbed models, and uniformly bounded weights, the memory function of the target decays exponentially [WLL24, p. 7, Thm. 3.9].

Changing the optimization method can improve training without changing the approximating kernel class [cf. LHEL22, p. 32, Fig. 4, and pp. 78–80, App. B]. For RNNs with full recurrent matrices and tanh activation, Li et al. observe numerically that Adam reduces plateauing to some extent, while the separation of timescales caused by long memory remains [LHEL22, p. 32, Fig. 4]. They also illustrate faster escape with momentum in a quadratic example [LHEL22, pp. 78–80, App. B]. The escape time is shorter, but it remains long for flat plateaus. A different optimizer can therefore reduce optimization difficulties, but it does not change which kernels the model can represent.

The results of Li et al. show that long term memory can be difficult to approximate and difficult to learn efficiently within the hypothesis class $\widehat{\mathcal{H}}$ [LHEL22]. One possible way to overcome these difficulties is to leave this hypothesis class. Mamba is a modern recurrent architecture in which the parameters of the recurrent dynamics depend on the input [GD24, Sec. 3.2]. In contrast, each model in $\widehat{\mathcal{H}}$ uses fixed parameters W, U, and c and has one fixed memory kernel. Gu and Dao call such models linear time invariant and argue that this restricts their ability to select information from the input [GD24, Sec. 3.1].

In Mamba, quantities that play the role of U and c, as well as the matrix used to update the hidden state, depend on the input [GD24, Sec. 3.2]. The model is therefore no longer linear in the input and can remember or ignore information depending on the input [GD24, Sec. 3.1]. Mamba shows strong empirical results on tasks that require long term memory [GD24, Sec. 4.1]. It therefore illustrates a possible way to overcome some of the limitations of the hypothesis class $\widehat{\mathcal{H}}$. However, the representation by one fixed memory kernel is lost, so the mathematical analysis used in this thesis does not directly apply to such models.

8 Appendix

8.1 Detailed Derivation of the BPTT Gradients

This subsection gives the detailed derivation of the gradients stated in Proposition 2.2. We follow the unrolled computational graph in Goodfellow et al. [GBC16, Sec. 10.2.2, pp. 378–380], the BPTT derivation in Salem [Sal22, Sec. 3.1.1, pp. 45–47], and the earlier BPTT formulation by Werbos [Wer90, Sec. III, pp. 1554–1555].

Our own contribution is the detailed derivation below. We define the remaining loss recursively and calculate the parameter gradients component by component.

Proof of Proposition 2.2. For $k = 1, \dots, T$, let

$$\theta^{(k)} = \big(W^{(k)}, U^{(k)}, b^{(k)}, c^{(k)}\big) \in \Theta$$

be an independent copy of the RNN parameters, where

$$W^{(k)} \in \mathbb{R}^{m\times m}, \qquad U^{(k)} \in \mathbb{R}^{m\times d}, \qquad b^{(k)}, c^{(k)} \in \mathbb{R}^m.$$

Starting from the fixed initial state $h_0 \in \mathbb{R}^m$, define

$$a_k \coloneqq W^{(k)} h_{k-1} + U^{(k)} x_k + b^{(k)} \in \mathbb{R}^m,$$

$$h_k = \sigma(a_k) \in \mathbb{R}^m, \qquad \widehat{y}_k = (c^{(k)})^\top h_k \in \mathbb{R},$$

and

$$D_k \coloneqq \operatorname{diag}\big(\sigma'((a_k)_1), \dots, \sigma'((a_k)_m)\big) \in \mathbb{R}^{m\times m}.$$

The matrix D_k contains the derivatives of the componentwise activation function evaluated at a_k.

The loss of the unrolled network is

$$\widetilde{\mathcal{L}} : \Theta^T \to [0, \infty),$$

defined by

$$\widetilde{\mathcal{L}}\big(\theta^{(1)}, \dots, \theta^{(T)}\big) \coloneqq \frac{1}{T} \sum_{k=1}^{T} \big((c^{(k)})^\top h_k - y_k\big)^2.$$

The original RNN is obtained by using the same parameter tuple at every time step. Define

$$\iota : \Theta \to \Theta^T, \qquad \iota(\theta) \coloneqq (\theta, \dots, \theta).$$

Then

$$\mathcal{L} = \widetilde{\mathcal{L}} \circ \iota : \Theta \to [0, \infty).$$

The parameter copies separate the different occurrences of a shared parameter in the unrolled computational graph. Goodfellow et al. use the same idea by introducing separate copies of W for each time step [GBC16, p. 380].

For fixed parameter copies, define

$$\widetilde{\mathcal{L}}_k : \mathbb{R}^m \to [0, \infty), \qquad \widetilde{\mathcal{L}}_k(z) \coloneqq \frac{1}{T}\big((c^{(k)})^\top z - y_k\big)^2.$$

To derive the backward recursion, we write the loss from time step k to T as a function of the hidden state at time k. We use the idea of the *ordered derivative* from

Werbos, which describes both the direct and indirect effects of a variable on the loss [Wer90, p. 1552, Eq. (7), and p. 1553, Eq. (8)].

Set

$$\Phi_T(z) \coloneqq \widetilde{\mathcal{L}}_T(z),$$

and, for $k < T$,

$$\Phi_k(z) \coloneqq \widetilde{\mathcal{L}}_k(z) + \Phi_{k+1}\left(\sigma\left(W^{(k+1)}z + U^{(k+1)}x_{k+1} + b^{(k+1)}\right)\right).$$

Starting with Φ_T, these functions are well defined recursively for $k = T, T-1, \ldots, 1$. The function $\Phi_k : \mathbb{R}^m \to [0, \infty)$ represents the loss accumulated from time step k onwards when the hidden state at time k is z.

Along the hidden states generated by the forward pass,

$$\widetilde{\mathcal{L}} = \sum_{j=1}^{k-1} \widetilde{\mathcal{L}}_j(h_j) + \Phi_k(h_k).$$

The sum over $j < k$ and the function Φ_k do not depend on $W^{(k)}$, $U^{(k)}$, or $b^{(k)}$. These parameter copies enter $\widetilde{\mathcal{L}}$ only through $h_k = \sigma(a_k)$.

We define

$$\nabla_{h_k}\widetilde{\mathcal{L}} \coloneqq \nabla\Phi_k(h_k) \in \mathbb{R}^m,$$

and

$$\frac{\partial\widetilde{\mathcal{L}}}{\partial(h_k)_i}$$

denotes its i-th coordinate.

Because $\sigma \in C^1(\mathbb{R})$, its componentwise extension is continuously differentiable. The remaining operations are additions, matrix vector products, inner products, and squaring. These operations are continuously differentiable. Finite sums and compositions preserve this property, so the gradients and Jacobians below exist.

The direct contribution of time step k satisfies

$$\nabla\widetilde{\mathcal{L}}_k(h_k) = \frac{2}{T}\big((c^{(k)})^\top h_k - y_k\big)c^{(k)} \in \mathbb{R}^m.$$

At the final time step,

$$\nabla_{h_T}\widetilde{\mathcal{L}} = \frac{2}{T}\big((c^{(T)})^\top h_T - y_T\big)c^{(T)} \in \mathbb{R}^m.$$

For $k < T$, the hidden state h_k contributes directly to the loss at time k and indirectly to all later loss terms through h_{k+1}. The multivariable chain rule gives the backward recursion [cf. GBC16, p. 379, Eqs. (10.19)–(10.21); cf. Wer90, Sec. III.C, p. 1555, Eq. (16)]

$$\nabla_{h_k}\widetilde{\mathcal{L}} = \nabla\widetilde{\mathcal{L}}_k(h_k) + \left(\frac{\partial h_{k+1}}{\partial h_k}\right)^\top \nabla_{h_{k+1}}\widetilde{\mathcal{L}}.$$

Since

$$h_{k+1} = \sigma\left(W^{(k+1)}h_k + U^{(k+1)}x_{k+1} + b^{(k+1)}\right),$$

the Jacobian is

$$\frac{\partial h_{k+1}}{\partial h_k} = D_{k+1}W^{(k+1)} \in \mathbb{R}^{m\times m}.$$

This gives

$$\nabla_{h_k}\widetilde{\mathcal{L}} = \frac{2}{T}\big((c^{(k)})^\top h_k - y_k\big)c^{(k)} + (W^{(k+1)})^\top D_{k+1}\nabla_{h_{k+1}}\widetilde{\mathcal{L}},$$

for $k = T-1, \dots, 1$.

Now set

$$\theta^{(1)} = \cdots = \theta^{(T)} = \theta.$$

Then a_k, h_k, $\widehat{y}_k$, and D_k coincide with the corresponding quantities of the original RNN in Section 2.2.2. We define

$$\nabla_{h_k}\mathcal{L}(\theta) \coloneqq \nabla_{h_k}\widetilde{\mathcal{L}}\Big|_{\theta^{(1)}=\cdots=\theta^{(T)}=\theta} \in \mathbb{R}^m,$$

with

$$\frac{\partial \mathcal{L}(\theta)}{\partial (h_k)_i}$$

denoting its i-th coordinate. The formulas above give the terminal value and the hidden state recursion stated in Proposition 2.2.

It remains to calculate the derivatives with respect to the shared parameters. For a scalar valued function of a matrix parameter, we define the gradient as the matrix of its scalar partial derivatives. In particular,

$$\nabla_W \mathcal{L}(\theta) \coloneqq \left(\frac{\partial \mathcal{L}(\theta)}{\partial W_{ij}}\right)_{i,j=1}^{m} \in \mathbb{R}^{m\times m}.$$

Gradients with respect to U, b, and c are defined analogously.

We show the componentwise calculation in detail for W. Fix $k \in \{1, \dots, T\}$ and an entry $W_{ij}^{(k)} \in \mathbb{R}$. The i-th coordinate of a_k is

$$(a_k)_i = \sum_{r=1}^{m} W_{ir}^{(k)}(h_{k-1})_r + \sum_{r=1}^{d} U_{ir}^{(k)}(x_k)_r + b_i^{(k)}.$$

The hidden state h_{k-1} only depends on parameter copies from earlier time steps, so it does not depend on $W^{(k)}$. We get

$$\frac{\partial (a_k)_i}{\partial W_{ij}^{(k)}} = (h_{k-1})_j.$$

The multivariable chain rule gives

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial W_{ij}^{(k)}} = \sum_{p=1}^{m} \frac{\partial \widetilde{\mathcal{L}}}{\partial (h_k)_p}\frac{\partial (h_k)_p}{\partial W_{ij}^{(k)}}.$$

Only the i-th coordinate of a_k, and therefore only the i-th coordinate of h_k, depends on $W_{ij}^{(k)}$. For $p \neq i$,

$$\frac{\partial (h_k)_p}{\partial W_{ij}^{(k)}} = 0,$$

while for $p = i$,

$$\frac{\partial (h_k)_i}{\partial W_{ij}^{(k)}} = \sigma'((a_k)_i)(h_{k-1})_j.$$

It follows that

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial W_{ij}^{(k)}} = \frac{\partial \widetilde{\mathcal{L}}}{\partial (h_k)_i} \sigma'((a_k)_i)(h_{k-1})_j.$$

The shared entry W_{ij} occurs at every time step. Changing W_{ij} changes all parameter copies $W_{ij}^{(1)}, \ldots, W_{ij}^{(T)}$, so their contributions are added. From $\mathcal{L} = \widetilde{\mathcal{L}} \circ \iota$,

$$\frac{\partial \mathcal{L}(\theta)}{\partial W_{ij}} = \sum_{k=1}^{T} \frac{\partial \widetilde{\mathcal{L}}}{\partial W_{ij}^{(k)}}.$$

After restricting the parameter copies,

$$\frac{\partial \mathcal{L}(\theta)}{\partial W_{ij}} = \sum_{k=1}^{T} \frac{\partial \mathcal{L}(\theta)}{\partial (h_k)_i} \sigma'((a_k)_i)(h_{k-1})_j, \qquad i, j = 1, \ldots, m.$$

For each time step k,

$$D_k \nabla_{h_k} \mathcal{L}(\theta) \in \mathbb{R}^m.$$

Because D_k is diagonal, its i-th coordinate is

$$\big(D_k \nabla_{h_k} \mathcal{L}(\theta)\big)_i = \sigma'((a_k)_i) \frac{\partial \mathcal{L}(\theta)}{\partial (h_k)_i}.$$

Taking the outer product with $h_{k-1}^\top \in \mathbb{R}^{1 \times m}$ gives

$$D_k \nabla_{h_k} \mathcal{L}(\theta) h_{k-1}^\top \in \mathbb{R}^{m \times m}.$$

Its (i, j)-th entry is

$$\big(D_k \nabla_{h_k} \mathcal{L}(\theta) h_{k-1}^\top\big)_{ij} = \frac{\partial \mathcal{L}(\theta)}{\partial (h_k)_i} \sigma'((a_k)_i)(h_{k-1})_j.$$

This is exactly the contribution of time step k to $\partial \mathcal{L}(\theta)/\partial W_{ij}$. Summing these matrices over all time steps gives

$$\left(\sum_{k=1}^{T} D_k \nabla_{h_k} \mathcal{L}(\theta) h_{k-1}^\top \right)_{ij} = \frac{\partial \mathcal{L}(\theta)}{\partial W_{ij}}.$$

By the definition of the matrix gradient,

$$\nabla_W \mathcal{L}(\theta) = \sum_{k=1}^{T} D_k \nabla_{h_k} \mathcal{L}(\theta) h_{k-1}^\top \in \mathbb{R}^{m \times m}.$$

For U, the same calculation uses

$$\frac{\partial (a_k)_i}{\partial U_{ij}^{(k)}} = (x_k)_j$$

in place of $(h_{k-1})_j$. This gives

$$\nabla_U \mathcal{L}(\theta) = \sum_{k=1}^{T} D_k \nabla_{h_k} \mathcal{L}(\theta) x_k^\top \in \mathbb{R}^{m \times d}.$$

For the bias parameter copies,

$$\frac{\partial (a_k)_i}{\partial b_j^{(k)}} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

The same calculation gives

$$\nabla_b \mathcal{L}(\theta) = \sum_{k=1}^{T} D_k \nabla_{h_k} \mathcal{L}(\theta) \in \mathbb{R}^m.$$

The parameter copy $c^{(k)}$ does not enter the state recursion, so the hidden state h_k does not depend on $c^{(k)}$. It enters directly through

$$\widehat{y}_k = (c^{(k)})^\top h_k.$$

For $i = 1, \ldots, m$,

$$\frac{\partial \widetilde{\mathcal{L}}}{\partial c_i^{(k)}} = \frac{2}{T}\big((c^{(k)})^\top h_k - y_k\big)(h_k)_i.$$

Adding the contributions of all parameter copies and collecting the coordinates gives

$$\nabla_c \mathcal{L}(\theta) = \frac{2}{T} \sum_{k=1}^{T} \big(c^\top h_k - y_k\big) h_k \in \mathbb{R}^m.$$

This proves all formulas in Proposition 2.2. □

8.2 Numerical Validation of the Plateau Experiment

This appendix gives additional details on the numerical computation used in Section 6.5. In our first implementation, we evaluated the integrals in the loss numerically. However, this was slow and led to numerical problems. Furthermore, we had to truncate the integration interval, although the loss is defined on $[0, \infty)$.

We therefore derive a closed form expression for the loss, which allows us to evaluate it without truncating the integration interval. We also check the numerical results using finite differences, independent quadrature, and stricter ODE solver tolerances. The derivation and the numerical checks are our own.

Recall that the model kernel is

$$\widehat{\rho}_{a,w}(t) = \sum_{i=1}^{m} a_i e^{-w_i t},$$

where

$$(a, w) \in \mathbb{R}^m \times (0, \infty)^m.$$

The target kernel used in the experiment is

$$\rho_\omega(t) = e^{-t/2} + e^{-2(t-L)^2}, \qquad L = \frac{1}{\omega}.$$

The reduced loss is

$$J_\omega(a, w) = \int_0^\infty \left(\widehat{\rho}_{a,w}(t) - \rho_\omega(t)\right)^2 \, \mathrm{d}t.$$

Expanding the square gives

$$J_\omega(a,w) = \int_0^\infty \widehat{\rho}_{a,w}(t)^2 \,\mathrm{d}t - 2\int_0^\infty \widehat{\rho}_{a,w}(t)\rho_\omega(t)\,\mathrm{d}t + \|\rho_\omega\|^2_{L^2([0,\infty))}.$$

The first term can be evaluated explicitly because $w_i, w_j > 0$ and

$$\int_0^\infty e^{-(w_i+w_j)t}\,\mathrm{d}t = \frac{1}{w_i+w_j}.$$

Defining the matrix $G \in \mathbb{R}^{m\times m}$ by

$$G_{ij} \coloneqq \frac{1}{w_i+w_j}, \qquad i,j = 1,\dots,m,$$

we get

$$\int_0^\infty \widehat{\rho}_{a,w}(t)^2\,\mathrm{d}t = \sum_{i=1}^m\sum_{j=1}^m a_i a_j \int_0^\infty e^{-(w_i+w_j)t}\,\mathrm{d}t = a^\top G a.$$

The cross term additionally contains integrals of the form

$$I(w,L) \coloneqq \int_0^\infty e^{-wt}e^{-2(t-L)^2}\,\mathrm{d}t, \qquad w, L > 0.$$

These can also be evaluated explicitly. Completing the square in the exponent gives

$$-wt - 2(t-L)^2 = -2\left(t - L + \frac{w}{4}\right)^2 - wL + \frac{w^2}{8}.$$

Therefore,

$$I(w,L) = e^{-wL+w^2/8}\int_0^\infty \exp\left(-2\left(t-L+\frac{w}{4}\right)^2\right)\,\mathrm{d}t.$$

To express the remaining Gaussian tail integral, we use the complementary error function $\operatorname{erfc}\colon \mathbb{R}\to(0,2)$ [Olv+10, p. 160, Eq. (7.2.2)],

$$\operatorname{erfc}(x) \coloneqq \frac{2}{\sqrt{\pi}}\int_x^\infty e^{-u^2}\,\mathrm{d}u.$$

For the numerical evaluation, we use `scipy.special.erfc` [VGO+20].

With the substitution

$$u = \sqrt{2}\left(t - L + \frac{w}{4}\right),$$

we obtain

$$I(w,L) = \sqrt{\frac{\pi}{8}}e^{-wL+w^2/8}\operatorname{erfc}\left(\frac{w-4L}{2\sqrt{2}}\right).$$

Consequently, the loss can be evaluated entirely in closed form as

$$J_\omega(a,w) = a^\top G a - 2\sum_{i=1}^m a_i\left[\frac{1}{w_i+1/2} + \sqrt{\frac{\pi}{8}}e^{-w_iL+w_i^2/8}\operatorname{erfc}\left(\frac{w_i-4L}{2\sqrt{2}}\right)\right] + 1 + 2\sqrt{\frac{\pi}{8}}e^{-L/2+1/32}\operatorname{erfc}\left(\frac{1/2-4L}{2\sqrt{2}}\right) + \frac{\sqrt{\pi}}{4}\operatorname{erfc}(-2L).$$

The loss can therefore be evaluated directly on the full interval $[0, \infty)$, without truncating the memory domain. Its gradient is obtained by differentiating these closed form expressions with respect to the model parameters.

The resulting gradient flow

$$\vartheta'_\omega(\tau) = -\nabla J_\omega\big(\vartheta_\omega(\tau)\big), \qquad \vartheta_\omega(0) = \vartheta_0,$$

is integrated using the adaptive explicit Runge–Kutta method DOP853 [HNW93, pp. 181–185], implemented in SciPy's `solve_ivp` [VGO+20]. The primary solver configuration uses

$$\mathrm{rtol} = 10^{-9}, \qquad \mathrm{atol} = 10^{-11},$$

with a maximum step size of 1. The solver chooses its step size adaptively, so the gradient flow is not evaluated on a fixed time grid.

As in Section 6.5, we use the fixed tolerance

$$\delta = 0.05.$$

The parameter hitting time τ_0 is determined by the first time at which

$$\|\vartheta_\omega(\tau) - \vartheta_0\|_2 = \delta,$$

whereas the loss hitting time τ'_0 is determined by the first time at which

$$J_\omega(\vartheta_0) - J_\omega\big(\vartheta_\omega(\tau)\big) = \delta.$$

These crossings are located using the event detection in SciPy's `solve_ivp` [VGO+20].

To check the numerical implementation, three consistency tests were performed. First, the analytically evaluated gradient was compared with a centered finite difference approximation at a test parameter point with $L = 12$. The relative discrepancy was approximately

$$4 \times 10^{-10}.$$

Second, the closed form loss was compared with an independent numerical quadrature of the original integral defining J_ω. We used SciPy's `quad` [VGO+20], splitting the integration domain into three intervals around the Gaussian peak. The relative difference was below 2×10^{-9}.

Finally, the complete gradient flow computation was repeated with stricter solver settings,

$$\mathrm{rtol} = 2 \times 10^{-11}, \qquad \mathrm{atol} = 2 \times 10^{-13},$$

and a reduced maximum step size of 0.5. Across all memory scales used in Section 6.5, the resulting hitting times agreed with those from the primary computation up to a maximum relative discrepancy of approximately

$$4 \times 10^{-10}.$$

These checks do not verify the asymptotic result of Theorem 6.5, which is a theoretical statement. They indicate, however, that the numerical plateau behavior reported in Section 6.5 is stable with respect to independent evaluations of the loss and gradient and to a substantial refinement of the ODE solver.

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Use of Generative AI

I mainly used ChatGPT (OpenAI), including the models GPT-5.4, GPT-5.5, GPT-5.6, GPT-6, and GPT-6.1. I also used Claude Opus 5.5 (Anthropic) for a final review of the thesis.

I used ChatGPT extensively to learn the mathematical and technical background needed for this thesis. In particular, I used it to learn about machine learning, neural networks, and recurrent neural networks, as well as concepts from functional analysis, measure theory, and approximation theory. I had not taken university courses in most of these topics before. This learning process also indirectly influenced my understanding of the mathematical topics discussed throughout the thesis.

I mainly used ChatGPT as a discussion partner when I was unsure about mathematical arguments or proofs. For example, when working on the inverse approximation theorem of Li et al. [LHEL22, pp. 12–13, Thm. 11], I could not justify one of the estimates used in their proof. I then used ChatGPT to discuss whether the estimate was correct. ChatGPT suggested the idea of using Jordan blocks to construct a counterexample, which I then worked out in Proposition 5.12.

I also used ChatGPT to find possible errors or missing steps in my proofs and mathematical arguments, particularly in Chapters 2–6 and the appendix. Furthermore, I used it for literature research, writing and debugging Python code for the numerical experiments in Chapters 2, 5, and 6, and improving the English language and structure of the thesis.

I used Claude Opus 5.5 for a final review, mainly to identify possible mathematical errors, missing assumptions, and problems with references and citations.

One difficulty I noticed was that AI sometimes gave incorrect mathematical arguments or suggested steps that did not follow from the assumptions. Furthermore, its explanations were often difficult to understand and not very helpful from a didactic perspective, even when the underlying arguments were simple. I therefore used the original literature as the main source for the mathematical results and ChatGPT to help me understand and discuss the arguments.

I take full responsibility for the final content and I am able to explain all AI influenced parts of this work.

Erklärung

Ich versichere wahrheitsgemäß, die Arbeit selbstständig verfasst, alle benutzten Hilfsmittel vollständig und genau angegeben und alles kenntlich gemacht zu haben, was aus Arbeiten anderer unverändert oder mit Abänderungen entnommen wurde, sowie die Satzung des KIT zur Sicherung guter wissenschaftlicher Praxis in der jeweils gültigen Fassung beachtet zu haben.

Karlsruhe, _____ _____

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